



Image Analysis

FROM PIXELS TO BIOLOGICAL INSIGHTS

GET STARTED



Basics &
Communities
(e.g. image.sc)

GET HELP



Claude, ChatGPT
& Gemini
(optional)

DENOISE & ANALYZE

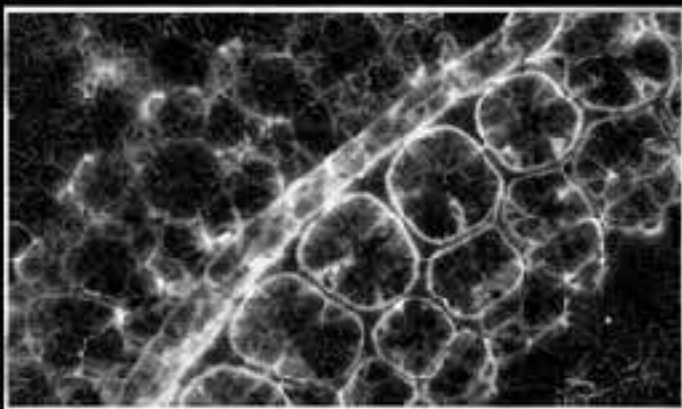


Python, CAREamics
(local or Colab)

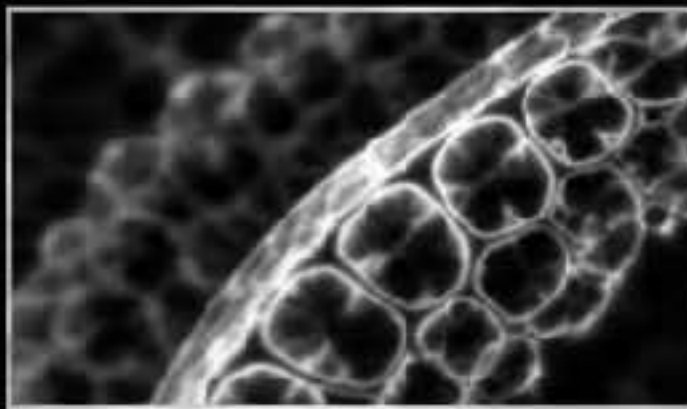
FROM IMAGES
TO INSIGHTS



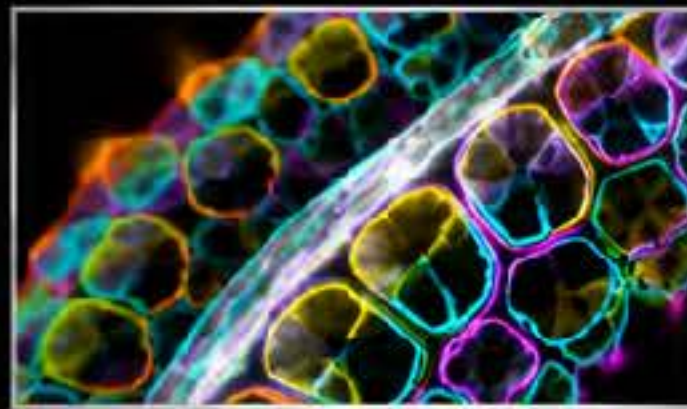
Quantification &
Biological Questions



Noisy image



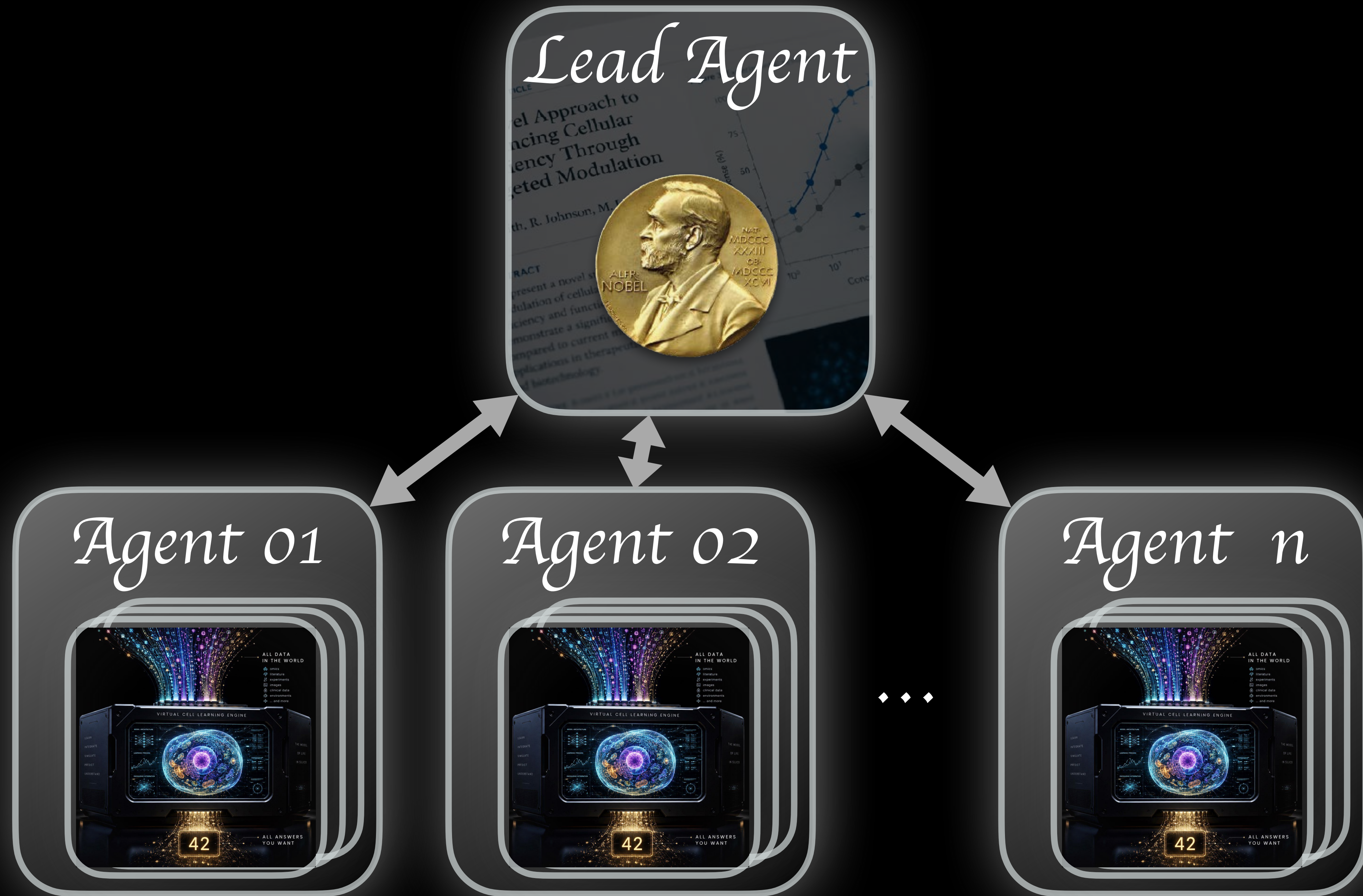
Denoised (CAREamics)

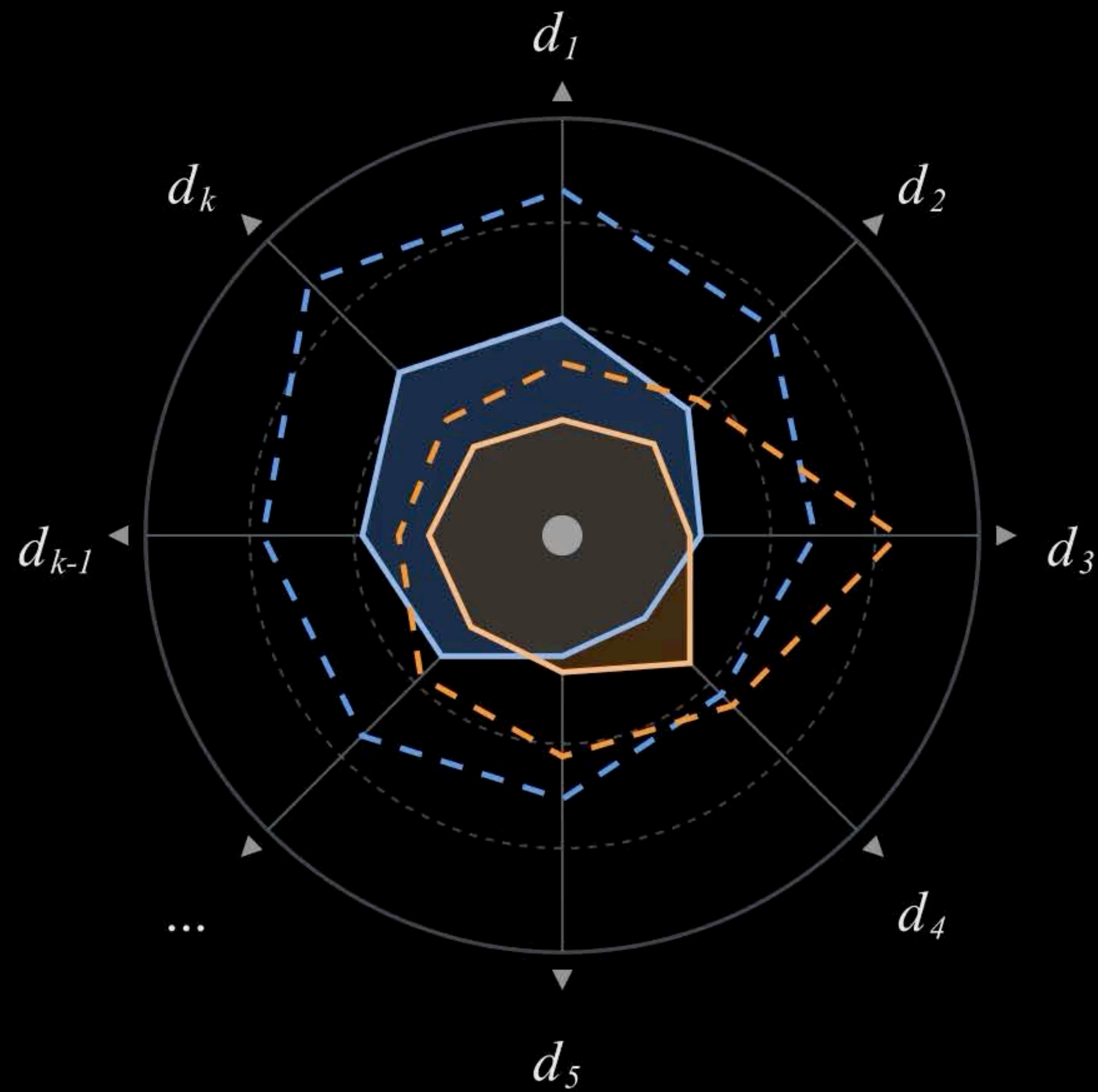


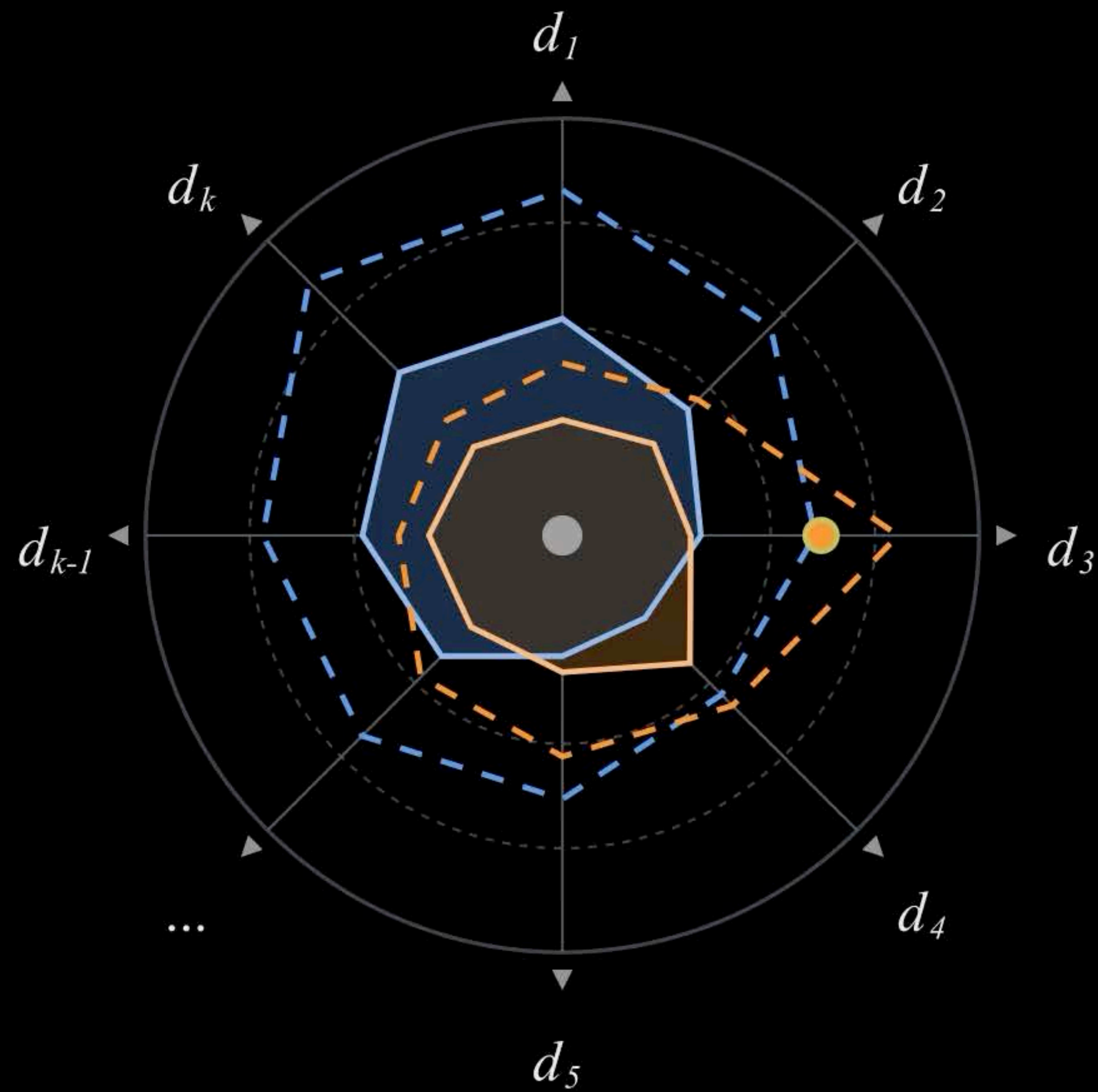
Analysis (segmentation, quantification)

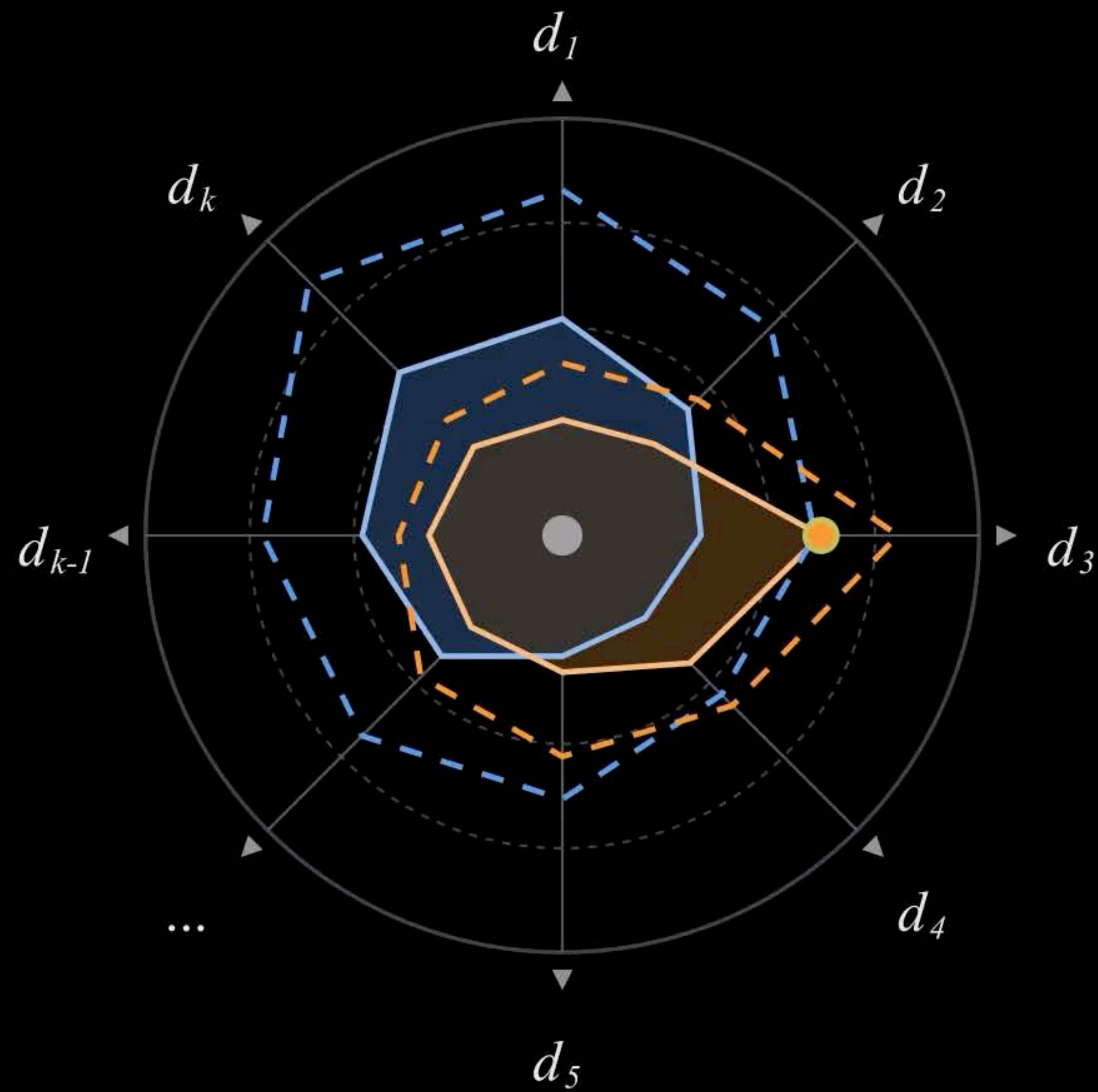
...AND HOW TO THINK ABOUT AGENTIC SUPPORT...
A SHORT COURSE FOR BEGINNERS AND MORE ADVANCED YOUNG SCIENTISTS

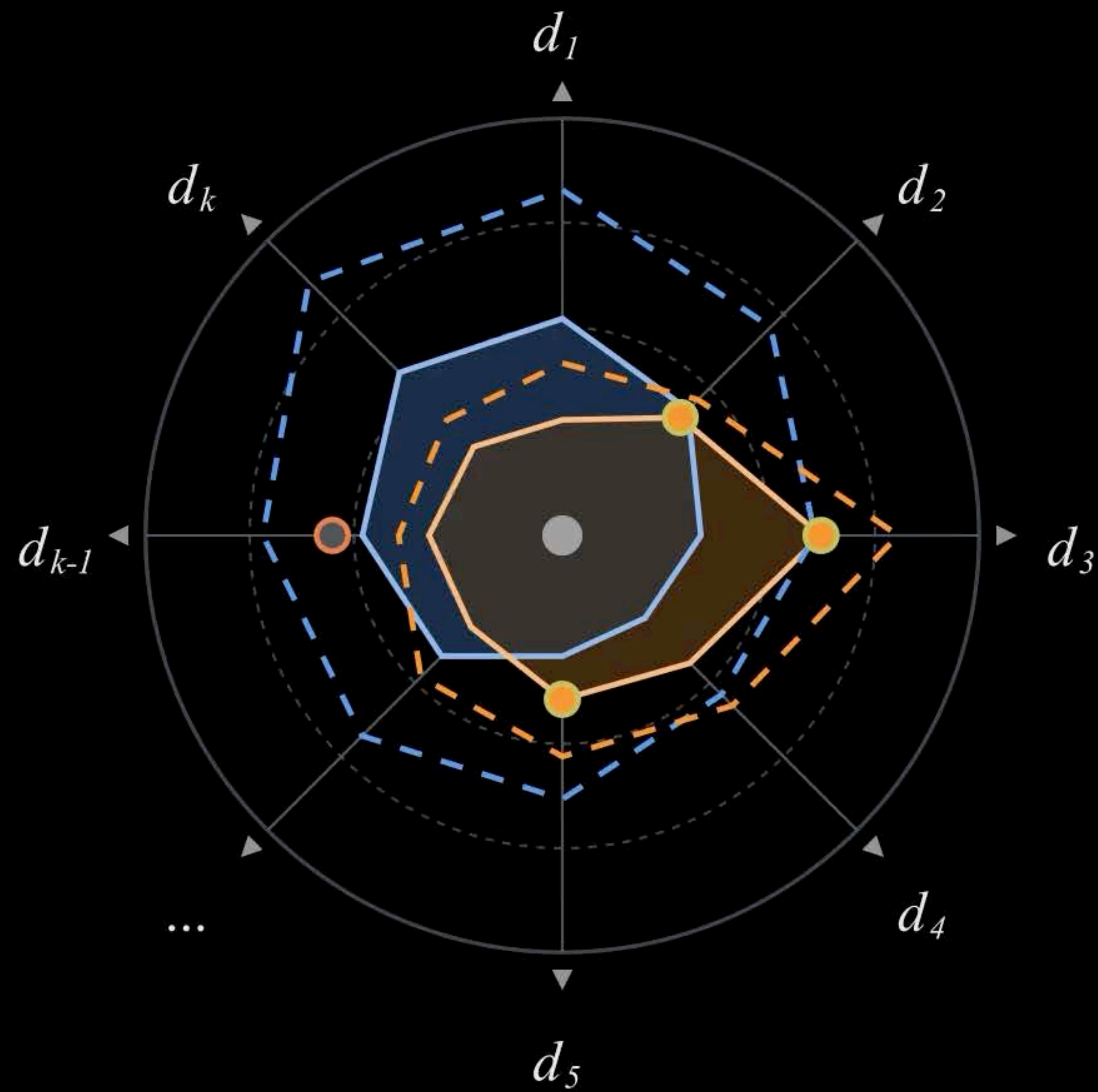


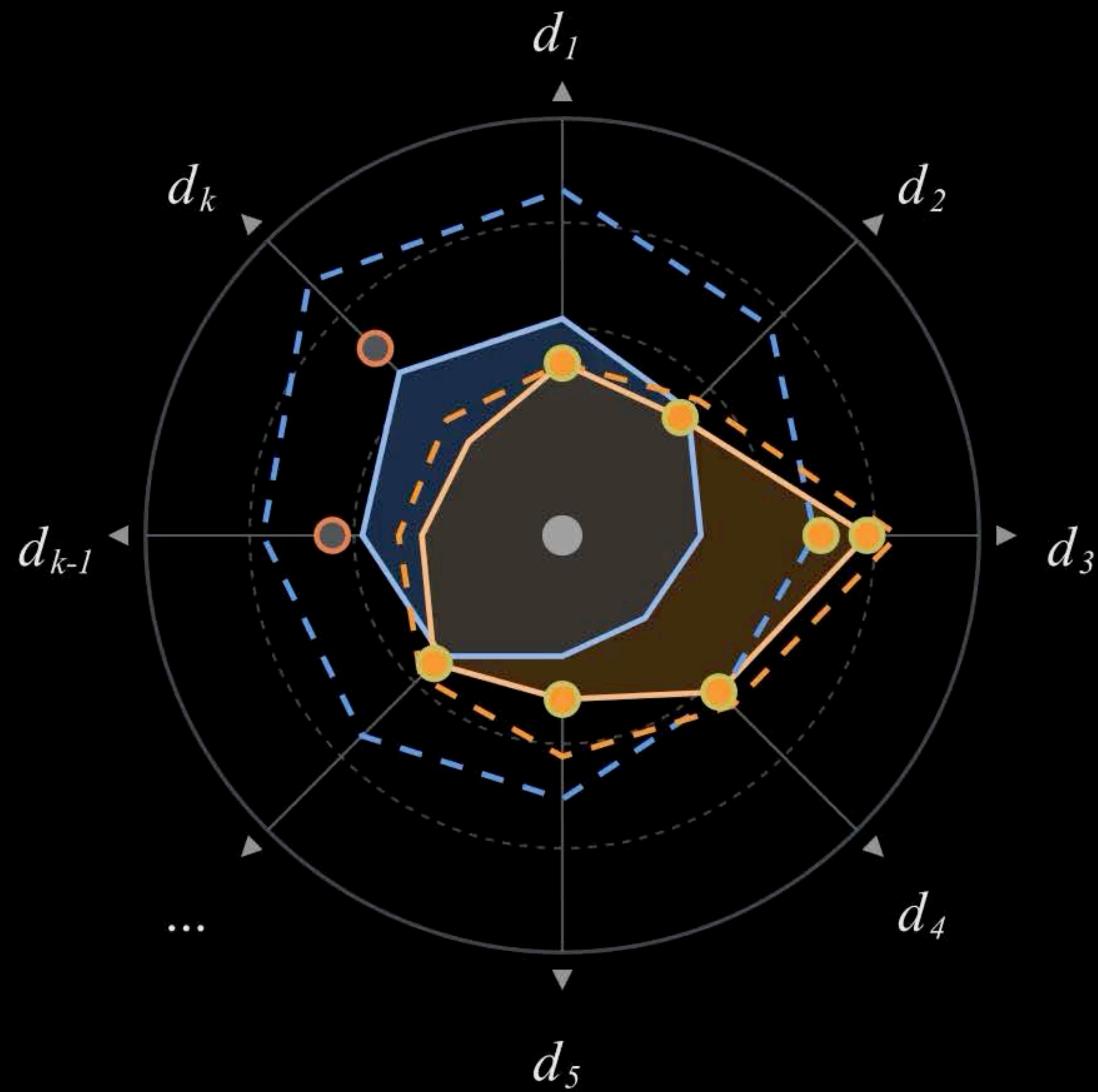












WELCOME TO THE COURSE!

- ▶ **Who we are, who you are.**

A show of hands: biologist, physicist, computational? Used Fiji? Written Python? Used ChatGPT or Claude for work this week? Note the answers, because they're useful for pairing.

- ▶ **Wednesday** (today): images, AI, denoising.

Thursday: recap, then segmentation, cilia segmentation, CiliaQ, ...

Friday: predicting ciliary protein structures and complexes with AlphaFold

- ▶ **One neutral question for the week:**

"How is AI changing data analysis, and what does that mean for you?"

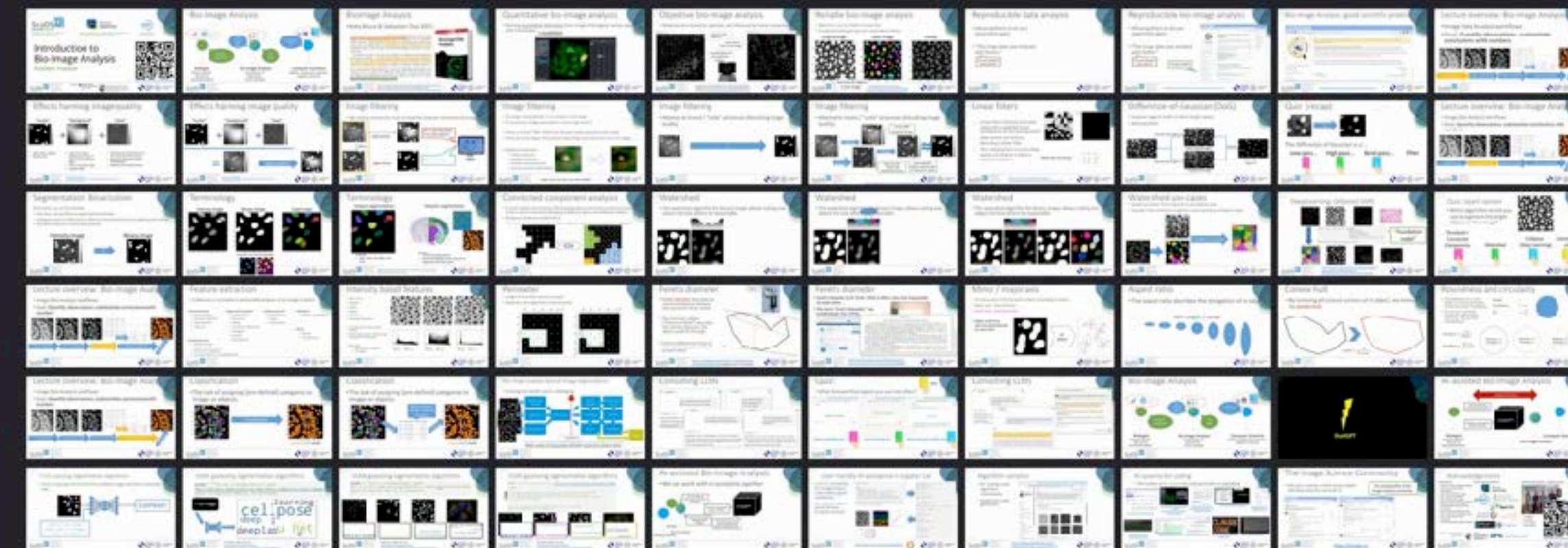
1. WHAT IS AN IMAGE? WHAT IS IMAGE ANALYSIS?
2. CLASSICAL ANALYSIS & MACHINE LEARNING
3. FROM DEEP LEARNING TO AGENTIC SYSTEMS
4. PRACTICAL EXERCISE: GETTING YOUR HANDS DIRTY WITH UNSUPERVISED DENOISING

1. WHAT IS AN IMAGE? WHAT IS IMAGE ANALYSIS?
2. CLASSICAL ANALYSIS & MACHINE LEARNING
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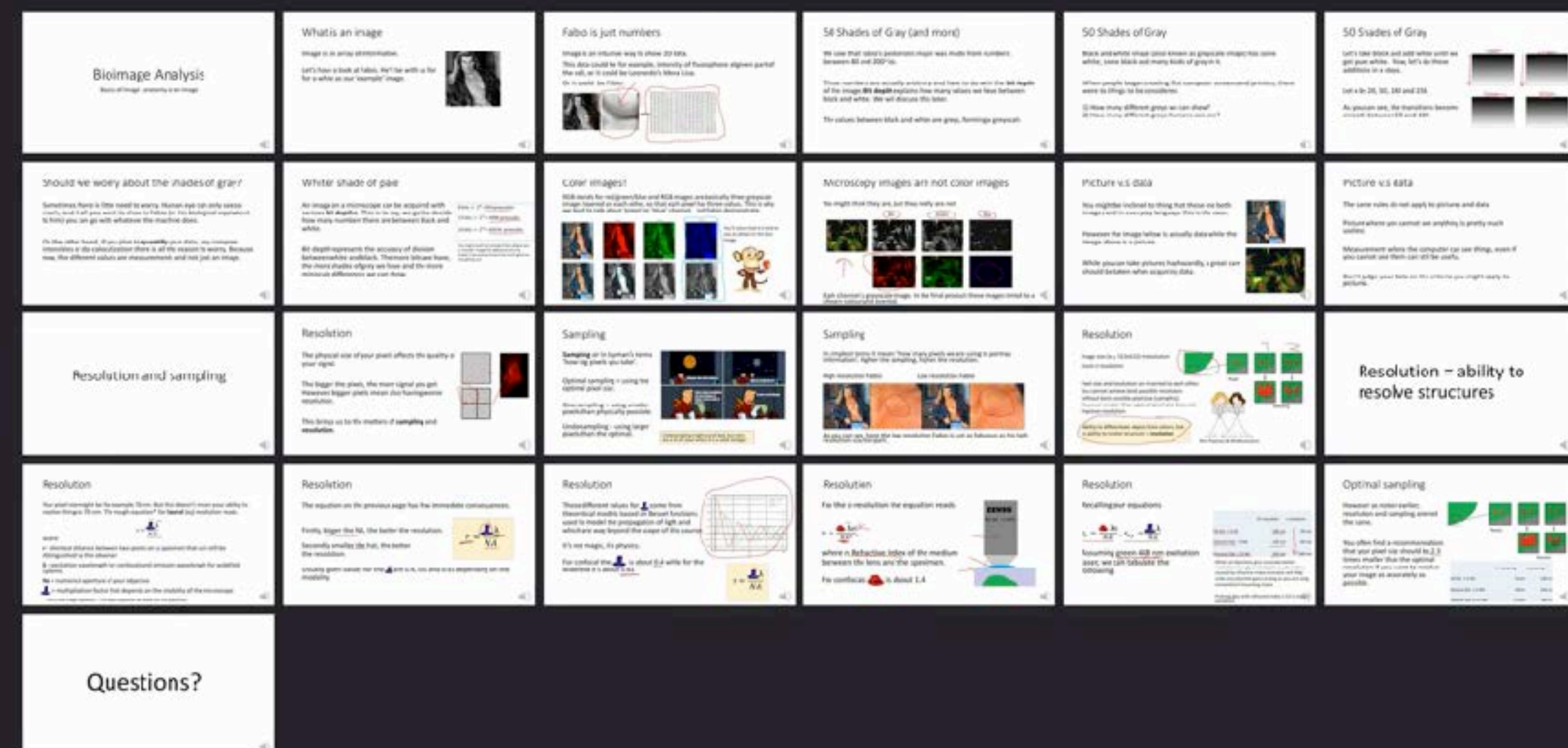
THESE SLIDES MUST HAVE EXISTED BEFORE...



Peter Bankhead · Analyzing fluorescence microscopy images with ImageJ (2014) · 195 pages



Robert Haase · Introduction to Bio-Image Analysis · doi.org/10.5281/zenodo.21370808



Marko Crivaro · Bioimage Analysis: anatomy of an image · Biomedicum Imaging Unit, University of Helsinki



Senft et al. · A biologist's guide to planning and performing quantitative bioimaging experiments · PLoS Biol 2023

THESE SLIDES MUST HAVE EXISTED BEFORE...

- ▶ **Pete Bankhead, Introduction to Bioimage Analysis (CC-BY 4.0)**

<https://bioimagebook.github.io/>

- ▶ **Robert Haase, Introduction to Bio-Image Analysis (CC-BY 4.0)**

<https://zenodo.org/records/21370808>

- ▶ **Helsinki Biomedicum Imaging Unit, "What is an image?":**

<https://wiki.helsinki.fi/xwiki/bin/download/biu/Biomedicum%20Imaging%20Unit/BIU%20courses/>

[Bioim%C2%ADage%20Analysis%20Course%202022/WebHome/What%20is%20an%20image%202020.pdf?rev=1.2](https://wiki.helsinki.fi/xwiki/bin/download/biu/Biomedicum%20Imaging%20Unit/BIU%20courses/Bioim%C2%ADage%20Analysis%20Course%202022/WebHome/What%20is%20an%20image%202020.pdf?rev=1.2)

- ▶ **Senft et al., A biologist's guide to planning and performing quantitative bioimaging experiments (PLoS Biol 2023)**

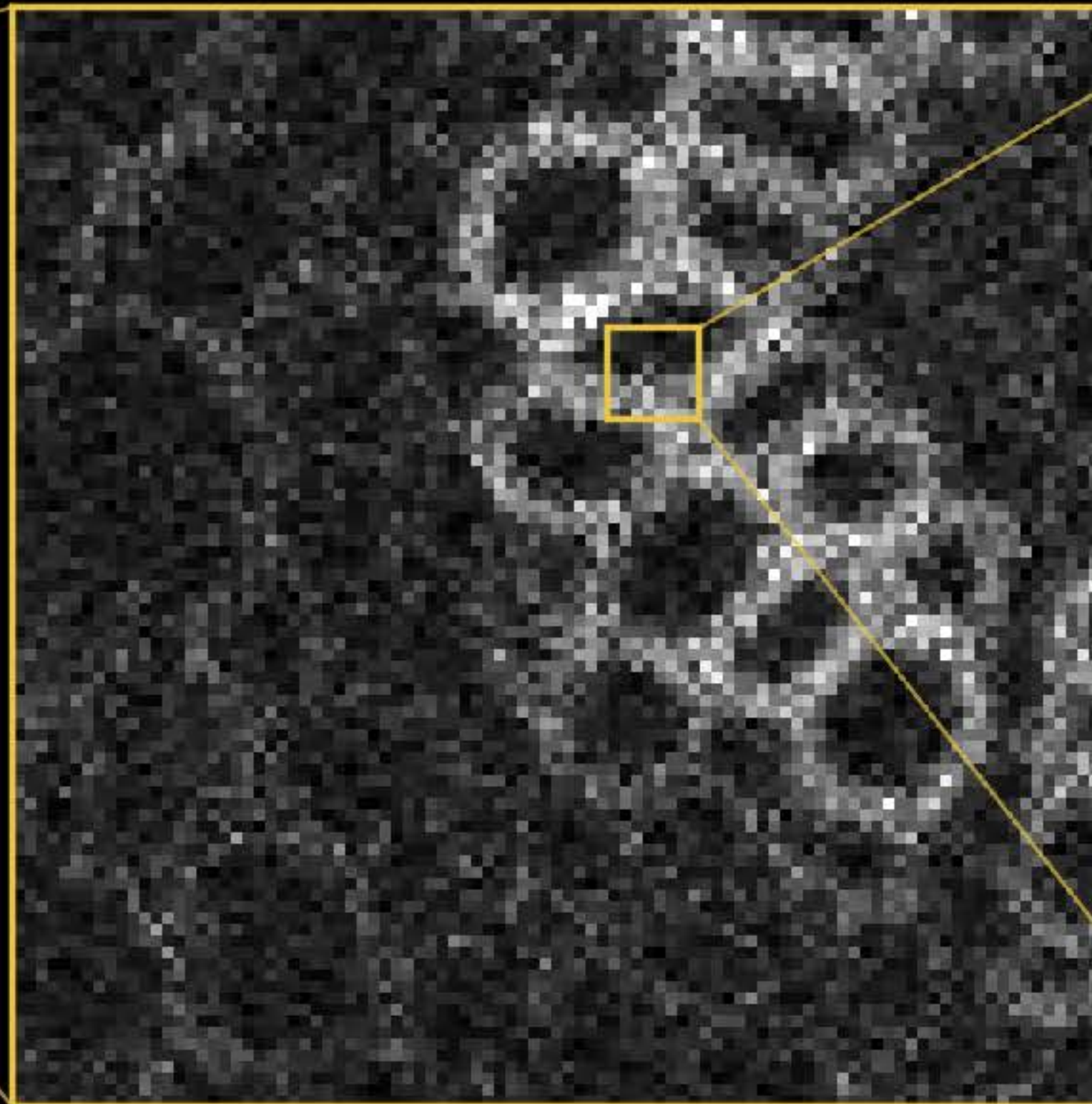
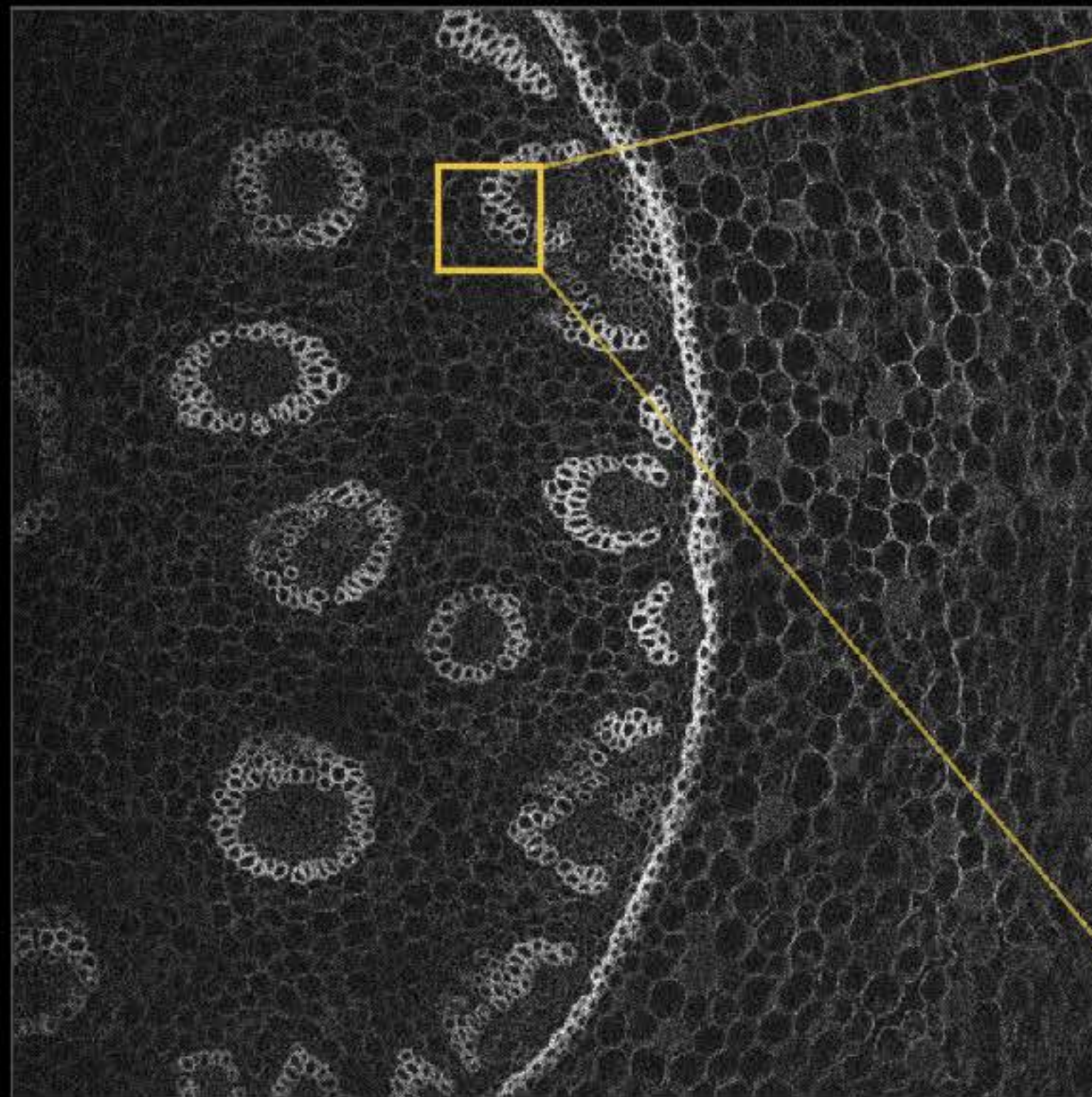
<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10298797/>

LESSON QUICKIE 1/6: IMAGES ARE ARRAYS OF NUMBERS

1024 × 1024 pixels

96 × 96 pixels

8 × 8 pixels = 64 numbers



568	516	512	486	418	465	542	510
567	531	506	501	462	512	490	469
455	493	517	571	550	480	490	474
553	434	496	746	505	571	534	606
776	584	903	570	683	779	770	841
758	857	727	997	679	709	710	862
937	703	734	976	759	836	714	566
673	547	987	783	991	942	921	838

Also look at: Bankhead Ch. 1 "Images & pixels" and the opening slides of Haase's course.

LESSON QUICKIE 2/6: YOUR EYES ARE AMAZING; YOUR EYES SUCK!



LESSON QUICKIE 2/6: YOUR EYES ARE AMAZING; YOUR EYES SUCK!

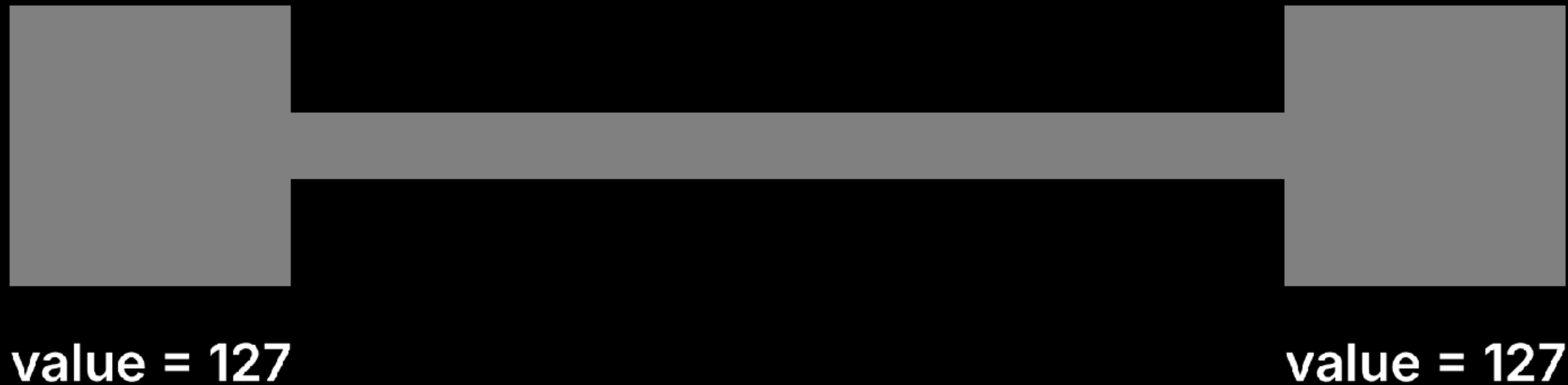


value = 127

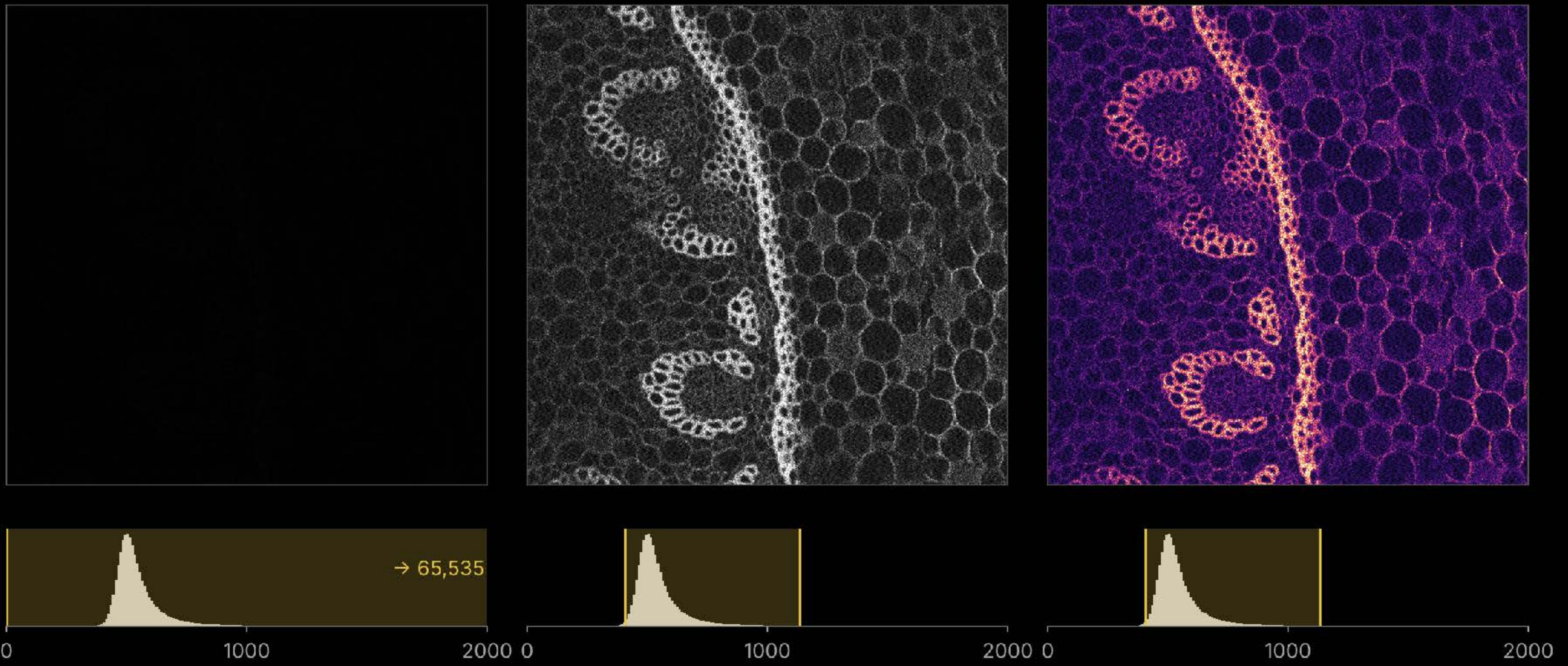


value = 127

LESSON QUICKIE 2/6: YOUR EYES ARE AMAZING; YOUR EYES SUCK!



LESSON QUICKIE 3/6: DISPLAY $\neq$ DATA



display range 0 – 65,535

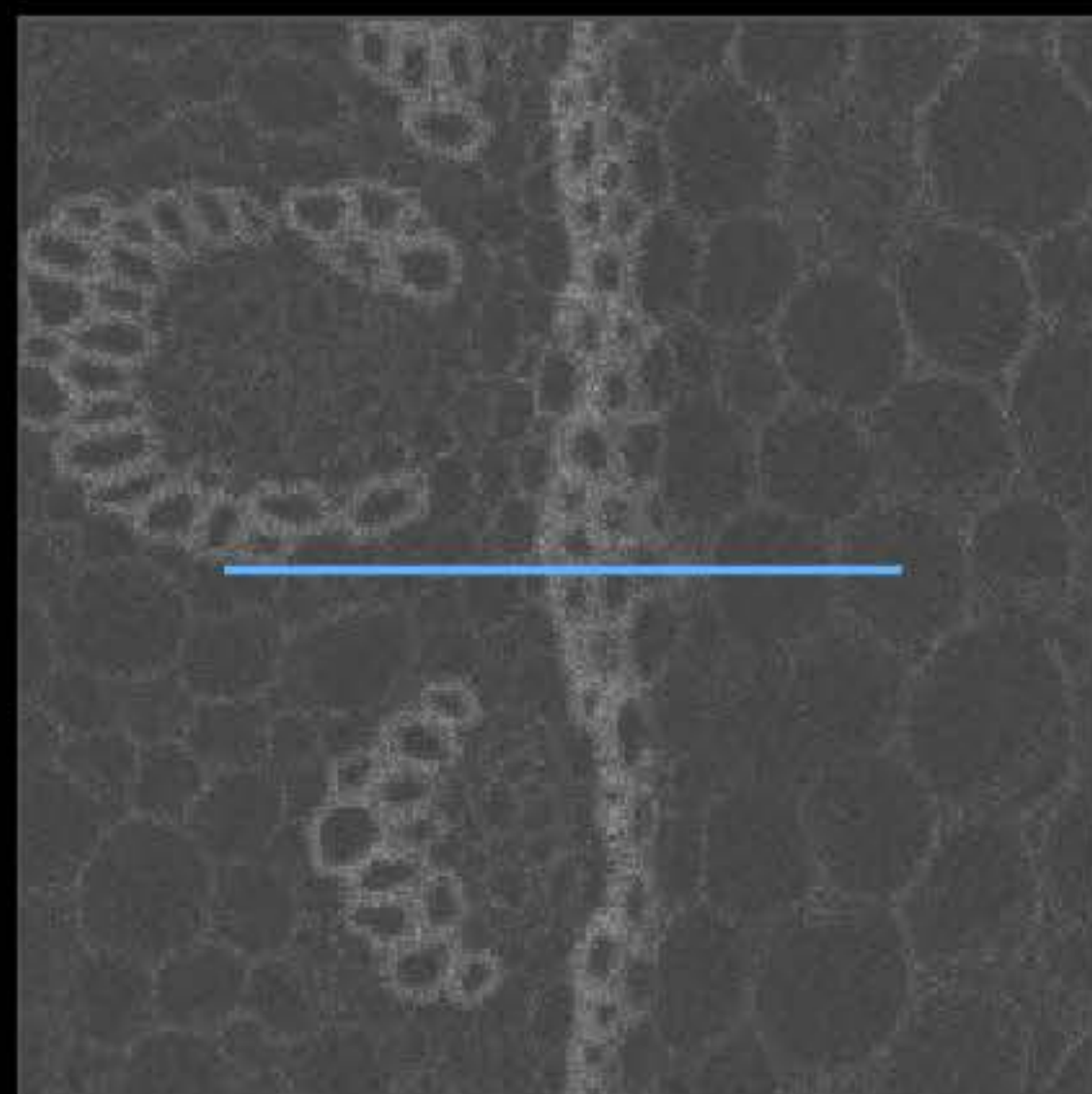
display range 409 – 1137

same range, other colour map

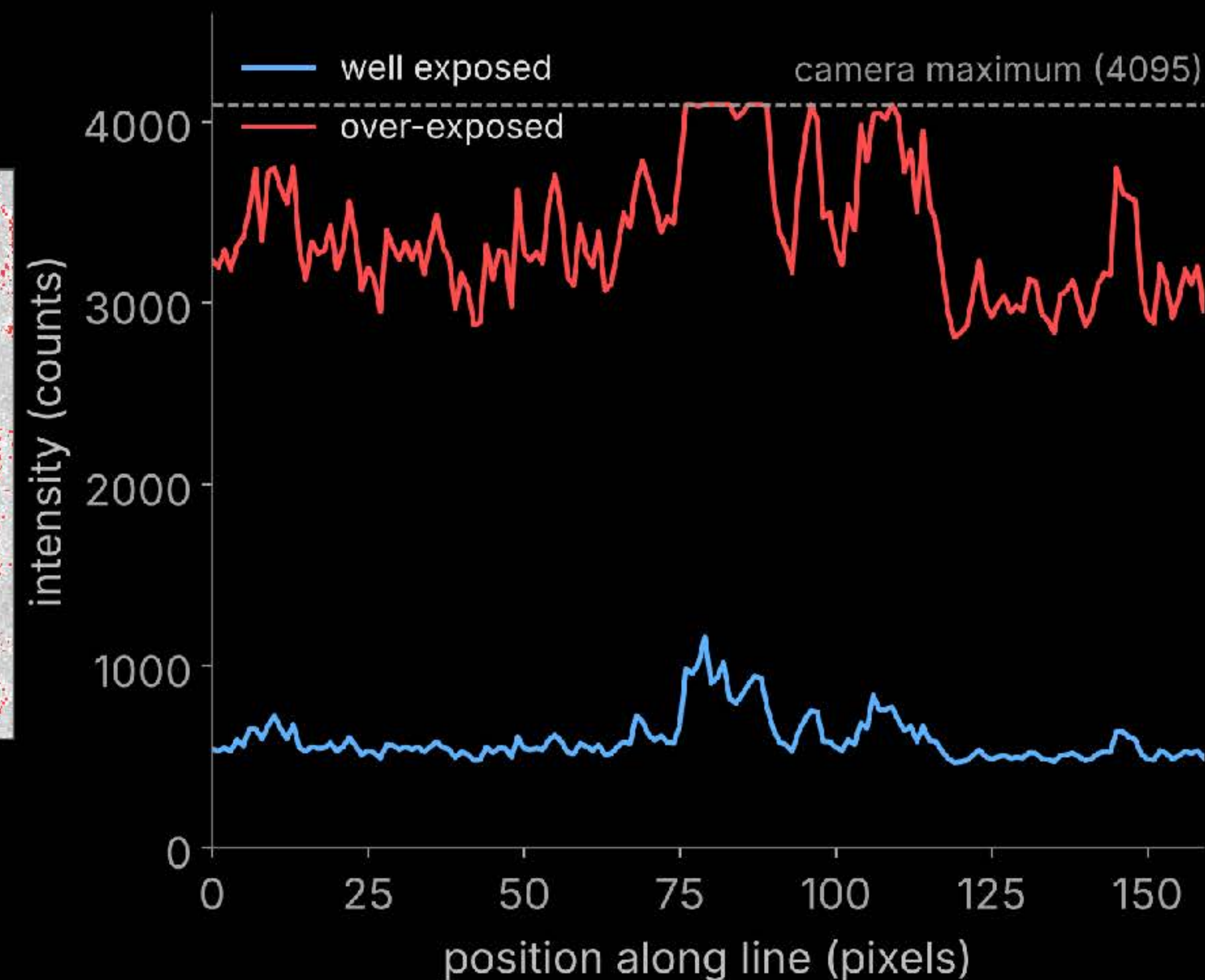
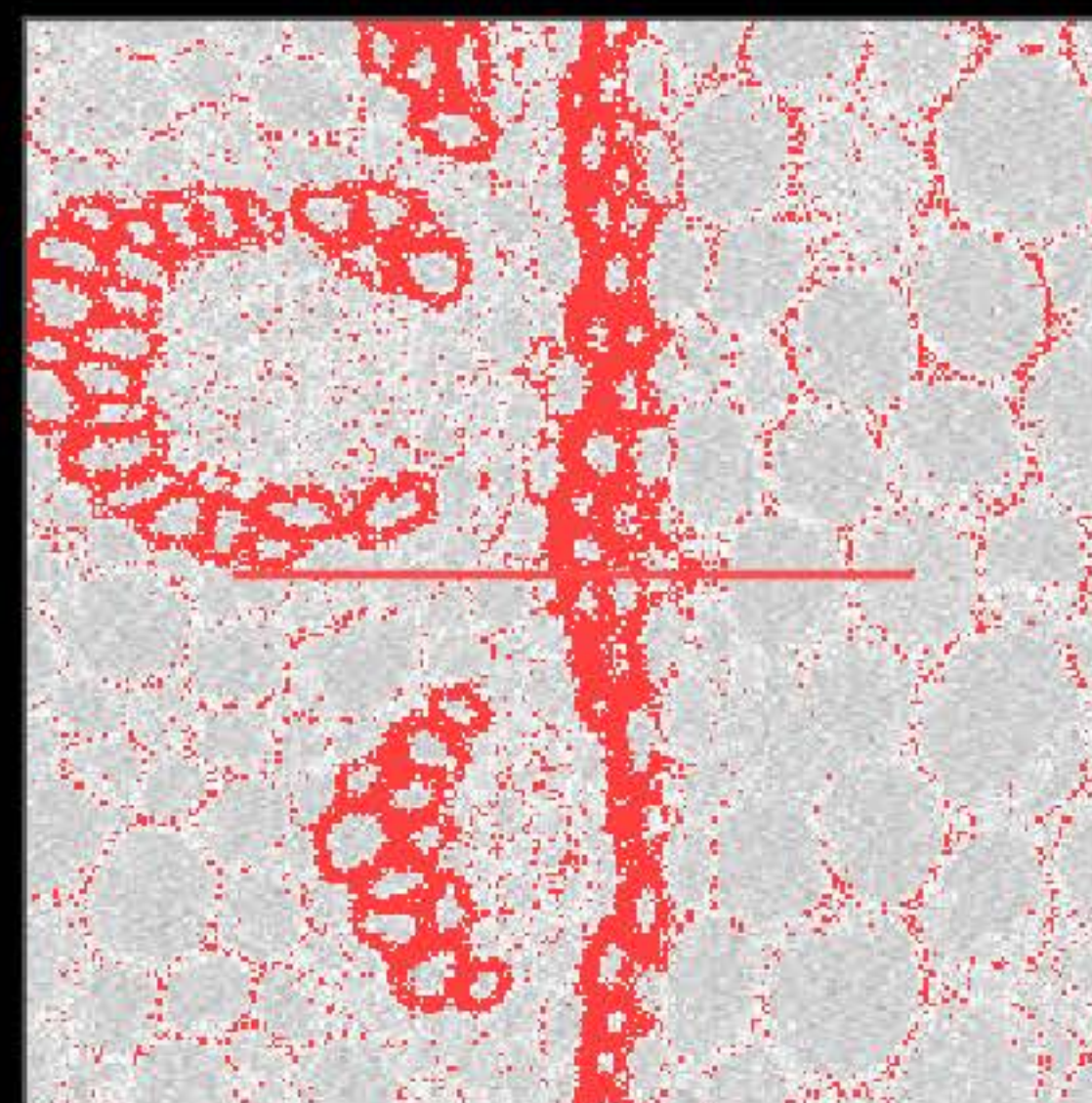
Also look at: Bankhead Ch. 1, the lookup-table section.

LESSON QUICKIE 4/6: SATURATION AND CLIPPING

well exposed



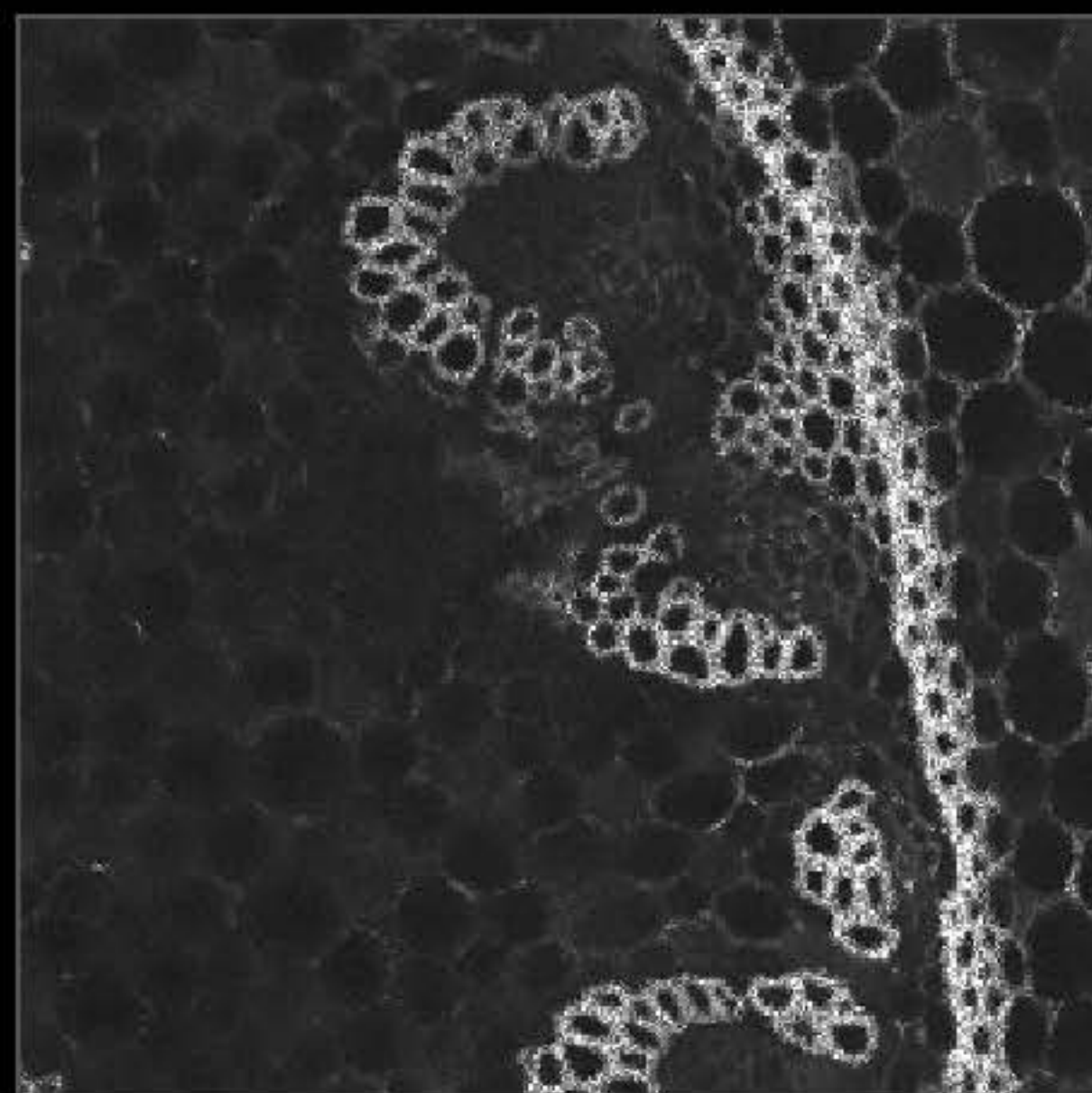
over-exposed (6× exposure)



red: 18% of pixels at 4095

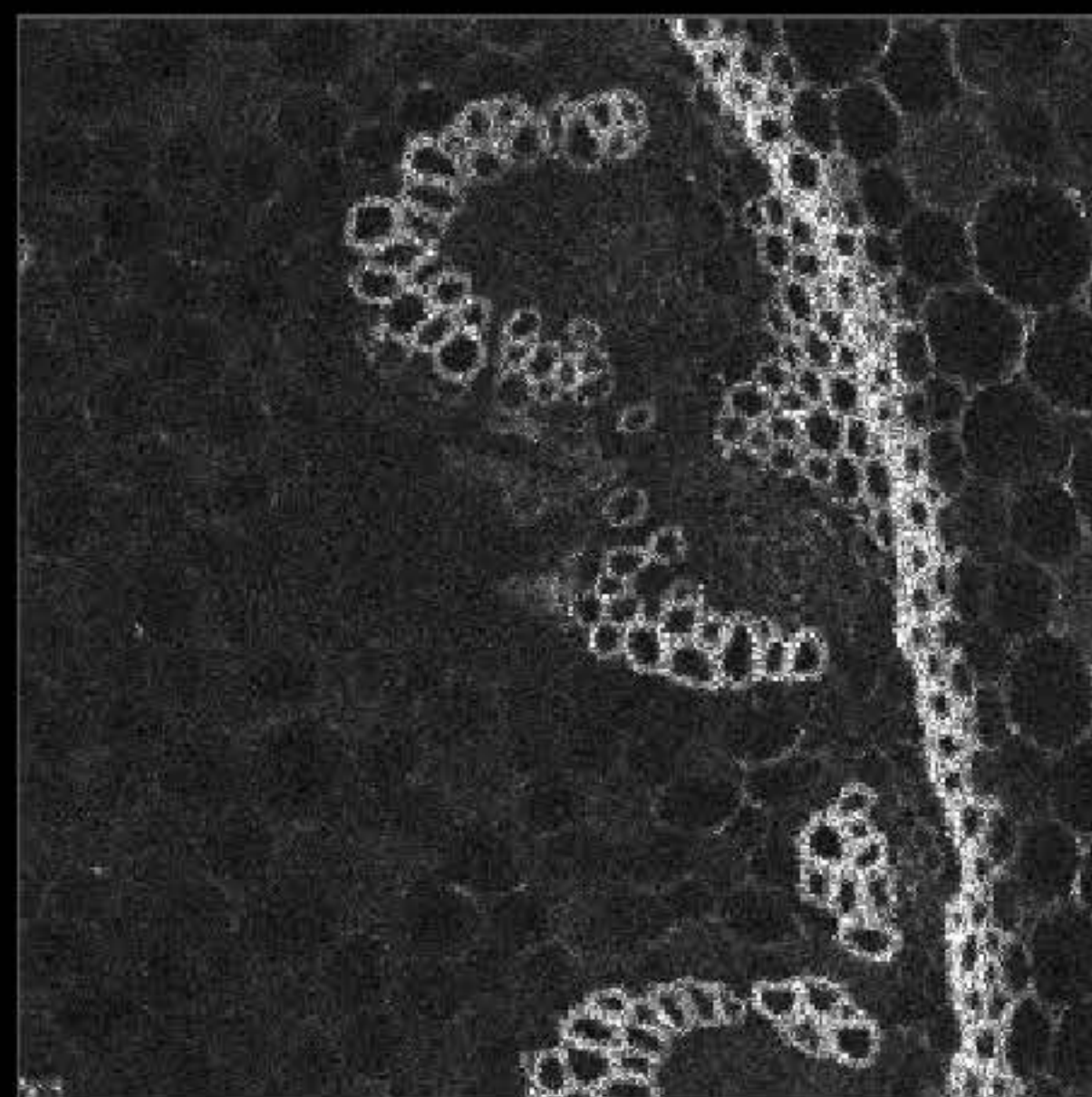
LESSON QUICKIE 5/6: NOISE IS PHYSICS, NOT A BAD CAMERA

≈ 1000 photons / bright pixel



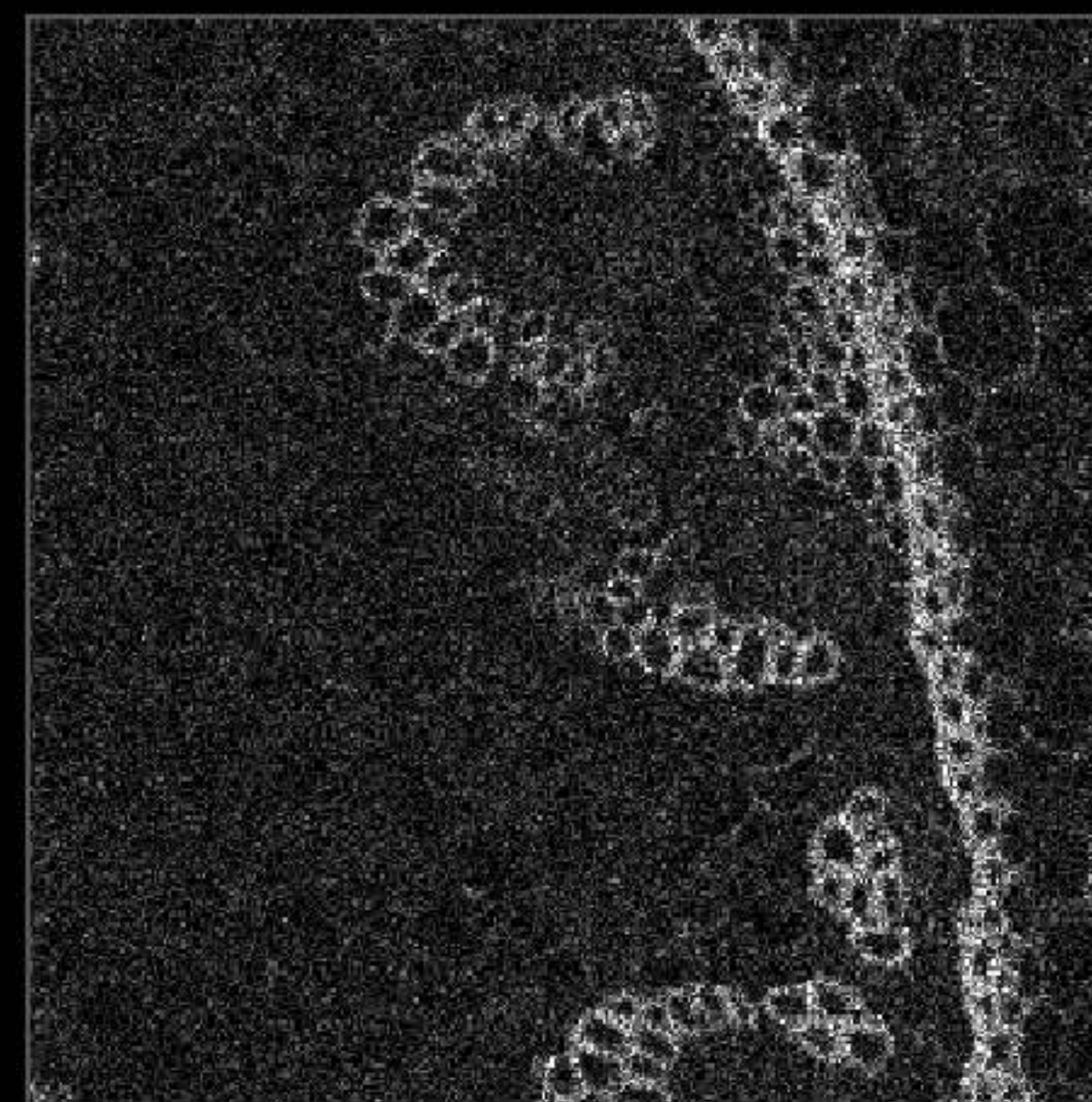
$\text{SNR} \approx \sqrt{1000} \approx 32$

≈ 100 photons / bright pixel



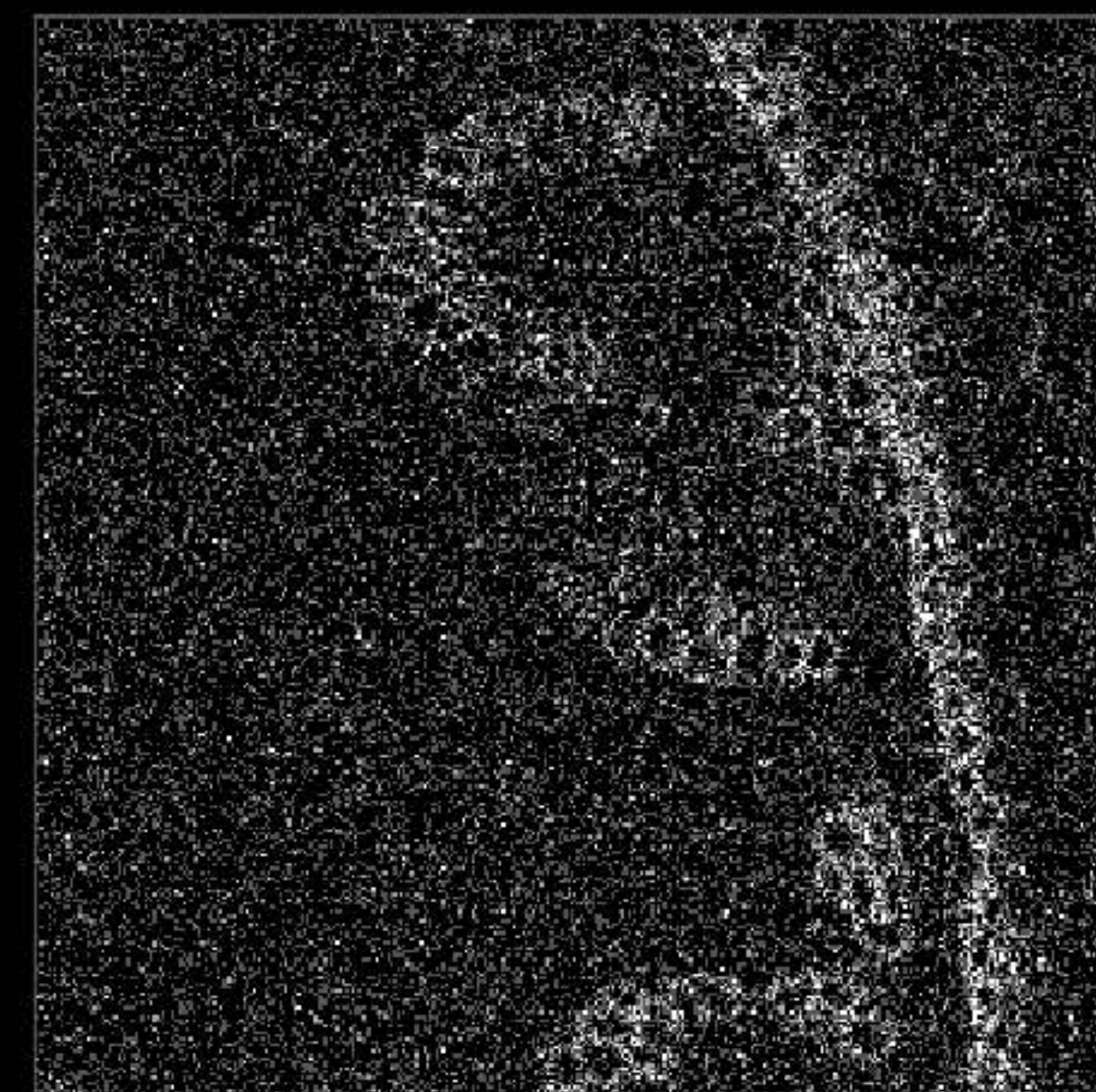
$\text{SNR} \approx \sqrt{100} \approx 10$

≈ 10 photons / bright pixel



$\text{SNR} \approx \sqrt{10} \approx 3.2$

≈ 2 photons / bright pixel

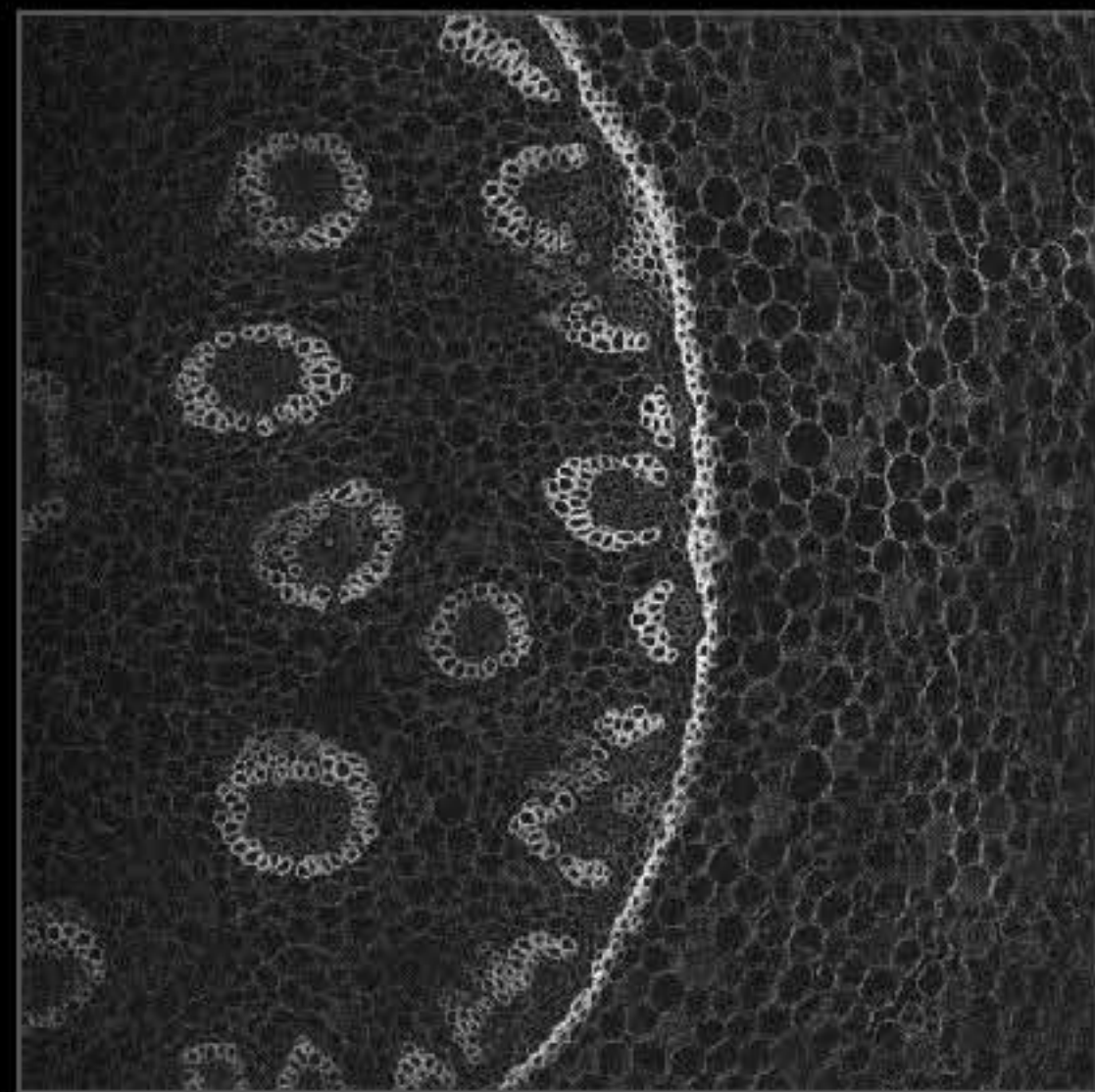


$\text{SNR} \approx \sqrt{2} \approx 1.4$

Also look at: Bankhead Ch. 16 "Noise", which covers photon noise and simulates exactly this.

LESSON QUICKIE 6/6: IMAGE ANALYSIS STARTS WITH A QUESTION!!!

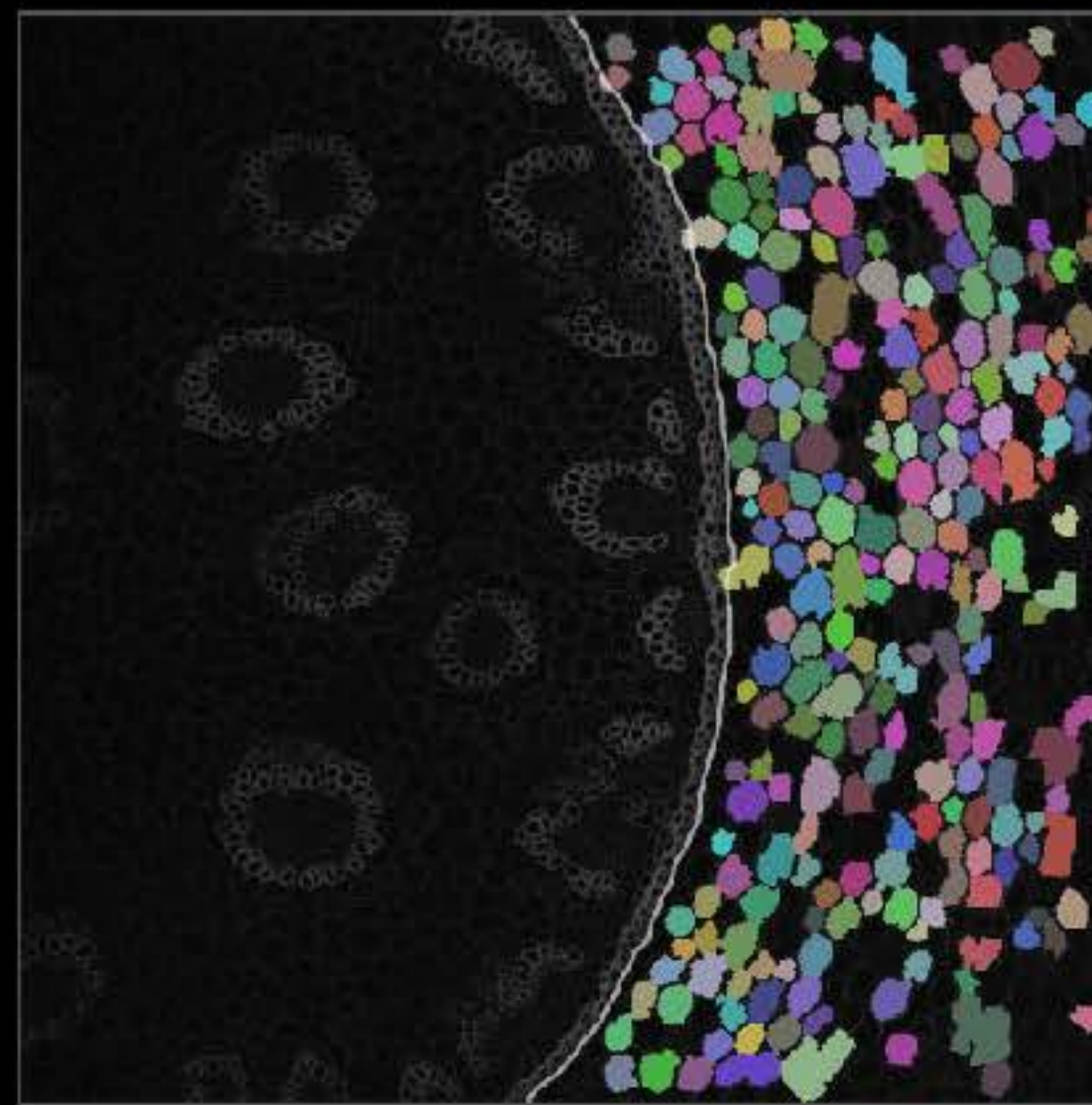
one image



1,048,576 numbers

no question
→ no answer

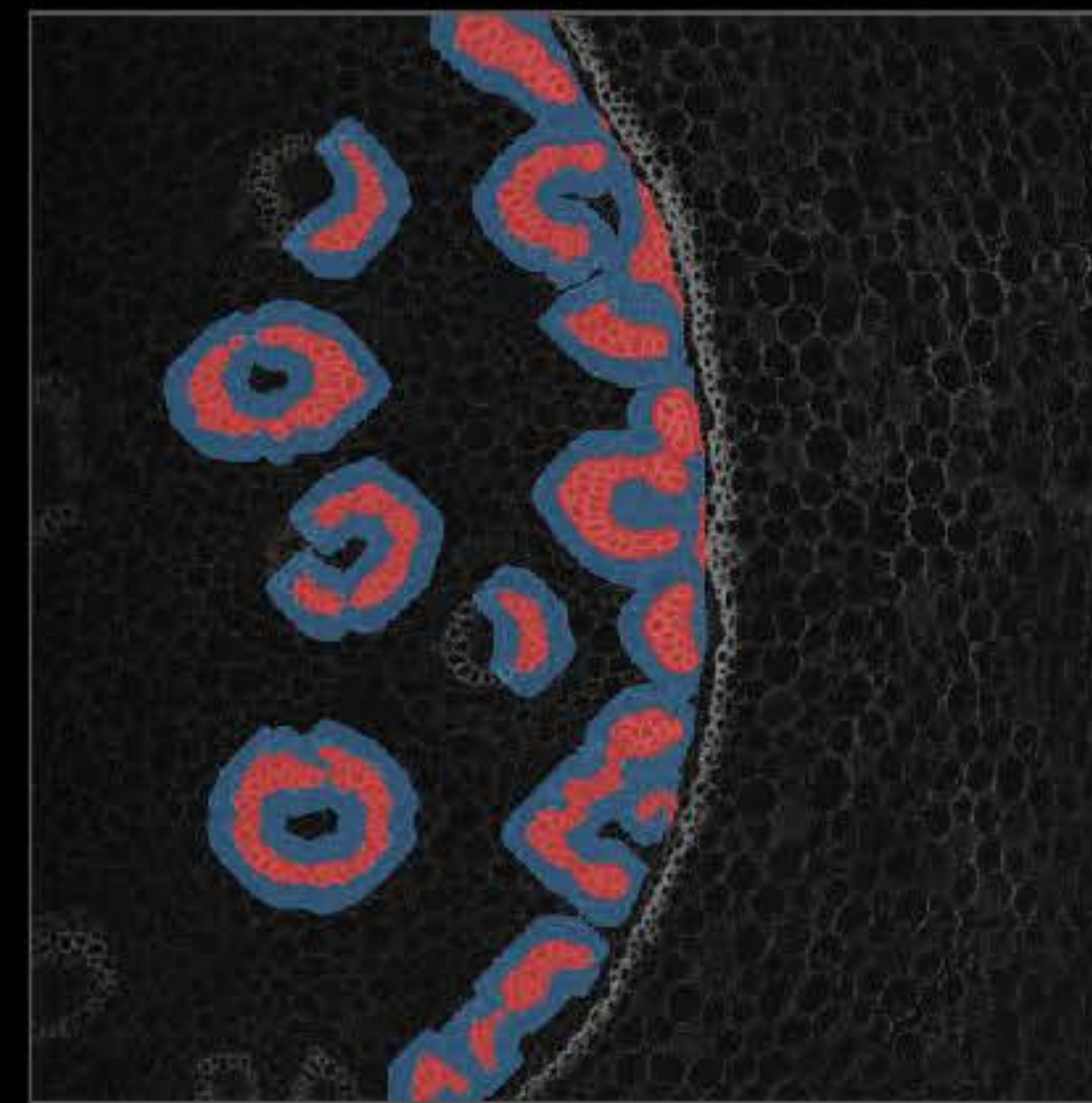
"How many cells are
in the cortex?"



≈ 260 cells

denoise → segment cells
→ count

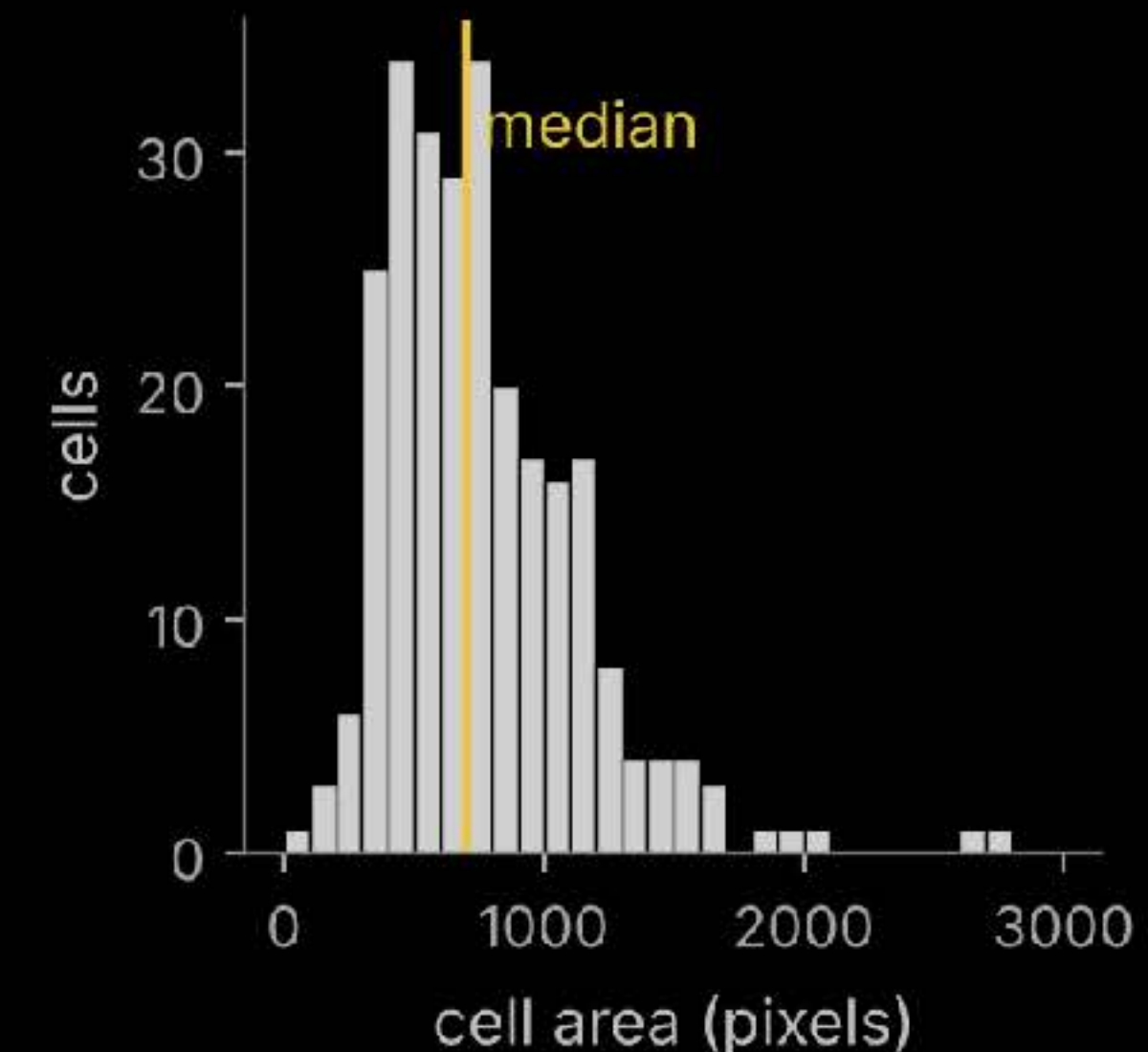
"Are the vascular bundles brighter
than the tissue around them?"



yes, ≈ 1.2× brighter

segment regions
→ measure on raw pixels

"How big are the
cortex cells?"

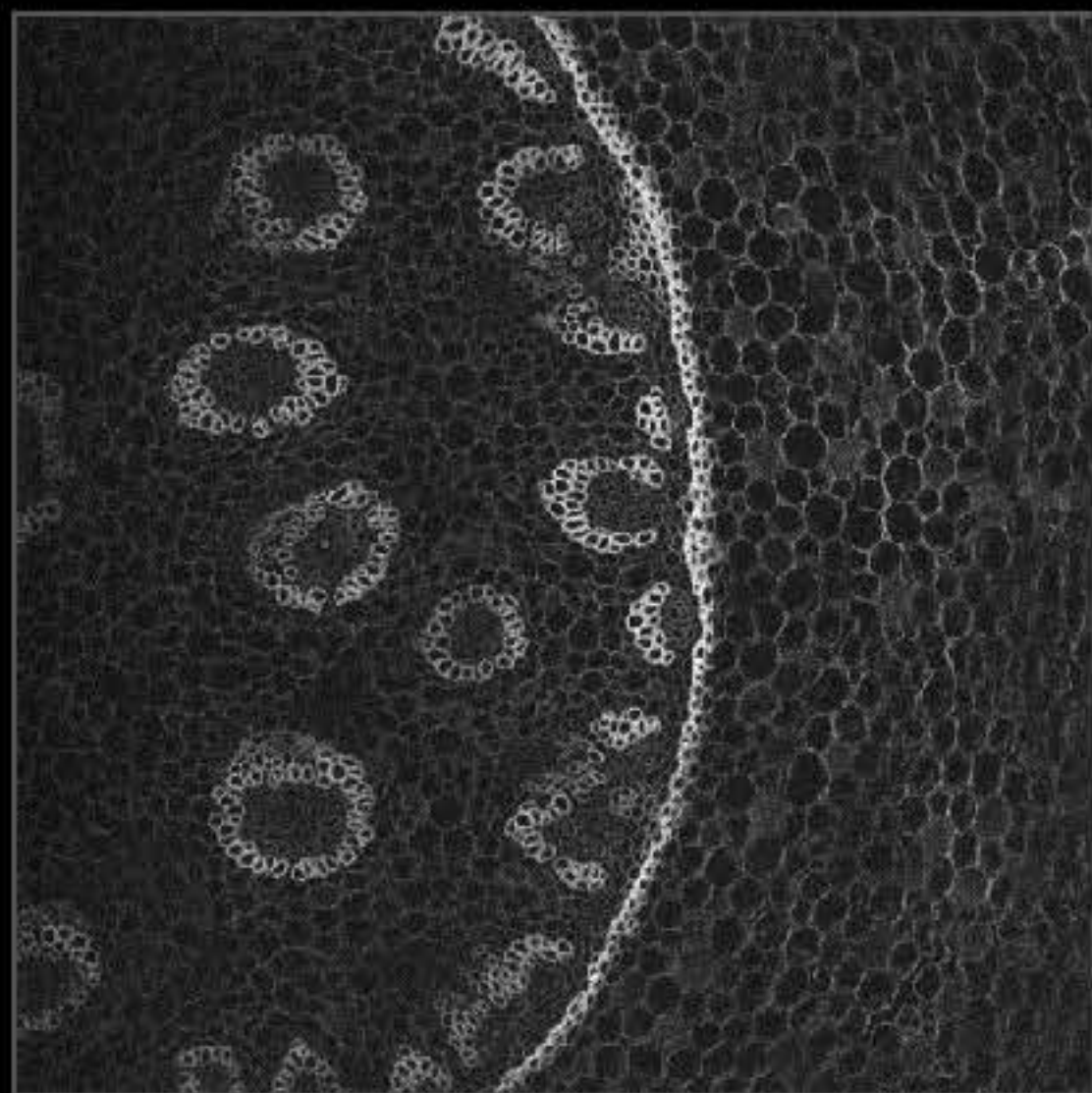


median ≈ 700 px²

segment cells → filter
→ measure shape

Also look at: Senft et al. Their main message is to plan acquisition and analysis around the question before imaging.

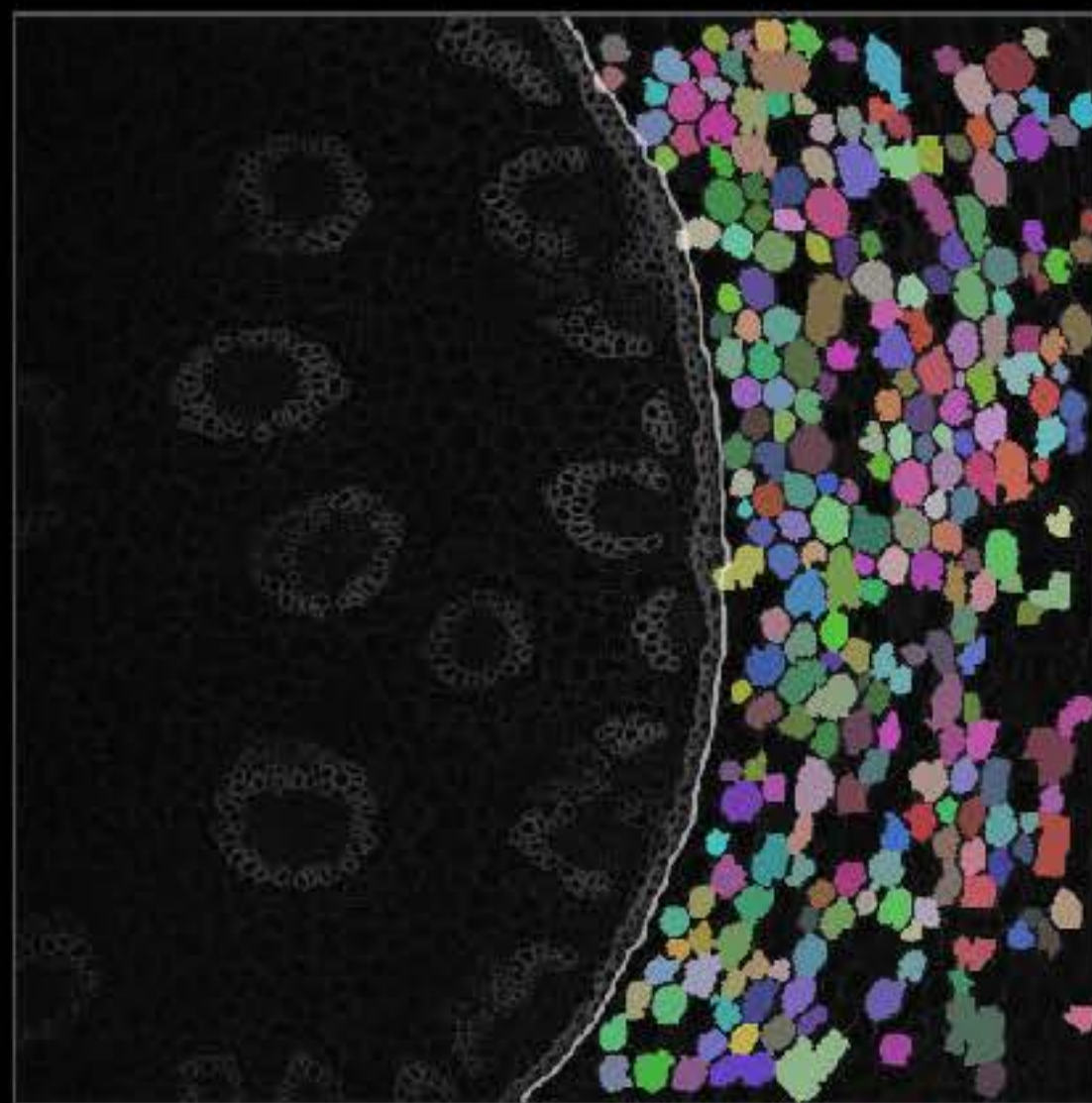
one image



1,048,576 numbers

no question
→ no answer

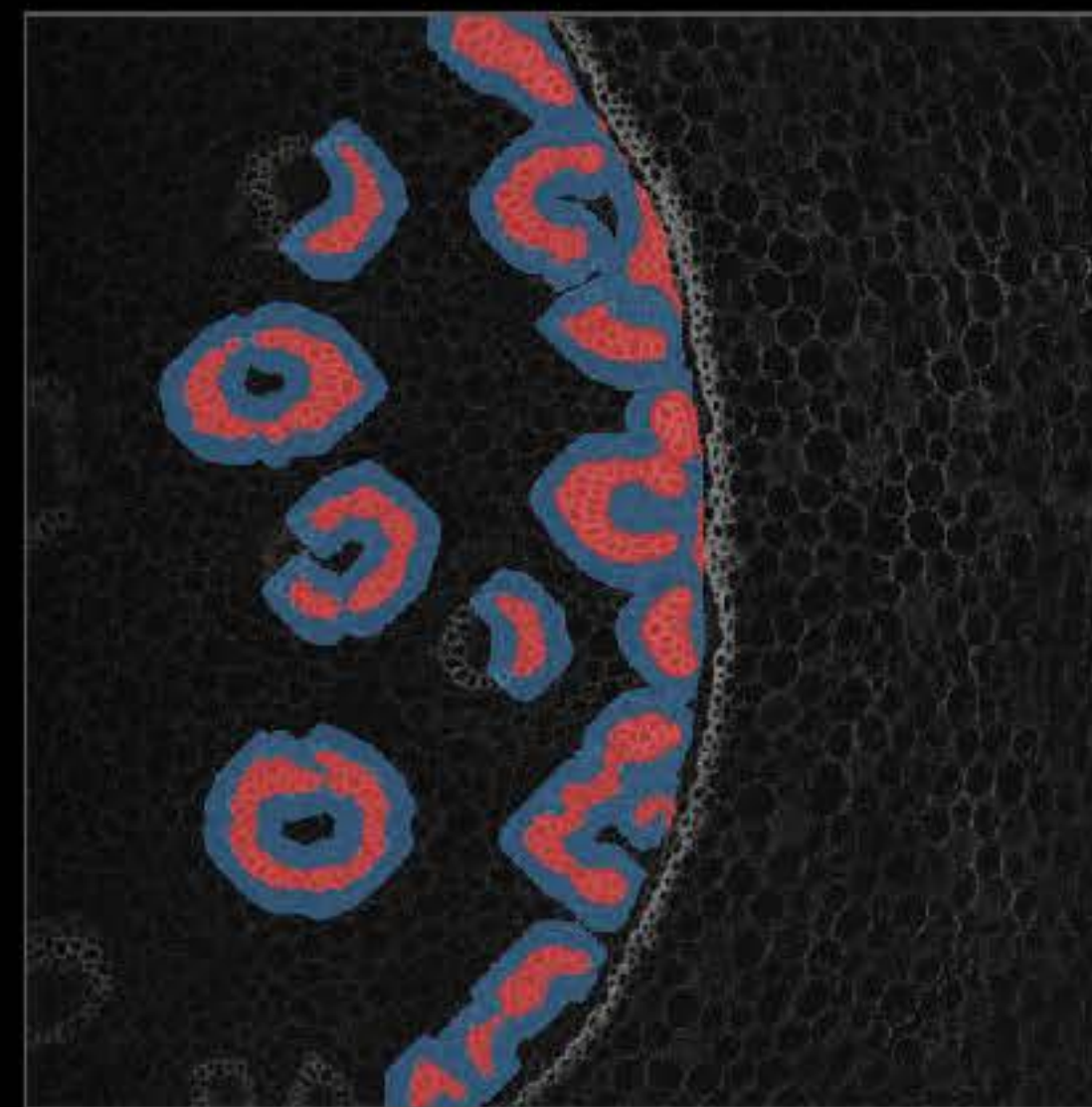
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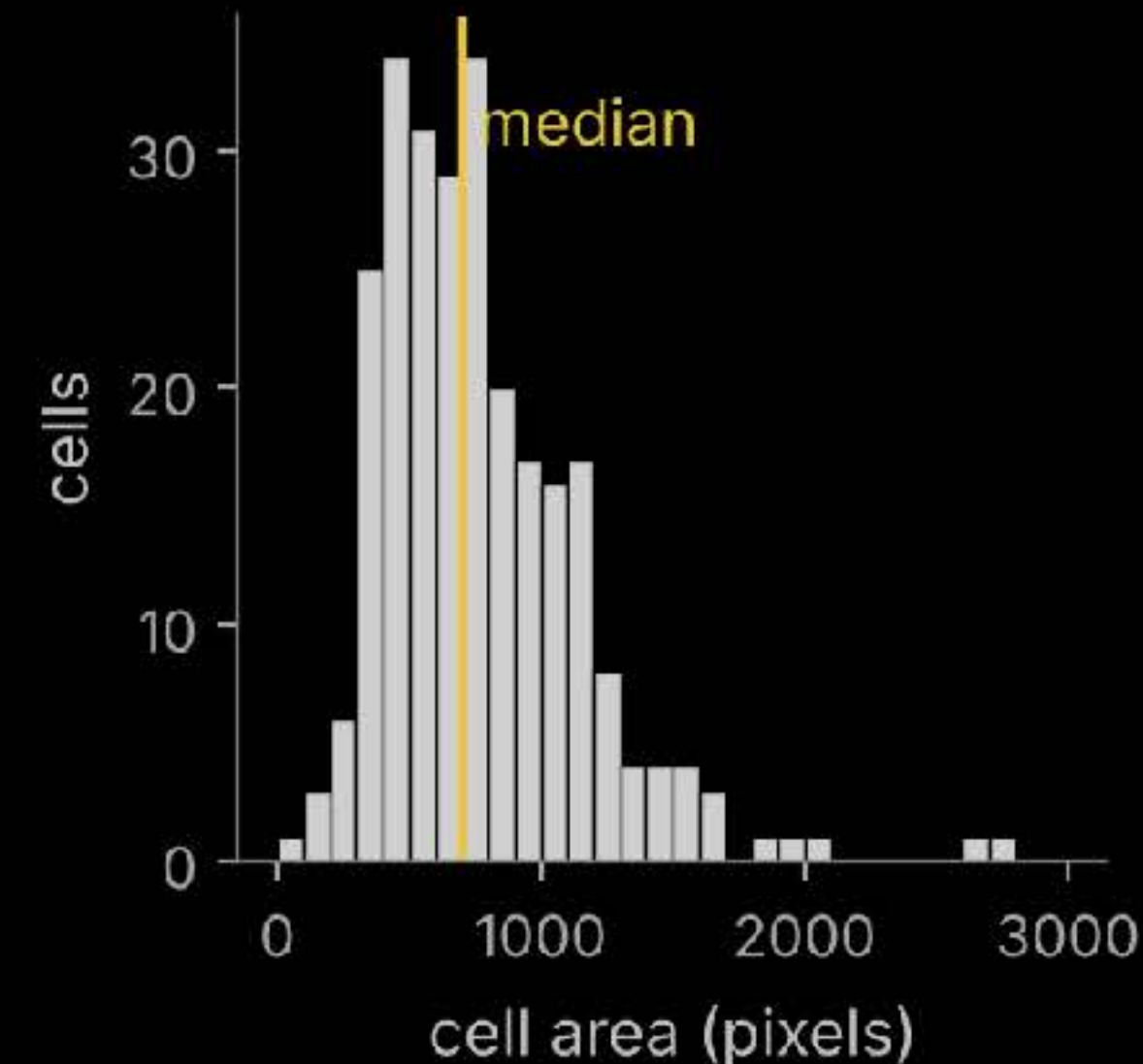
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segment regions
→ measure on raw pixels

"How big are the
cortex cells?"



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segment cells → filter
→ measure shape

Too common

image



image



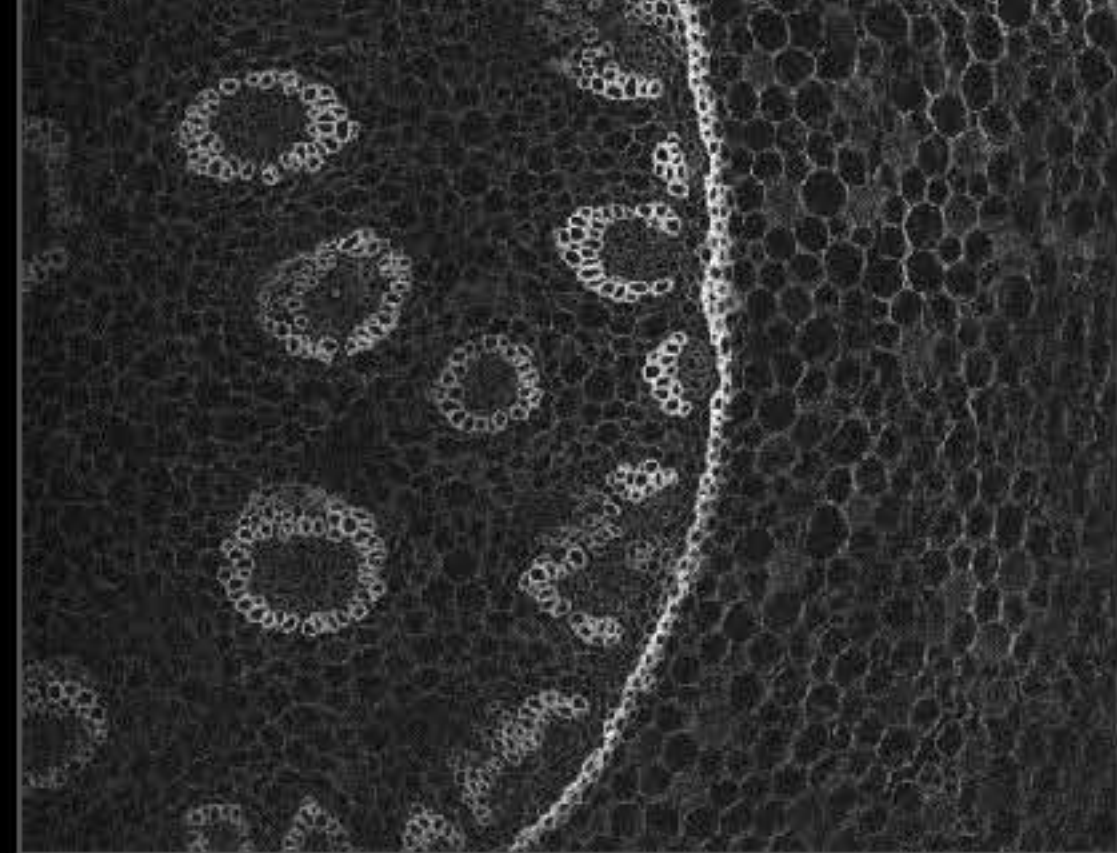
image



more images

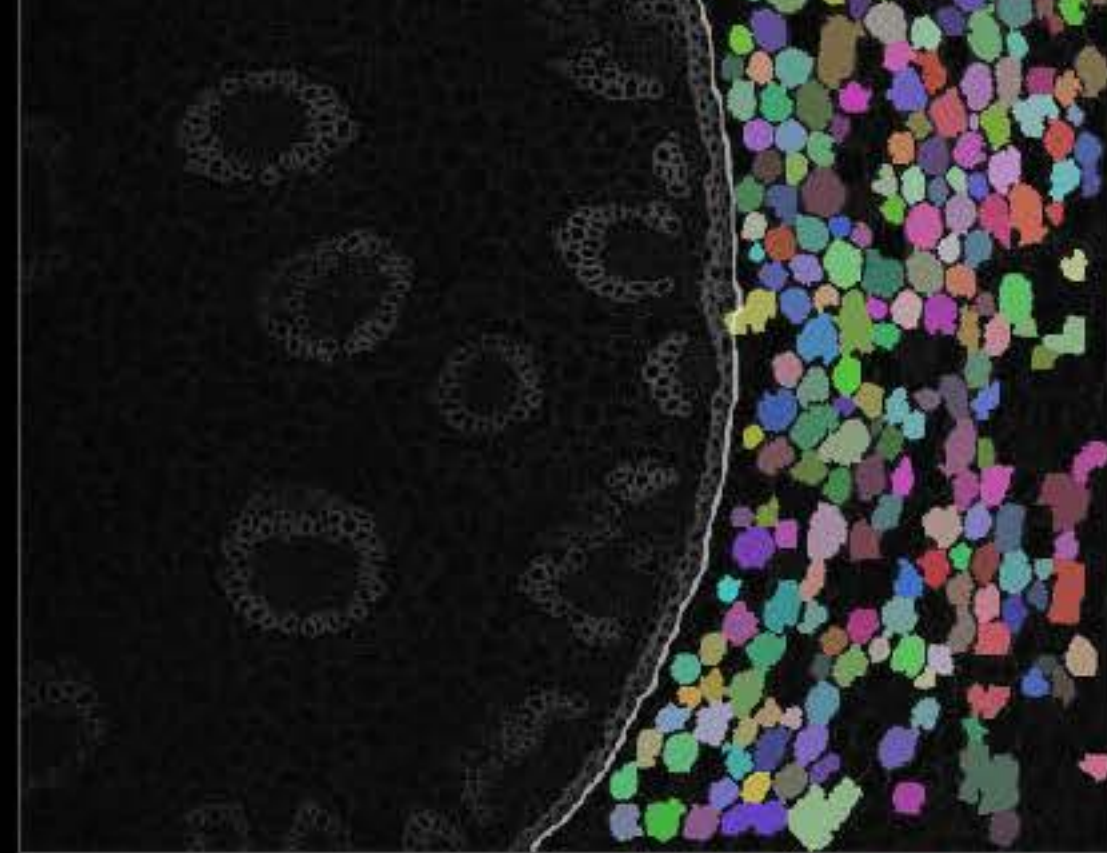


"So... what do I
do with all this?"



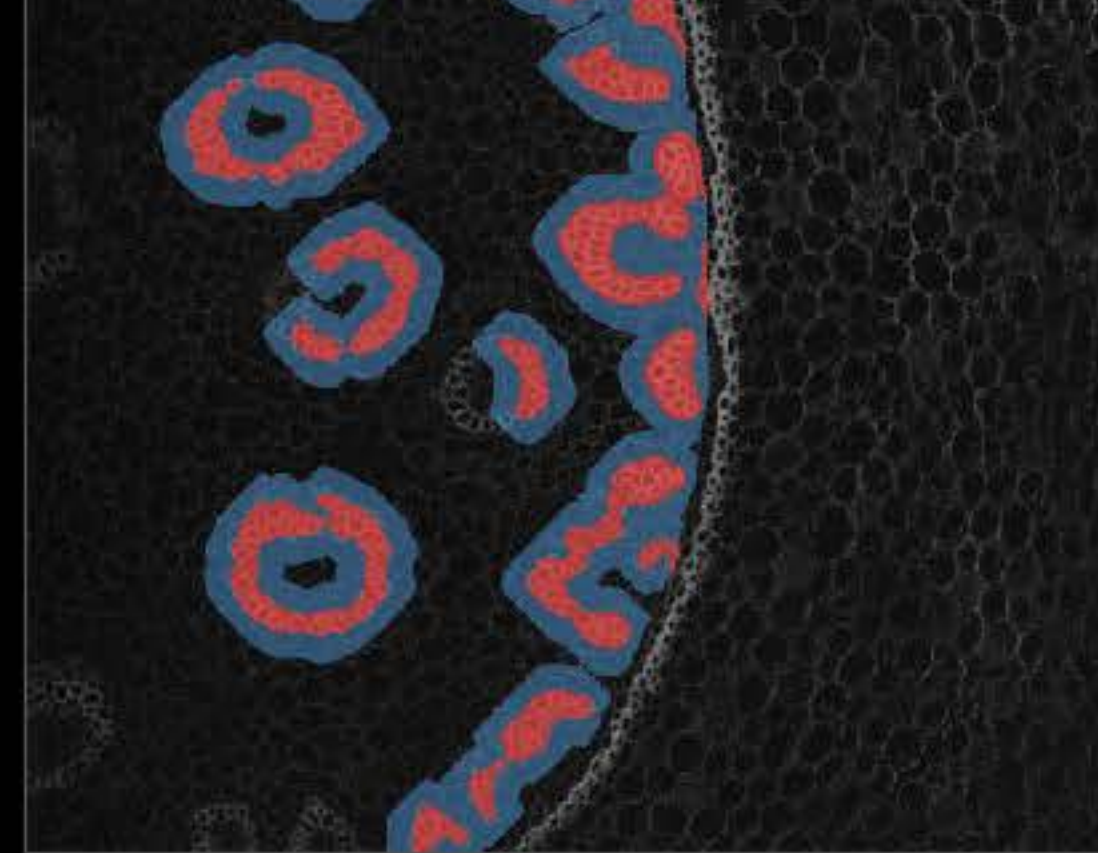
1,048,576 numbers

no question
→ no answer



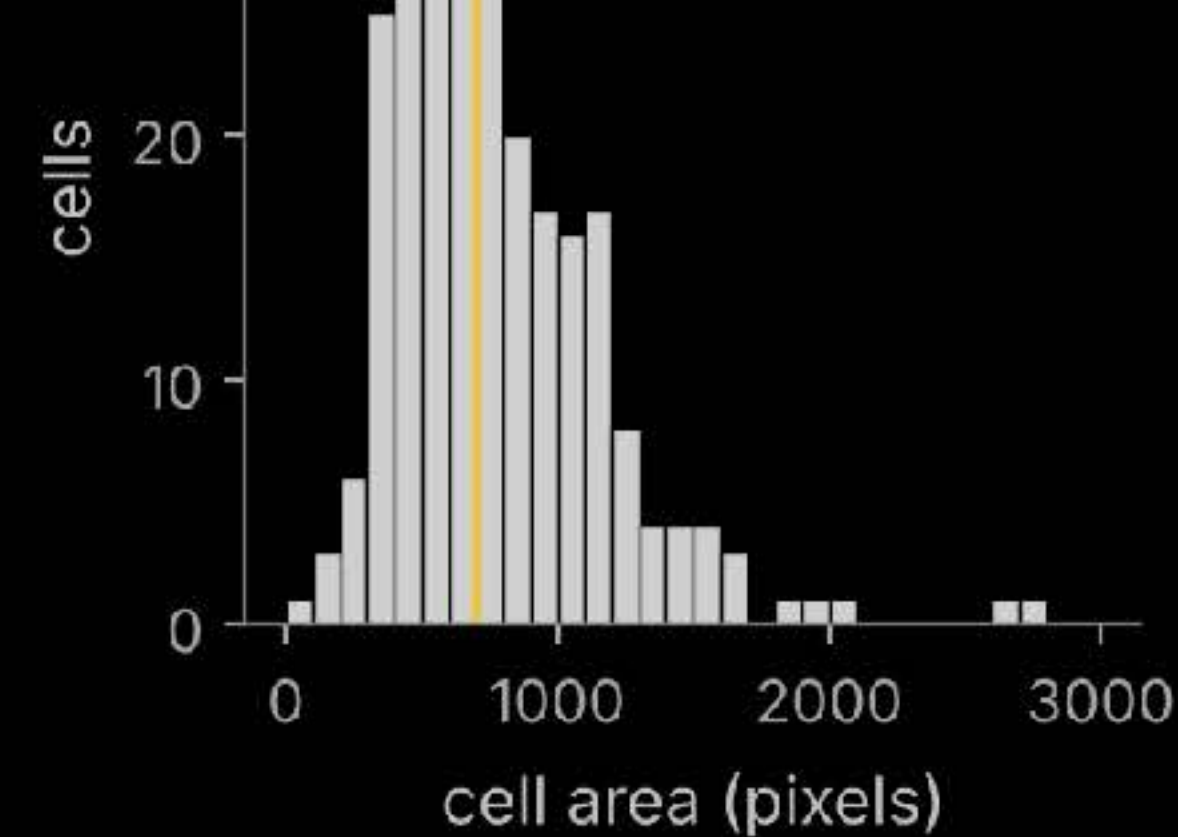
≈ 260 cells

denoise → segment cells
→ count



yes, ≈ 1.2× brighter

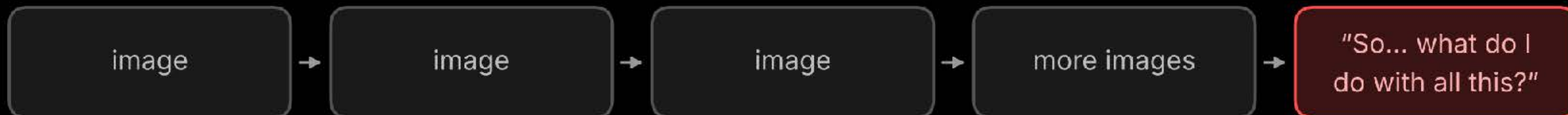
segment regions
→ measure on raw pixels



median ≈ 700 px²

segment cells → filter
→ measure shape

Too common



all of this happens BEFORE you touch the microscope

Better



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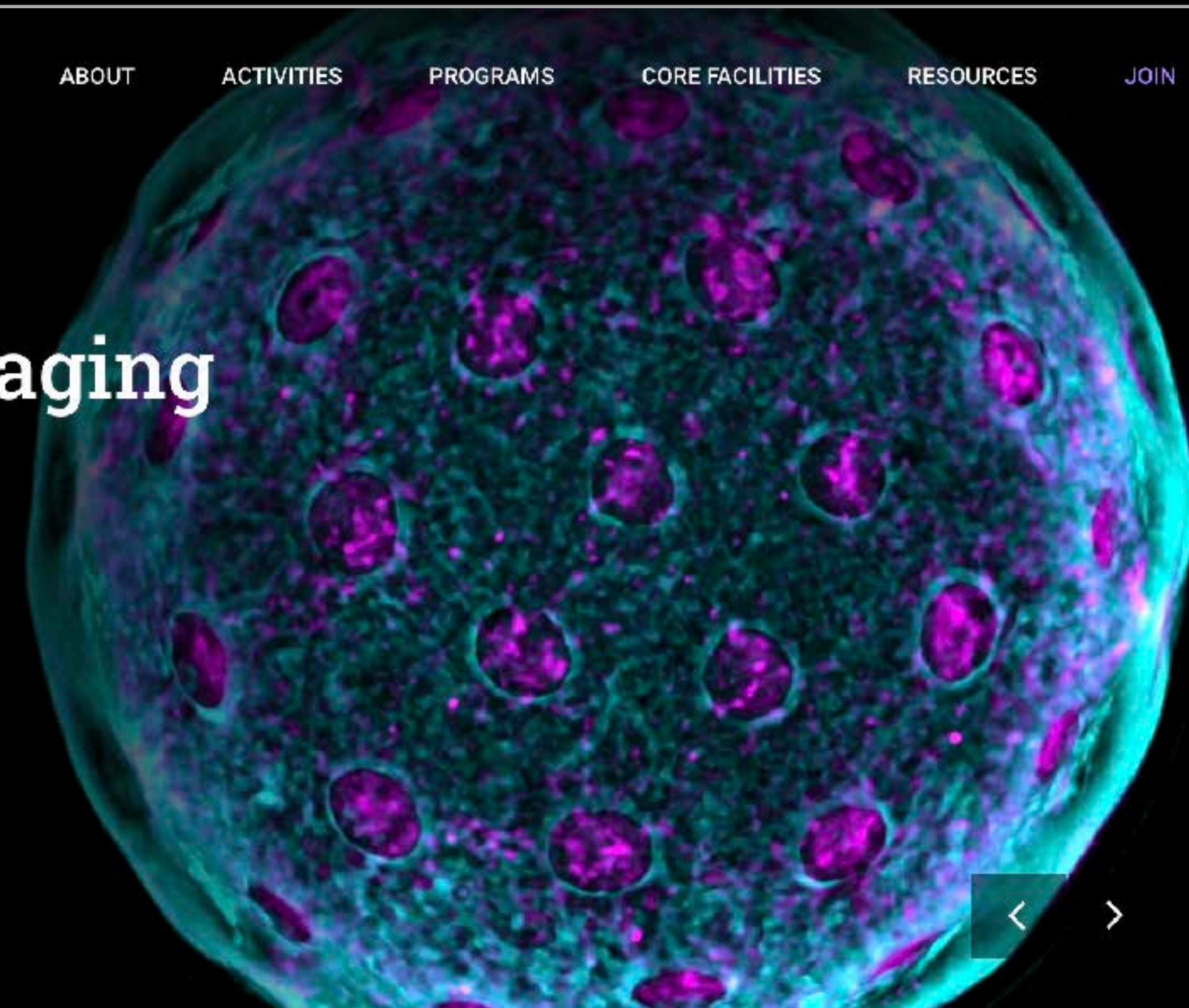


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Connecting, empowering and giving a unified voice to the bioimaging community to advance science.

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Quinoa Pollen Grain – Anastasiya Klebanovych

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image.sc

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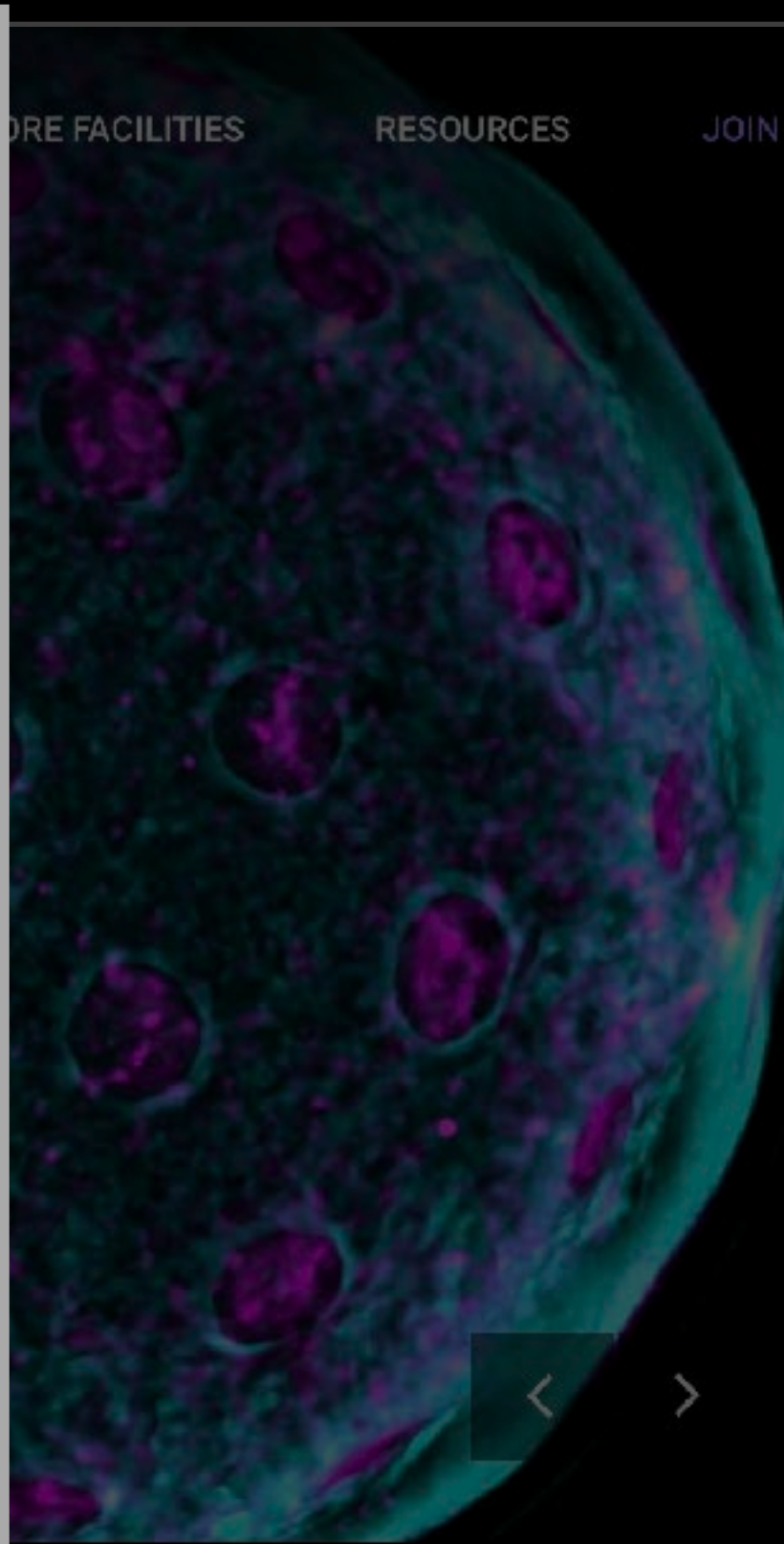
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Quinoa Pollen Grain – Anastasiya Klebanovych

THERE IS A COMMUNITY... AND WE ARE HAPPY TO WELCOME YOU!

nature methods

Perspective

Community-developed checklists for publishing images and image analyses

<https://doi.org/10.1038/s41592-023-01987-9>

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 Check for updates

Christopher Schmied^{1,53}✉, Michael S. Nelson², Sergiy Avilov³, Gert-Jan Bakker⁴, Cristina Bertocchi^{5,6}, Johanna Bischof⁷, Ulrike Boehm⁸, Jan Brocher⁹, Mariana T. Carvalho¹⁰, Catalin Chiritescu¹¹, Jana Christopher¹², Beth A. Cimini¹³, Eduardo Conde-Sousa¹⁴, Michael Ebner¹⁵, Rupert Ecker^{16,17,18}, Kevin Eliceiri¹⁹, Julia Fernandez-Rodriguez²⁰, Nathalie Gaudreault²¹, Laurent Gelman²², David Grunwald²³, Tingting Gu²⁴, Nadia Halidi²⁵, Mathias Hammer²³, Matthew Hartley²⁶, Marie Held²⁷, Florian Jug¹, Varun Kapoor²⁸, Ayse Aslihan Koksoy²⁹, Judith Lacoste³⁰, Sylvia Le Dévédec³¹, Sylvie Le Guyader³², Penghuan Liu³³, Gabriel G. Martins³⁴, Aastha Mathur⁷, Kota Miura³⁵, Paula Montero Llopis³⁶, Roland Nitschke³⁷, Alison North³⁸, Adam C. Parslow³⁹, Alex Payne-Dwyer⁴⁰, Laure Plantard²², Rizwan Ali⁴¹, Britta Schroth-Diez⁴², Lucas Schütz⁴³, Ryan T. Scott⁴⁴, Arne Seitz⁴⁵, Olaf Selchow⁴⁶, Ved P. Sharma³⁸, Martin Spitaler⁴⁷, Sathya Srinivasan⁴⁸, Caterina Strambio-De-Castilla⁴⁹, Douglas Taatjes⁵⁰, Christian Tischer⁵¹✉ & Helena Klara Jambor⁵²✉

Images document scientific discoveries and are prevalent in modern biomedical research. Microscopy imaging in particular is currently undergoing rapid technological advancements. However, for scientists wishing to publish obtained images and image-analysis results, there are currently no unified guidelines for best practices. Consequently, microscopy images and image data in publications may be unclear or difficult to interpret. Here, we present community-developed checklists for preparing light microscopy images and describing image analyses for

Checklist for image publishing

Image format

-  Focus on relevant image content (e.g., crop, rotate, resize) ☐ Minimal
-  Separate individual images ☐
-  Show example image used for quantifications ☐
-  Indicate position of zoom view/inset in full-view/original image ☐
-  Show images of the range of the described phenotype ☐

Image colors and channels

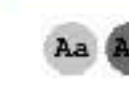
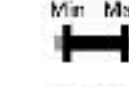
-  Annotation of channels (staining, marker, etc.) visible ☐ Minimal
-  Adjust brightness/contrast, report adjustments, use uniform color scales ☐
-  Image comparison: use the same adjustments ☐
-  Channel colors: high visibility on the background
Best visibility: grayscale ☐
-  Multicolors: provide grayscale for each color channel ☐
-  Multicolor: if channels are merged, make accessible to color-blind individuals ☐
-  Provide intensity scales (calibration bar) for grayscale, color, pseudocolor etc. ☐ Recommended
-  Pseudocolored images: additionally provide grayscale version for comparison ☐ Ideal
-  Gamma adjustments: additionally provide linear-adjusted image for comparison ☐

Image annotation

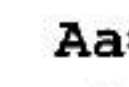
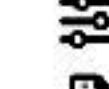
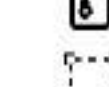
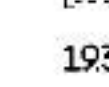
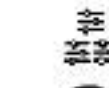
-  Add scale information (scale bar, image length in figure/figure legend) ☐ Minimal
-  Explain all annotations (in figure/figure legend) ☐
-  Annotations should be legible (line width, size/point size, color) ☐
-  Annotations should not obscure key data ☐
-  Annotate imaging details important for interpreting the figure (depending on the main message and imaging technique, this may be, e.g., image pixel size, imaging intervals (time-lapse in movies), exposure time or anatomical section) ☐ Recommended

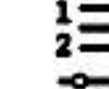
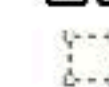
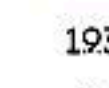
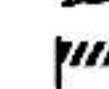
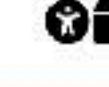
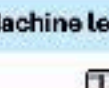
Image availability

Checklists for publication of image-analysis workflows

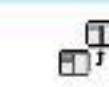
Established workflows

-  Cite workflow and platform ☐ Minimal
-  Key settings ☐
-  Example data ☐
-  Manual ROI ☐
-  Exact version ☐
-  All settings ☐ Recommended
-  Public example ☐
-  Document usage (e.g., screen recording or tutorial) ☐ Ideal
-  Cloud hosted or container ☐

New workflows

-  Cite components and platform ☐ Minimal
-  Describe sequence ☐
-  Key settings ☐
-  Example data and code ☐
-  Manual ROI ☐
-  Exact versions ☐
-  All settings ☐ Recommended
-  Public example data and code ☐
-  Rationale ☐
-  Limitations ☐
-  Screen recording or tutorial ☐ Ideal
-  Easy install and usage, container ☐

Machine learning workflows

-  Cite original method ☐ Minimal (all models)
-  Access to model ☐
-  Example or validation data ☐
-  Training and testing data and metadata ☐ Recommended (pretrained and



1. WHAT IS AN IMAGE? WHAT IS IMAGE ANALYSIS?
2. CLASSICAL ANALYSIS & MACHINE LEARNING
3. FROM DEEP LEARNING TO AGENTIC SYSTEMS
4. PRACTICAL EXERCISE: GETTING YOUR HANDS DIRTY WITH UNSUPERVISED DENOISING

WHAT IS THE CLASSICAL WAY?

- ▶ **You design the rules:** filters, thresholds, morphology, watershed.
- ▶ **Live demo:** Gaussian blur on Convallaria.
The noise goes down, but so do the membranes.
- ▶ **Tools:** *Fiji* (plugins and macros; CiliaQ is one, LabKit, ...), *napari* (Python), *ilastic* (fully open, Kreshuk Lab), *Imaris*, *Aivia*, *Amira* (commercial), *QuPath* (Pete Bankhead, again), and many more.
- ▶ **Community:** *image.sc* and *I2K*.
You're not alone, reach out and find friends/answers.

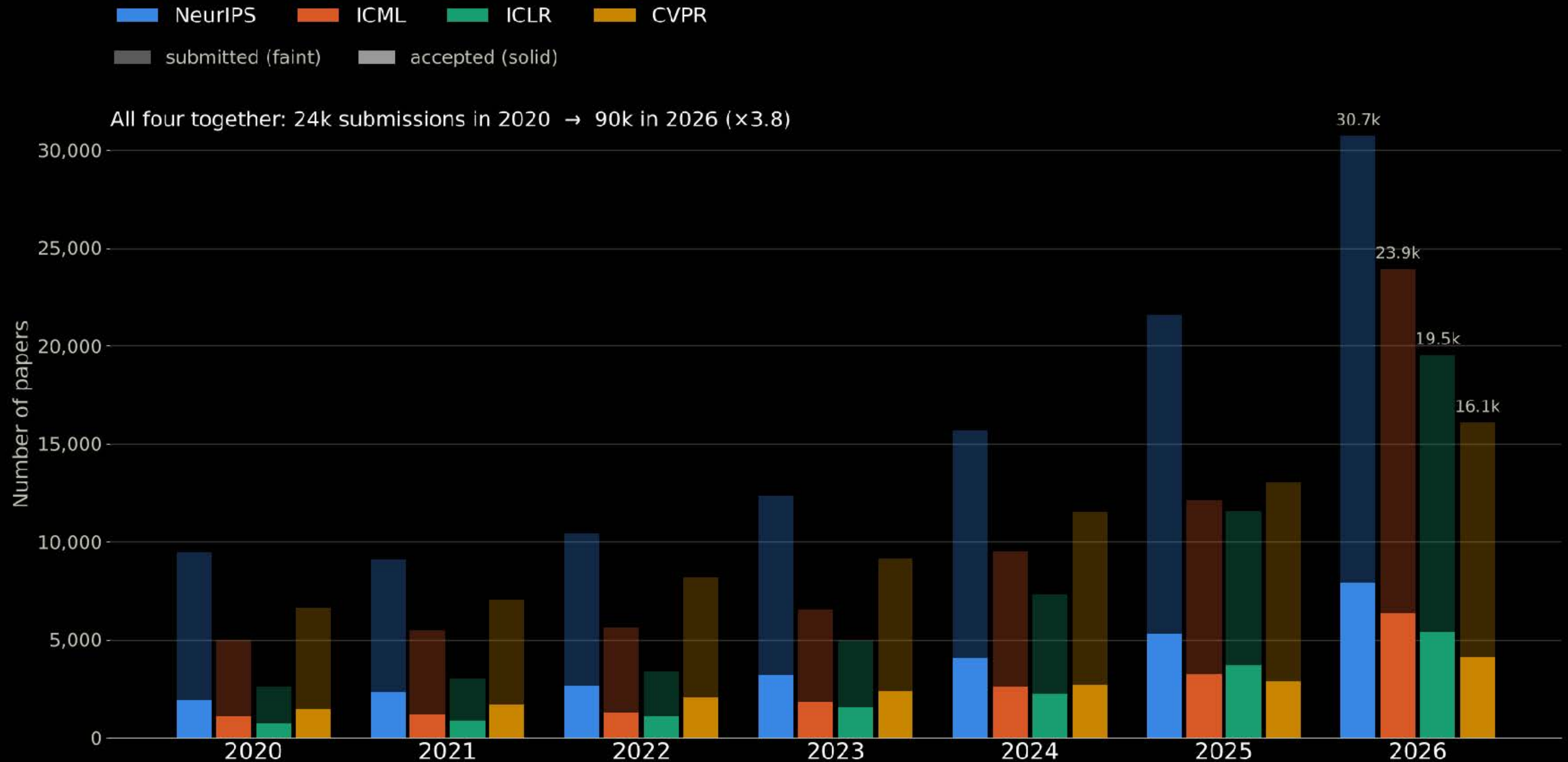
QUESTIONS TO YOU:

- ▶ **What are advantages and disadvantages of "classical image analysis"?**
- ▶ **What are alternatives for YOU doing it?**
- ▶ **How about reproducibility?**
- ▶ **How about reusability?**

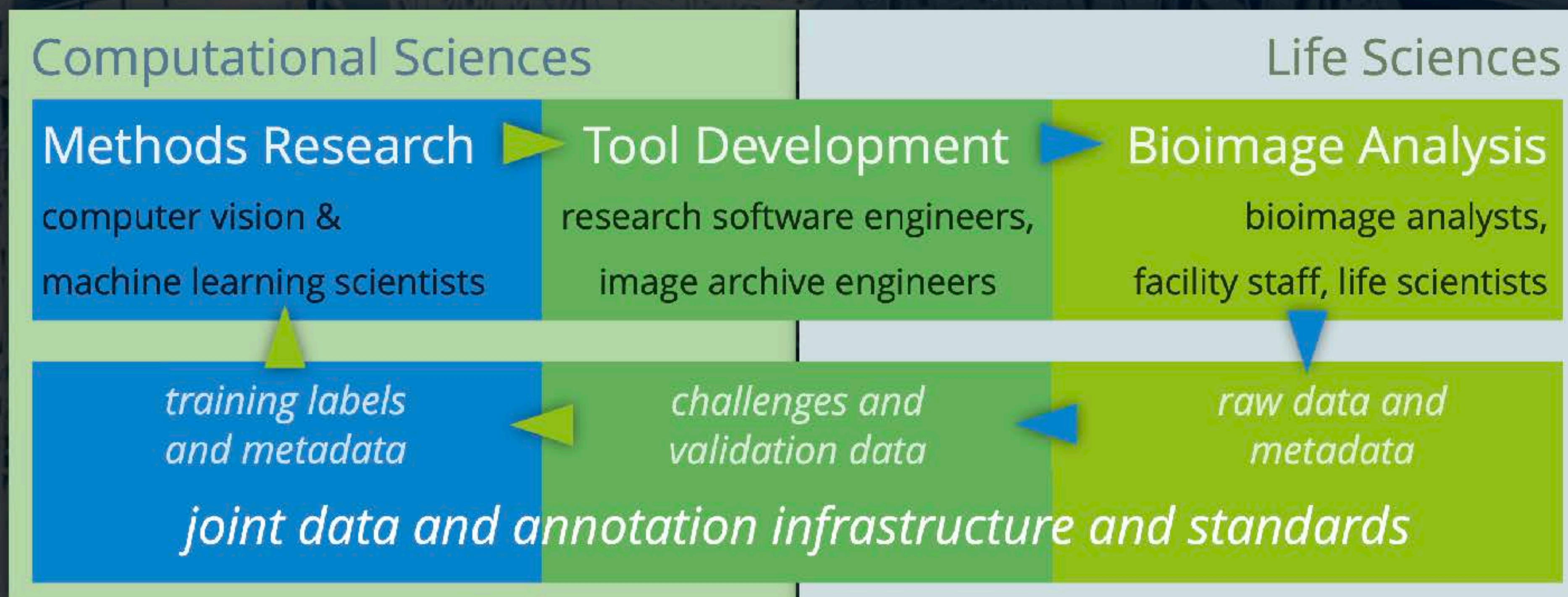
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1. WHAT IS AN IMAGE? WHAT IS IMAGE ANALYSIS?
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4. PRACTICAL EXERCISE: GETTING YOUR HANDS DIRTY WITH UNSUPERVISED DENOISING

Submissions and acceptances at major AI conferences, 2020–2026



Elevating the rate of scientific discovery via AI in bioimage analysis!



Dalle Nogare et al., Nature Methods, 2023

arXiv:2303.16743





SCAN ME

Elevating the rate of scientific discovery via AI in bioimage analysis

Computational Sciences



SCAN ME

Life Sciences

Comment

<https://doi.org/10.1038/s41592-023-01929-5>

Using AI in bioimage analysis to elevate the rate of scientific discovery as a community

Damian Dalle Nogare, Matthew Hartley, Joran Deschamps, Jan Ellenberg & Florian Jug



The future of bioimage analysis is increasingly defined by the development and use of tools that rely on deep learning and artificial intelligence (AI). For this trend to continue in a way most useful for stimulating scientific progress, it will require our multidisciplinary community to work together, establish FAIR (findable, accessible, interoperable and reusable) data sharing and deliver usable and reproducible analytical tools.

Bioimage analysis is in the midst of a revolution that will profoundly shape the future of the field for decades, spurred by recent developments in deep learning and AI. When thinking about the future of this change, it is tempting to imagine one where current limitations and pain points have been resolved. Although imagining such a future is easy, it is less easy to imagine the transition from today's status quo to that desired future state: what it will require and how it will be enabled. Here we discuss how to facilitate scientific progress in the life sciences, identify necessary changes, and provide ideas for how these changes might best be realized.

From a method developer perspective, a wider range of open and standardized data, metadata and ground truth labels need to become available to advance the state of the art. These data should be chosen or generated in such a way as to enable method developers to tackle challenging analyses that are current impediments to scientific progress in the life sciences.

From a user perspective, finding appropriate models to analyze a dataset is currently not a straightforward task. Even if many models are publicly available to users, choosing a suitable one requires a way to evaluate the quality of model predictions on the user's own data. Indeed, predictions generated by a given model need to be critically assessed and carefully interpreted to ensure the responsible use of AI-driven tools. Method developers must enable this by offering suitable training opportunities and providing tools that deliver interpretable quality metrics.

Ground-breaking AI research requires a fair amount of FAIR data

The reason that AI-based methods outperform classical approaches in so many analysis tasks is that they can distill the most relevant priors from a given body of training data. (A data prior is a task-specific clue extracted from previously seen examples that can later be used to make better decisions when new images are processed.) Trained networks are therefore precisely tailored to solving a specific analysis task in the context of a specific type of data; that is, the kind of data on which they were trained. These training data must be of sufficient quality



Frontiers in Bioinformatics

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Better research software tools to elevate the rate of scientific discovery or why we need to invest in research software engineering

Joran Deschamps*, Damian Dalle Nogare and Florian Jug*

Fondazione Human Technopole, Milan, Italy

KEYWORDS

research software engineering, image analysis, image analysis software, community support, opinion

Introduction

In the past decade, enormous progress has been made in advancing the state-of-the-art in bioimage analysis—a young computational field that works in close collaboration with the life sciences on the quantitative analysis of scientific image data. In many cases, tremendous effort has been spent to package these new advances into useable software tools and, as a result, users can nowadays routinely apply cutting-edge methods to their analysis problems using software tools such as ilastik (Berg et al., 2019), cellprofiler (McQuin et al., 2018), Fiji/

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Where biology meets microscopy

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Welcome to FocalPlane

FocalPlane is a community site for microscopy research. It is a place where you, as an imaging scientist, engineers, clinicians, and developers can share your work and experience. FocalPlane is your site and once you post, add an image to our gallery listing.

Create an account to share your latest news.

Register here

Recent posts

VolumeEM: An Interview with Alan Agre

Posted by Daniel Doucet, on 14 Dec 2023

I had the pleasure of interviewing the Alan Agre, a pioneer in Microscopy, and his involvement with the community.

image.sc

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DAIM

GloBIAS

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OMERO

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BoneJ

CellProfiler

DeepLabCut

Icy

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OmeSliCC

PolusAI

scenery

SpotMAX

vedo

Arkitekt

BIII

BrainGlobe

CLIJ

Fiji

IDR

ImJoy

Mars

MoBIE

NEUBIAS

OpenIRIS

PYME

SCIFIO

SR-Tesseler

VVDViewer

Ay Aydin

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ilastik

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ity to

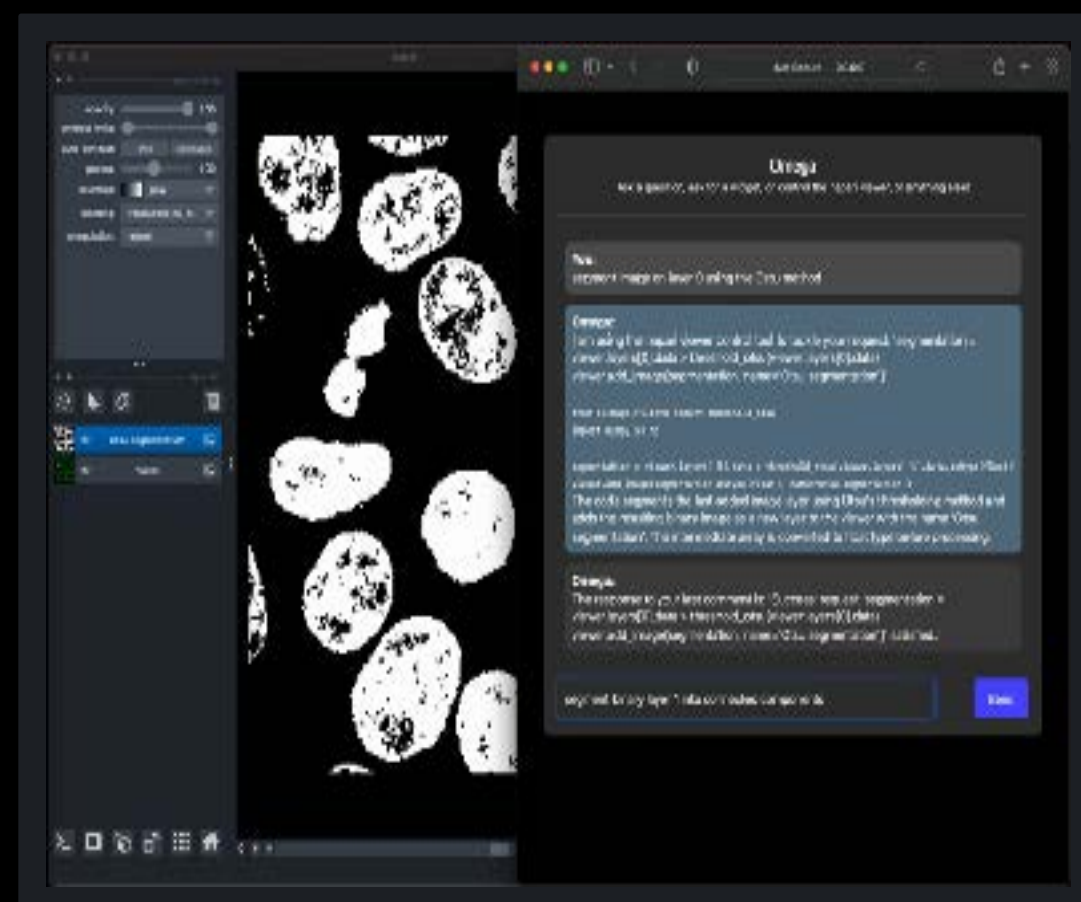
INCLUSION

stering a welcoming and diverse environment to encourage participation and individual contributions



“JUST TELL IT WHAT YOU WANT”

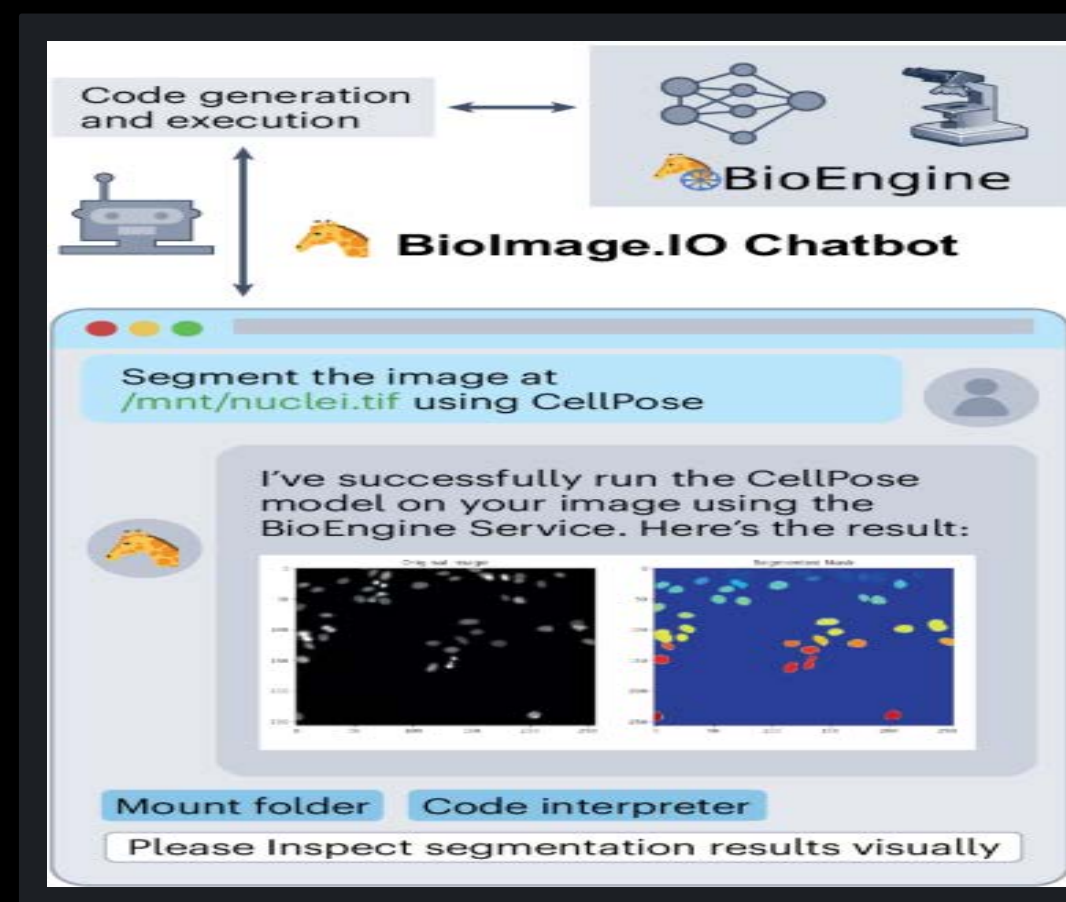
Natural-language image analysis, as advertised (2024–2026)



Omega

Royer lab · Nature Methods 2024

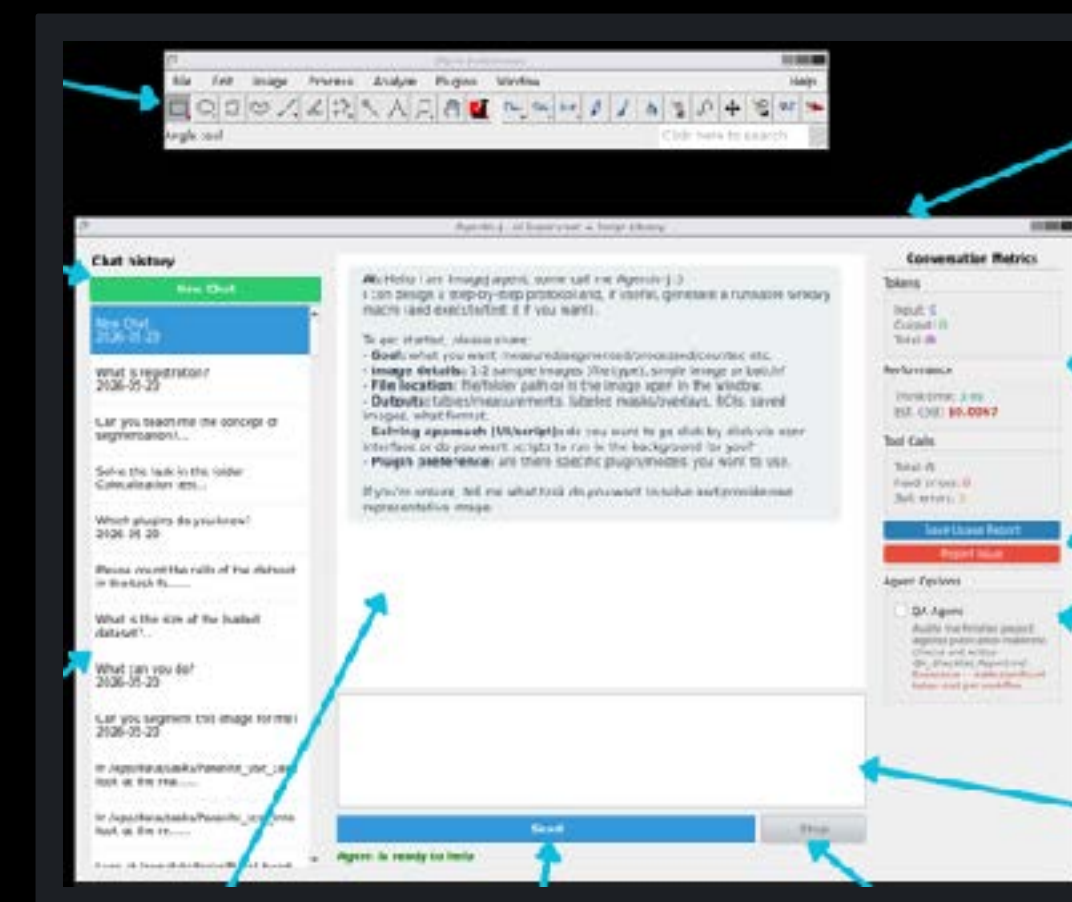
“process and analyze biological images through natural language conversations”



BioImage.IO Chatbot

Ouyang lab · Nature Methods 2024

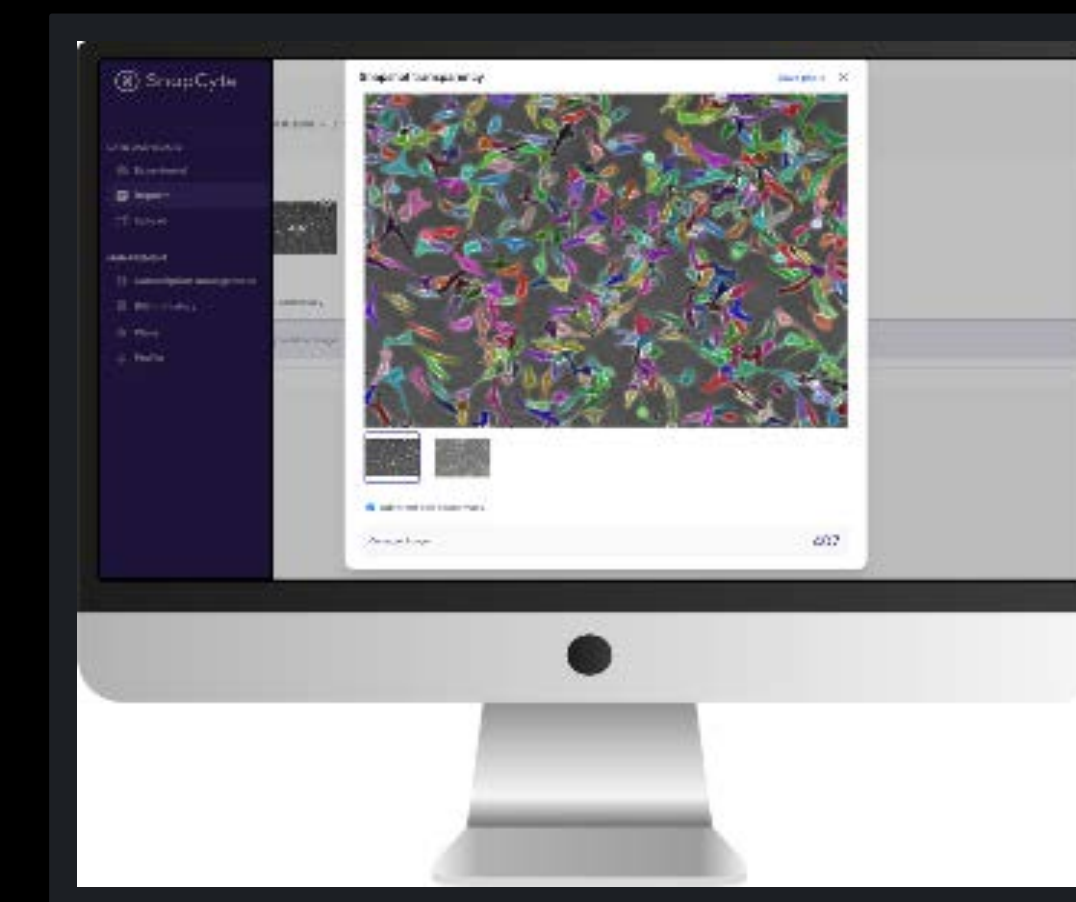
“Segment the image ... using CellPose” → “I’ve successfully run the CellPose model”



Agentic-J

Chen lab et al. · arXiv 2026

“from nuclei segmentation and cell tracking to multi-condition quantification”



SnapCyte

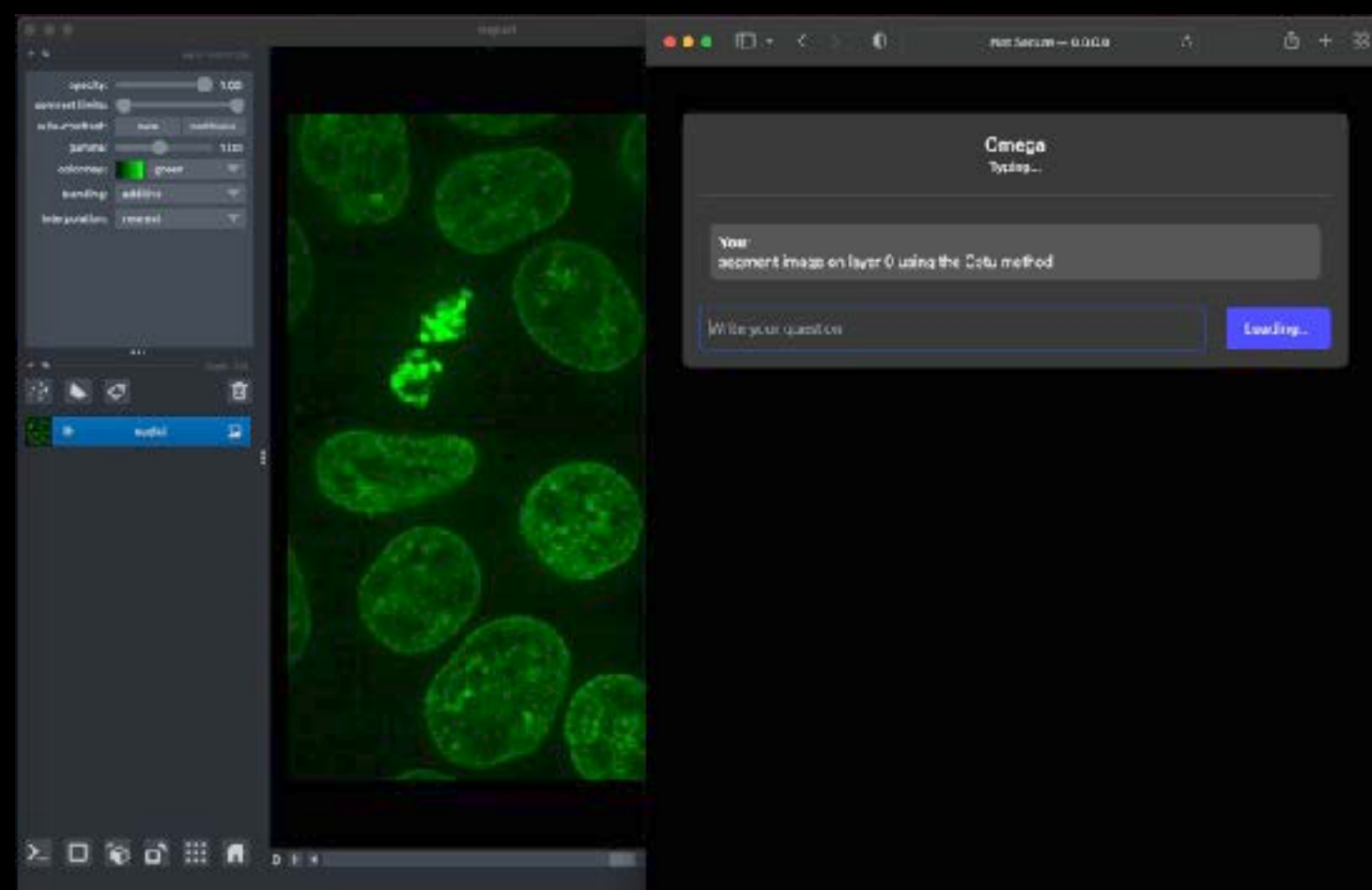
commercial

“One click. No plugins, macros, or manual setup required.”

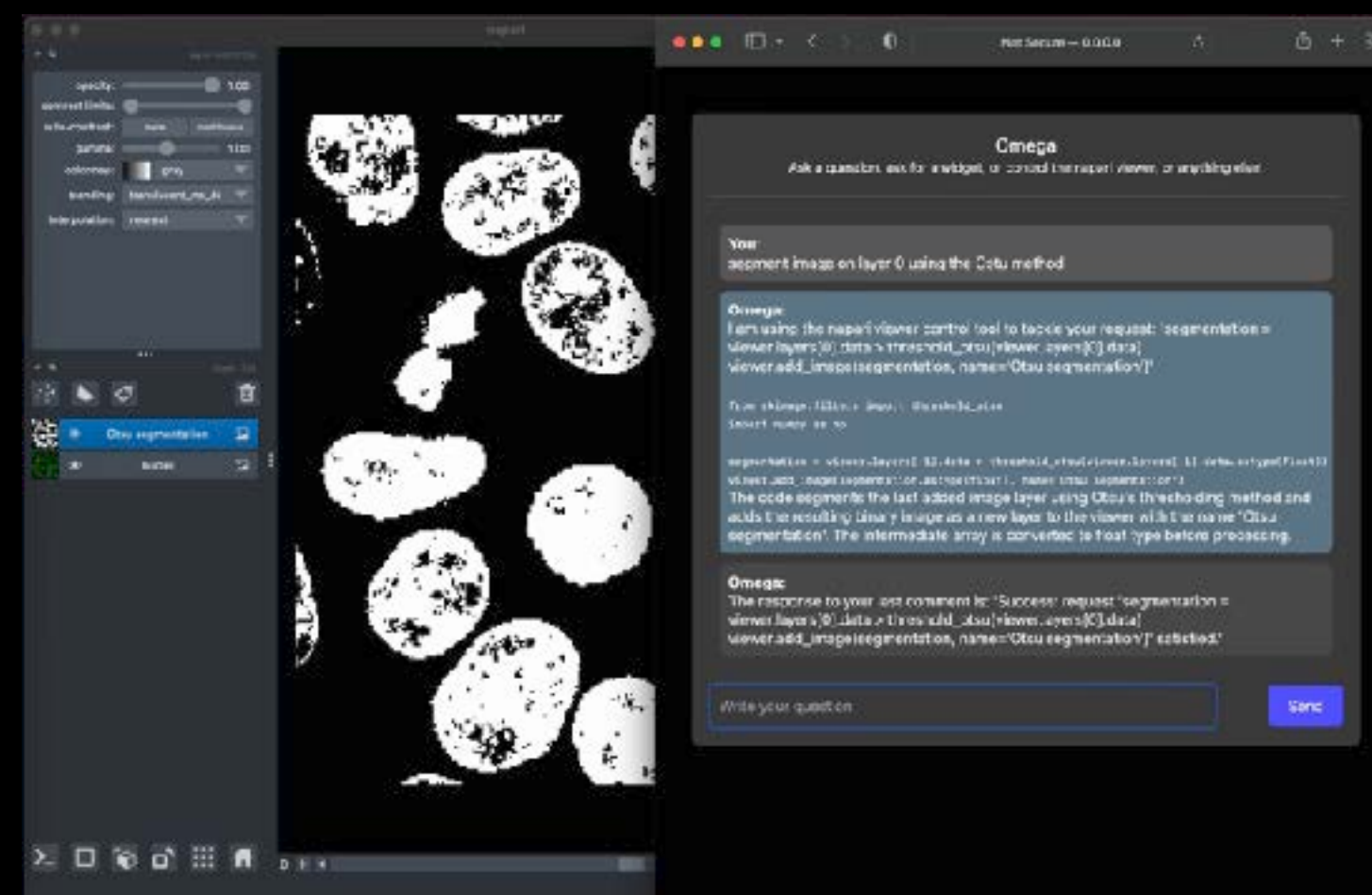
Also: Leica Aivia 15 promises to “transform images into insights effortlessly”, “without needing to code or train their own models”.

HOW IT WORKS: ASK → WRITE CODE → RUN → FIX

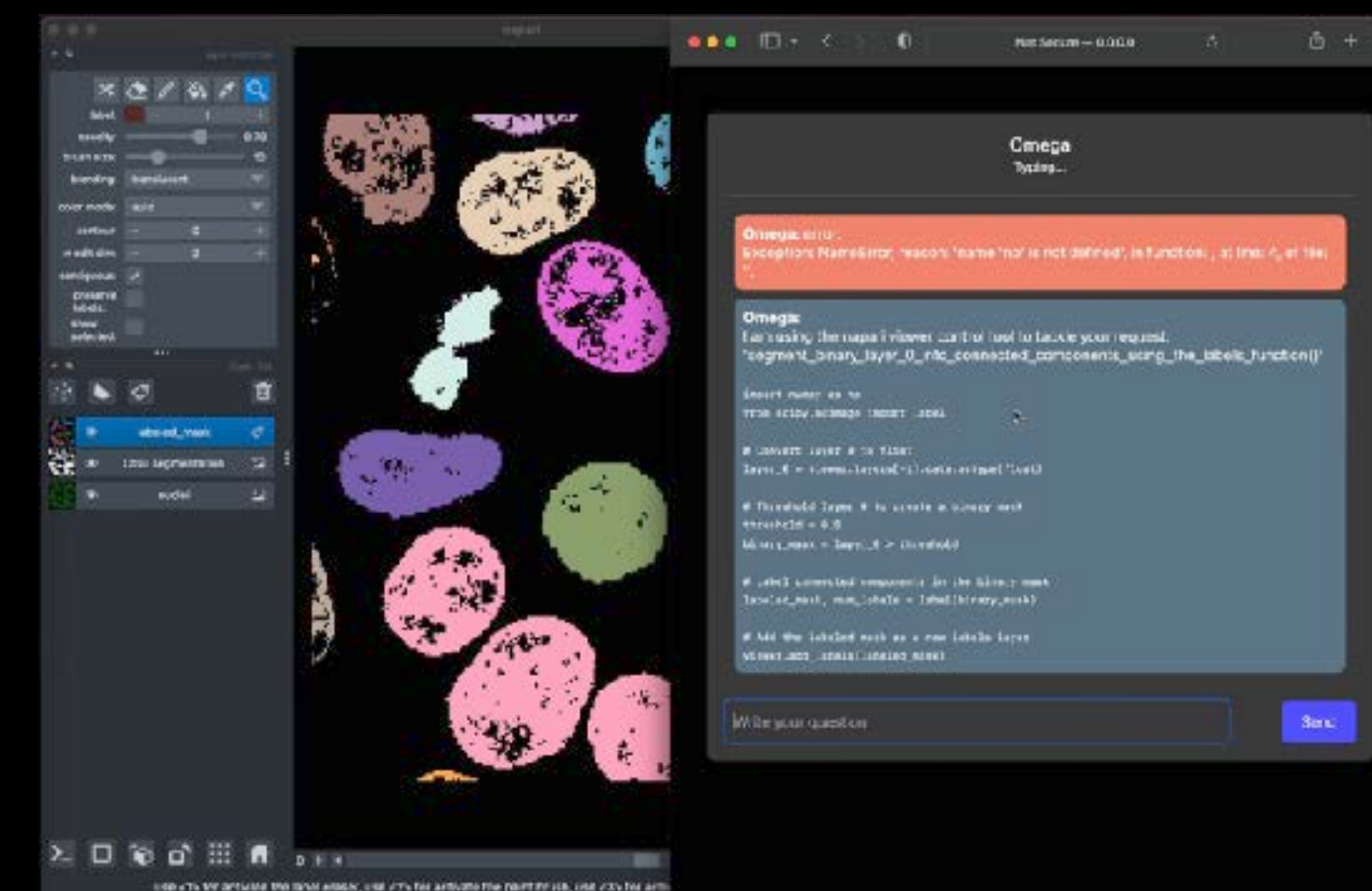
Omega in napari, three moments from the authors' demo video



1 You ask in plain words: “segment image on layer 0 using the Otsu method”



2 The LLM writes Python (scikit-image), runs it inside napari and adds the mask as a new layer.



3 Next request crashes (NameError). The agent reads the error, rewrites its code and labels the nuclei.

Same loop everywhere: BioImage.IO Chatbot, Agentic-J (Fiji), napari-mcp or Fiji-MCP + Claude/ChatGPT. The LLM never “sees” like you do: it writes code, runs it, reads the output.

AGENTS THAT PLAN, SEARCH AND CHECK THEMSELVES

GenCellAgent: "Help me segment the Golgi in the provided image"

1 Ask

2 Plan

3 Search

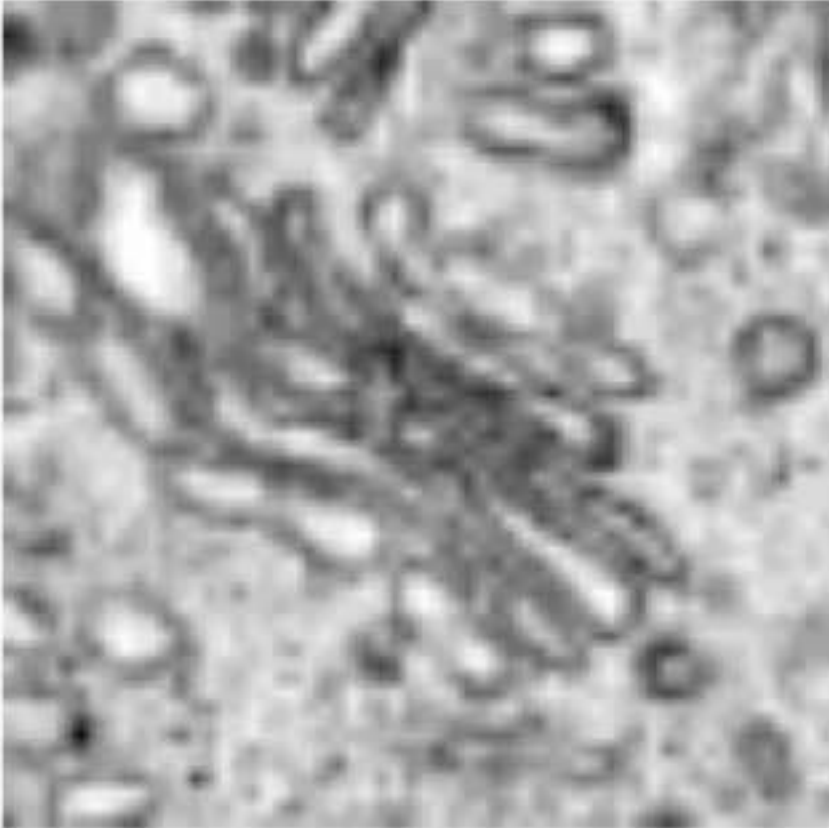
4 Segment, check

Cell Image Segmentation Agent

Upload your main image

Drag and drop file here
Up to 20MB per file - PNG, JPG, JPEG

go_golgi_image_53.png (1.0M)



Discarded main image

Upload file path: /home/klies/workspace/Storage/yul/persistent/LISA/imgs/golgi_image_53.png

Do you have an example image and mask for one-shot segmentation?

☐ I have an example image/mask

☒ I don't have any example image/mask

You have indicated you don't have any example image/mask.

Enter a short query:

Help me segment the Golgi in the provided image

Send

[Iteration 1] Agent Planning

Planning Steps:

```

[
  {
    "step": 1,
    "tool": "google",
    "condition": "",
    "next_step_if_success": 2,
    "next_step_if_failure": NULL
  },
  {
    "step": 2,
    "tool": "imagesegmentation",
    "condition": "",
    "next_step_if_success": 3,
    "next_step_if_failure": NULL
  },
  {
    "step": 3,
    "tool": "segmentationevaluation",
    "condition": "",
    "next_step_if_success": 4,
    "next_step_if_failure": NULL
  },
  {
    "step": 4,
    "tool": "imagesegmentation",
    "condition": "segmentation score < satisfactory threshold",
    "next_step_if_success": 5,
    "next_step_if_failure": 6
  },
  {
    "step": 5,
    "tool": "segmentationevaluation",
    "condition": "",
    "next_step_if_success": 4,
    "next_step_if_failure": NULL
  },
  {
    "step": 6,
    "tool": "summarizer",
    "condition": "",
    "next_step_if_success": NULL,
    "next_step_if_failure": NULL
  }
]

```

[Iteration 2] Agent Action

Action Details

- Name: google
- Reason: The Golgi apparatus is a complex organelle, and I need to gather information about its visual characteristics to create an effective segmentation prompt. Google can provide this information.

Input Parameters

```

{
  "search_query": "Golgi apparatus shape features",
  "location": ""
}

```

Action: Using google tool

Observation from google:

```

{
  "page_results": [
    {
      "title": "Golgi apparatus",
      "url": "https://en.wikipedia.org/wiki/Golgi_apparatus",
      "snippet": "The Golgi apparatus (or Golgi body) is a series of flattened, membrane-bound organelles (cisternae) located near the cell nucleus and endoplasmic reticulum (ER). It is involved in the transport, sorting, and modification of proteins. It is an essential organelle in eukaryotic cells."
    },
    {
      "title": "Golgi apparatus",
      "url": "https://www.britannica.com/science/Golgi-apparatus",
      "snippet": "The Golgi apparatus is a series of flattened, membrane-bound organelles (cisternae) located near the cell nucleus and endoplasmic reticulum (ER). It is involved in the transport, sorting, and modification of proteins. It is an essential organelle in eukaryotic cells."
    }
  ]
}

```

[Iteration 3] Agent Thought

Thought: Now that I have gathered information about the Golgi apparatus, I can use the image segmentation tool to segment the Golgi in the provided image using the segmentation prompt I generated from the Google search results.

[Iteration 3] Agent Action

Action Details

- Name: imagesegmentation
- Reason: I need to segment the Golgi apparatus in the image, and the image segmentation tool is designed for this purpose. I will use the segmentation prompt derived from the Google search to guide the segmentation process.

Input Parameters

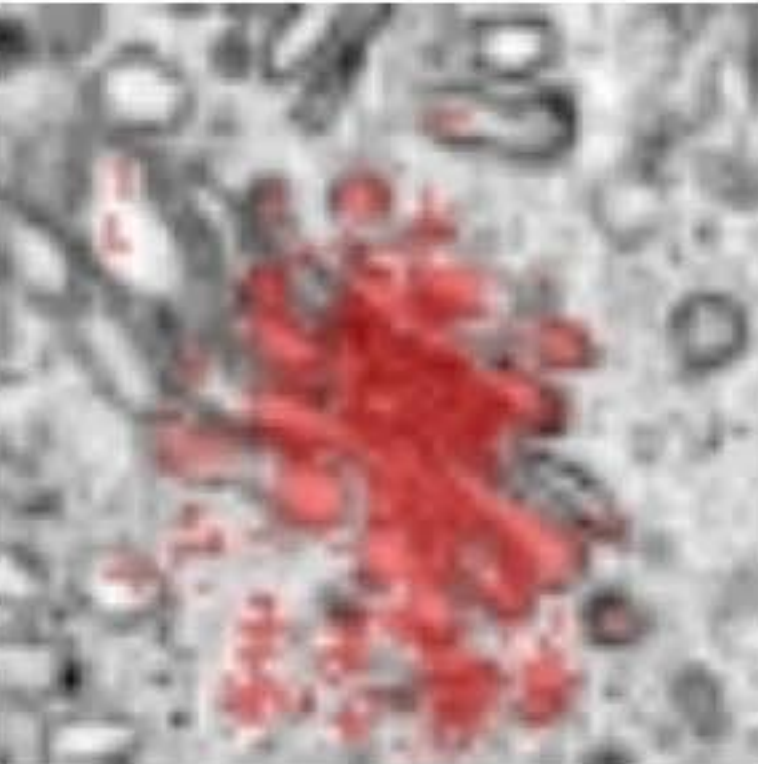
```

{
  "image_path": "/home/klies/workspace/Storage/yul/persistent/LISA/imgs/golgi_image_53.png",
  "prompt": "Segment the Golgi apparatus, identified by stacks of flattened, membrane-bound cisternae and associated vesicles located near the cell nucleus and endoplasmic reticulum (ER). Focus on identifying sac-like structures with a particular location, distinguish light from darker cellular components based on the distinct cisternal morphology and proximity to the cell nucleus."
}

```

Action: Using imagesegmentation tool

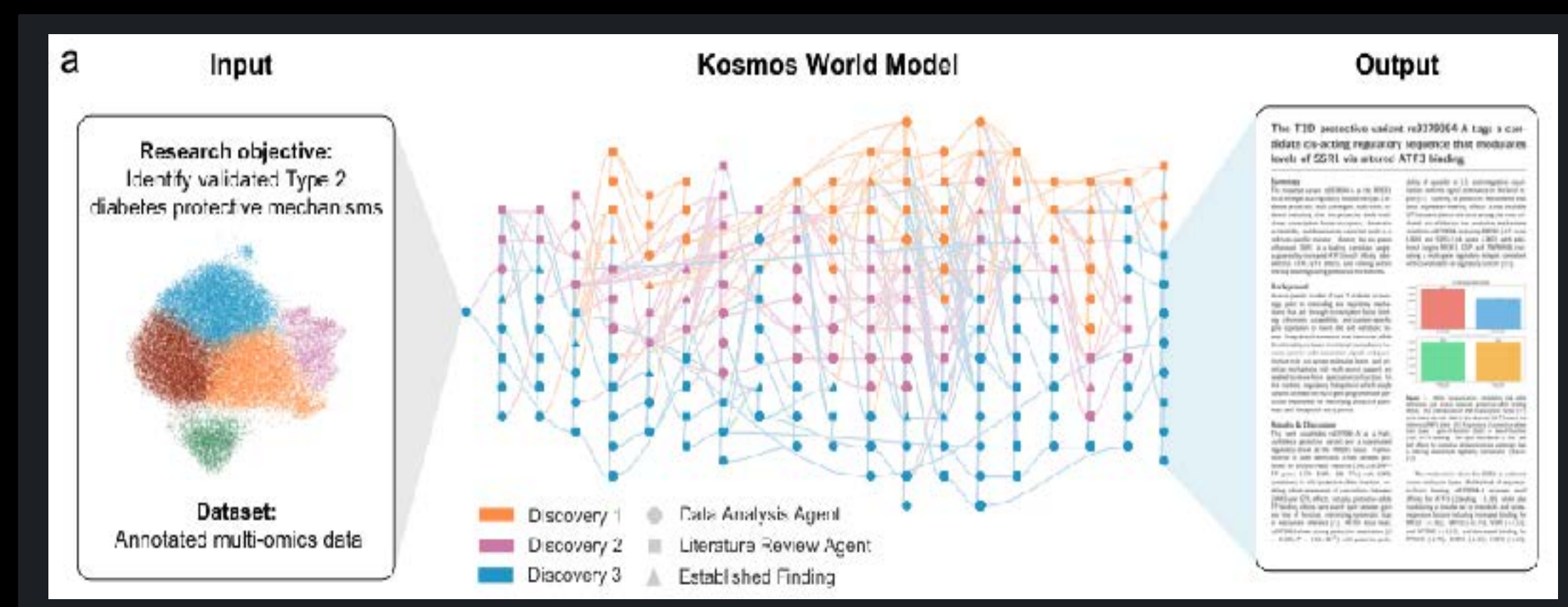
Observation from imagesegmentation: Segmentation completed successfully in segment_mask_path/home/klies/workspace/Storage/yul/persistent/LISA/imgs/golgi_image_53.png. The resulting mask is saved in segment_mask_path/home/klies/workspace/Storage/yul/persistent/LISA/imgs/golgi_image_53_mask.png.



Processed image [Iteration 3]

NEXT CLAIM: THE AI SCIENTIST

From analysing your images to running the whole study



Kosmos

Edison Scientific · arXiv 2511.02824 (Nov 2025)

Claim One 20-cycle run did “the equivalent of 6 months of their own research time” (collaborators’ estimate); 42,000 lines of code and 1,500 papers per run.

Fine print 79.4% of statements accurate overall, but only 57.9% of synthesis statements.

Also: EMSeek (Cornell, Science Advances 2026): “AI turns electron microscopy into materials insights in minutes”.



AILA

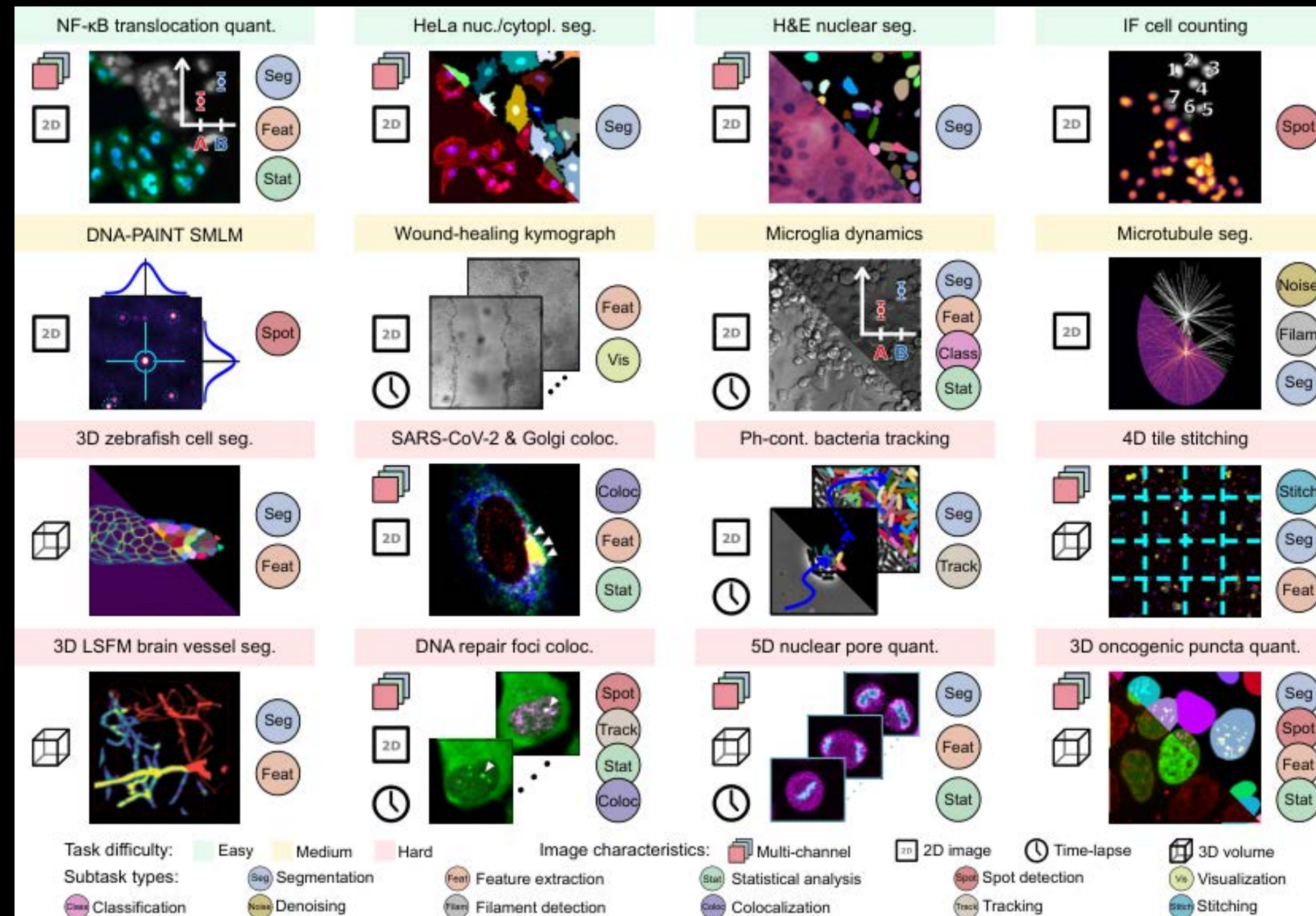
Mandal et al. · Nature Communications 2025

Claim “The first microscopy experiment performed by complete AI”: calibrates the AFM, picks modes, saves and analyses the images.

Fine print Agents “sleepwalk”: they deviate from instructions and carry out unauthorised steps.

REALITY CHECK: BIABENCH (SEPTEMBER 2026)

6 agents, 16 tasks rebuilt from published studies. Built by the Agentic-J authors.



0.85–0.96

best agents on easy 2D tasks (counting, nucleus segmentation)

0.19 / 0.05

on 5D nuclear-pore kinetics / 3D puncta quantification

40%

of agent-task pairs: three runs differ by more than 0.2

$r = 0.29$

process score vs. outcome: a correct run cannot be told from a wrong one

Tested: Claude Code, Codex, Biomni, Agentic-J, CopilotJ and others. General-purpose agents tied or beat biology-specific ones on 14 of 16 tasks.



...OR WILL CLAUDE AND CHATGPT SIMPLY GET BETTER?

Michael J. Black on X, 9 Sep 2026, on his way to ECCV 2026

His ECCV paper, VIGA

"After it was accepted at ECCV, it was quickly surpassed by people using Claude Code for the same purpose. Today GPT-6 Astra blows away all previous results."

"But we still head off to ECCV to tell the community about our invention that is now fully out of date."

The whole field

"Any paper I see at ECCV is likely two years out of date. In AI today, two years means your work is likely irrelevant."

"Another group of papers focuses on very niche problems where large models likely fail because of insufficient data or lack of business interest."

What to do now

"Every project should start by trying really hard to solve the problem with existing tools."

"If you haven't tried using Astra (or whatever comes next) to solve your problem, then you have not done your homework."

His proposal: a new "related work" section in every paper, on how current large models do on the task and why they fail.

SOURCES

[Omega: Royer, Nature Methods 2024](#)

[Omega code and demo videos](#)

[BioImage.IO Chatbot: Lei et al., Nature Methods 2024](#)

[Agentic-J: Johanns et al., arXiv 2606.02080](#)

[GenCellAgent: Yu et al., arXiv 2510.13896](#)

[napari-mcp \(Royer lab\)](#)

[Fiji MCP server](#)

[SnapCyte](#)

[Leica Aivia 15 press release \(May 2025\)](#)

[Kosmos: Edison Scientific, arXiv 2511.02824](#)

[AILA: Mandal et al., Nature Communications 2025](#)

[EMSeek: Cornell, Science Advances 2026](#)

[Biomni: Huang et al., bioRxiv 2025](#)

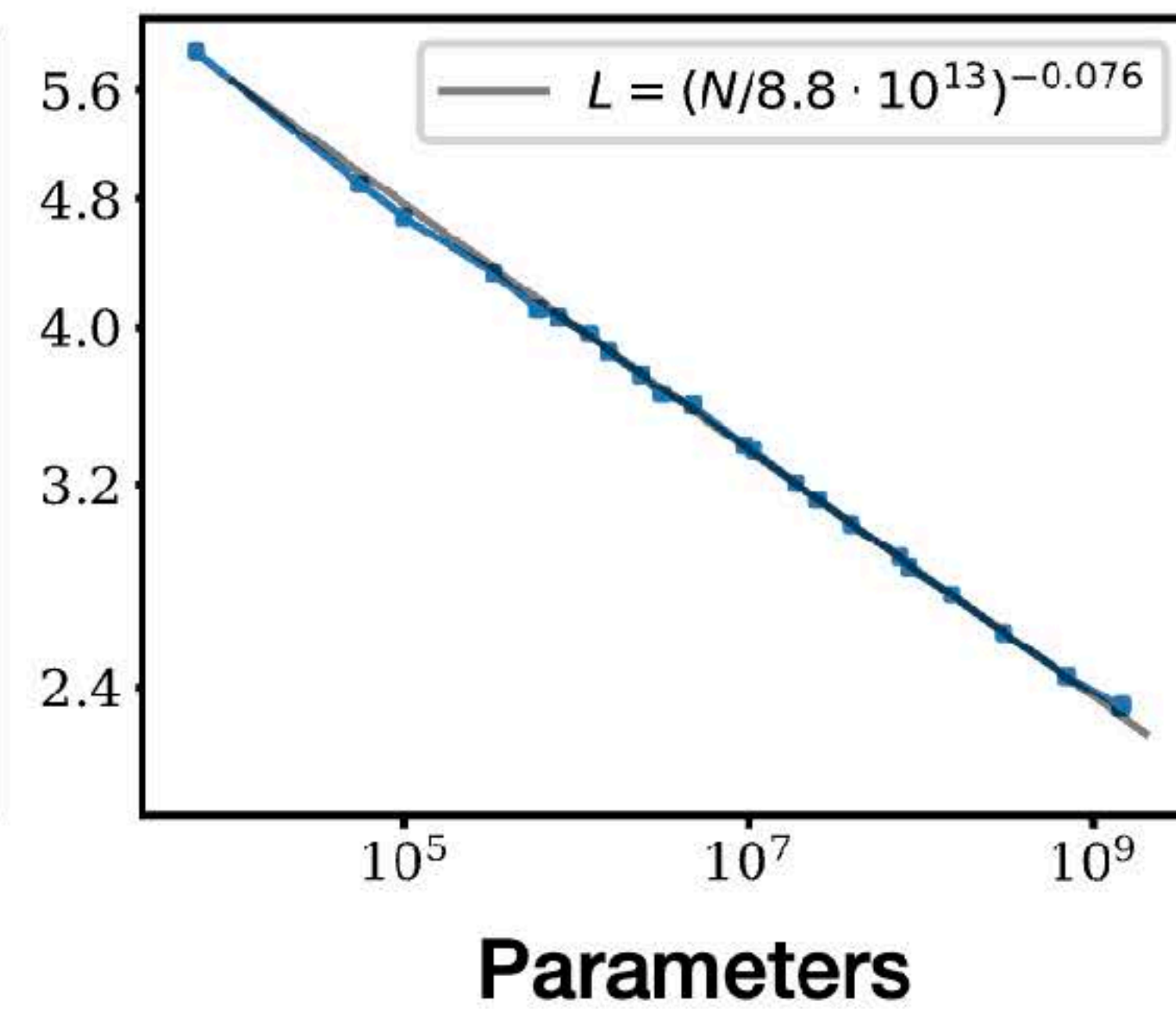
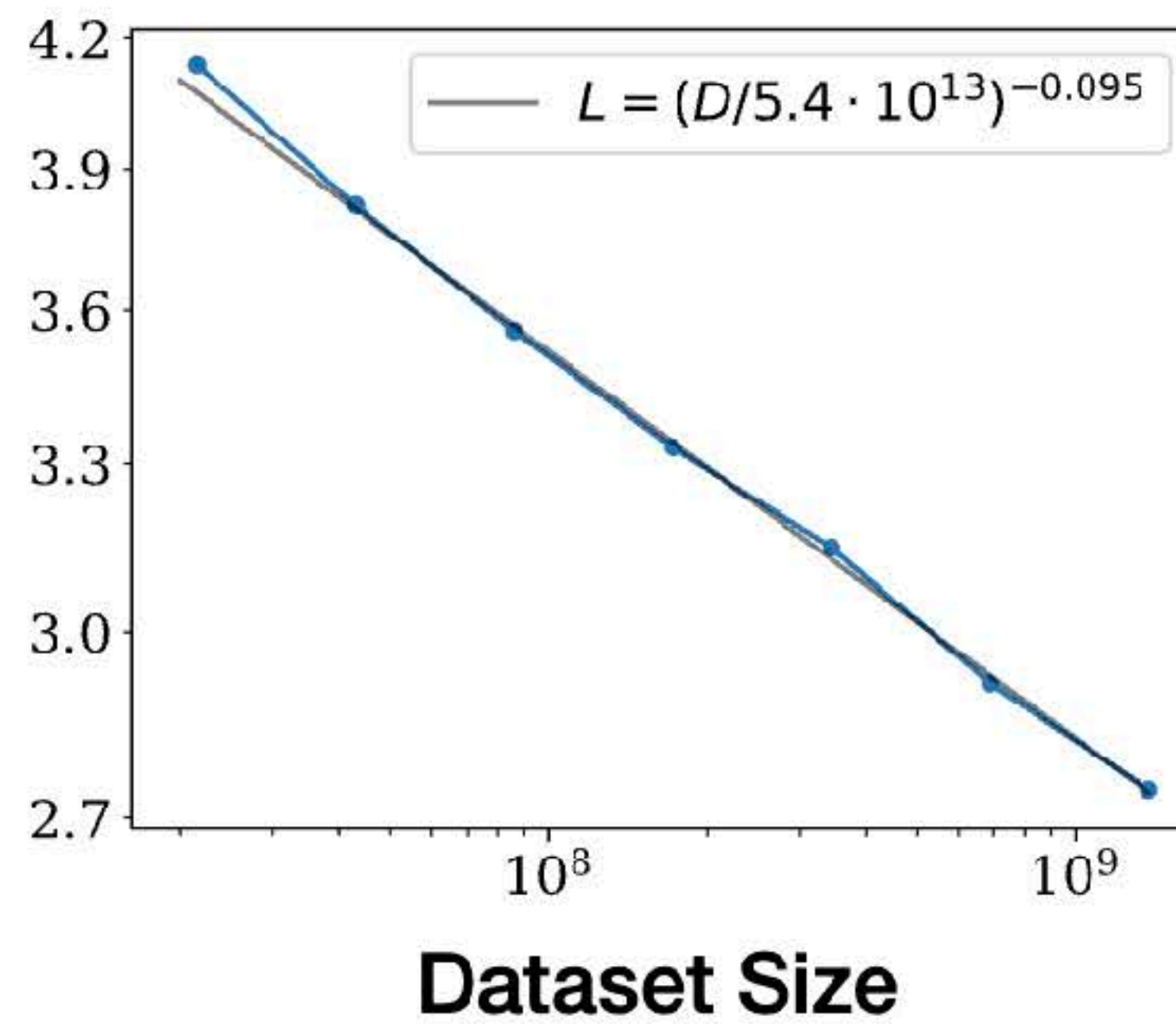
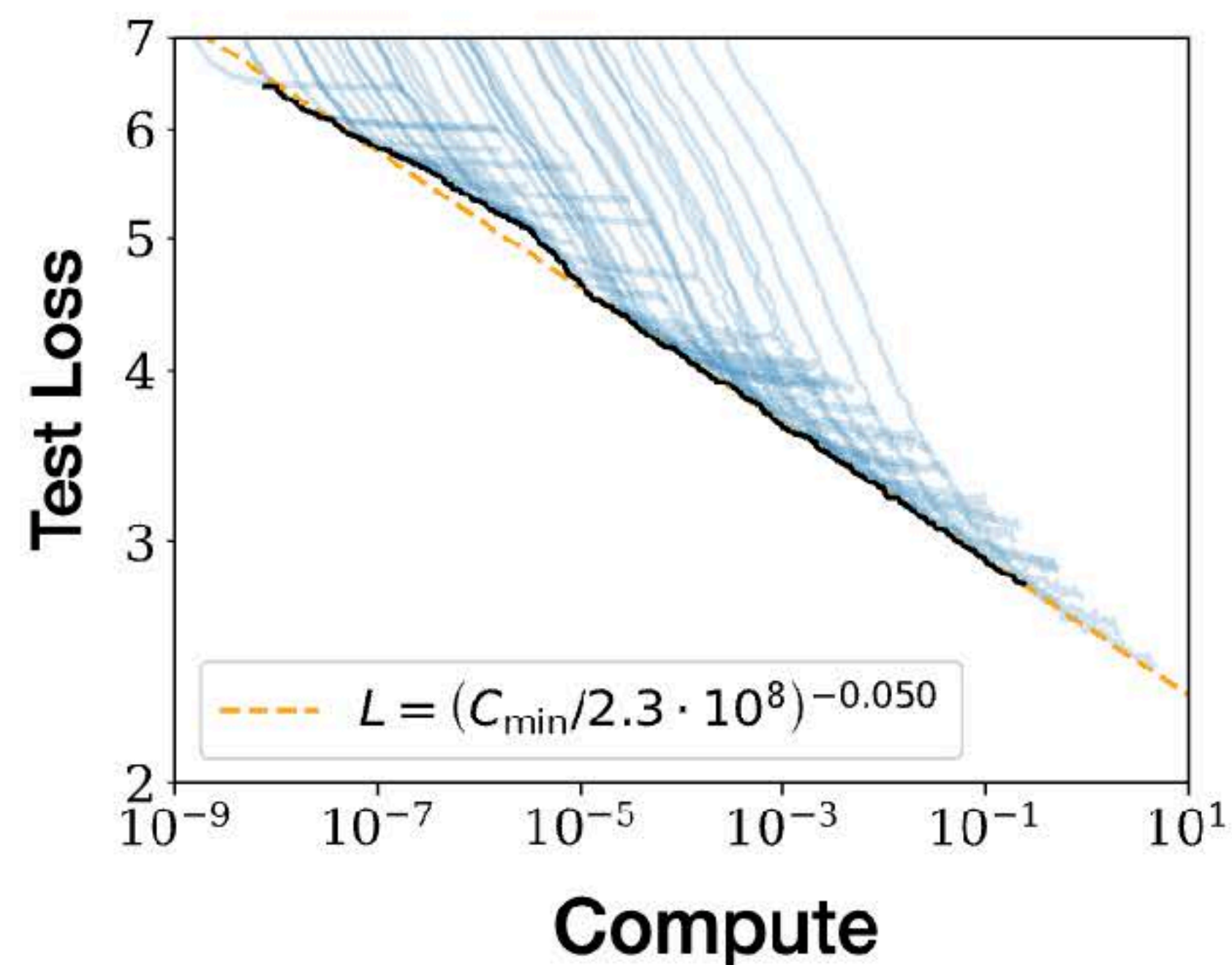
[CellVoyager: Zou group, bioRxiv 2025](#)

[BIABench: Pan et al., arXiv 2609.34274](#)

[BIABench project page](#)

[Michael J. Black on X \(9 Sep 2026\)](#)

[Hard Vision, Easy Vision: Rasheed et al., arXiv 2609.35718](#)



Kaplan J, McCandlish S, Henighan T, Brown TB, Chess B, Child R, et al. **Scaling laws for neural language models**. arXiv [cs.LG]. 2020

METHODS, DATA, OR COMPUTE?

THE SCALE OF MONEY · ALL IN EUROS, TO SCALE

One company's year, our decades

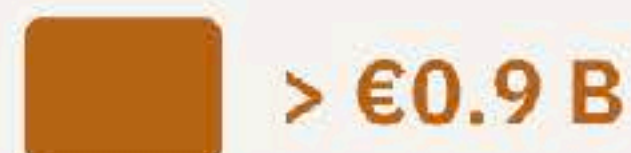
Anthropic, 2025 · one year

Operating expenses \$12.65 B



Anthropic, training environments

Reportedly discussed: over \$1 B in one year



Human Technopole · ten years

€140.3 M per year of state funding



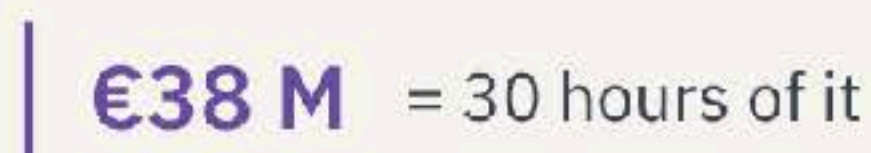
All cilia research in Europe · ten years

Our estimate, see methods



EU-funded cilia projects · 2015–2026

30 projects, Horizon 2020 + Horizon Europe



Cilia-AI doctoral network · 2025–2029

15 doctoral candidates, 8 partners



METHODS, DATA, OR COMPUTE?

THE SCALE OF MONEY · ALL IN EUROS, TO SCALE

One company's year, our decades

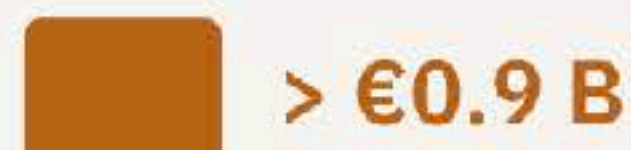
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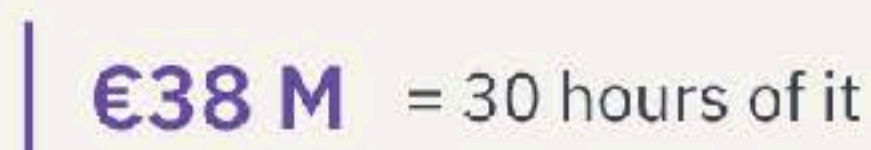
All cilia research in Europe · ten years

Our estimate, see methods



EU-funded cilia projects · 2015–2026

30 projects, Horizon 2020 + Horizon Europe



Cilia-AI doctoral network · 2025–2029

15 doctoral candidates, 8 partners



ALL OF IT, PURPOSEFULLY DEVELOPED, COLLECTED, AND UTILIZED!

nature methods

Perspective

Community-developed checklists for publishing images and image analyses

<https://doi.org/10.1038/s41592-023-01987-9>

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Accepted: 26 July 2023

Published online: 14 September 2023

 Check for updates

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Images document scientific discoveries and are prevalent in modern biomedical research. Microscopy imaging in particular is currently undergoing rapid technological advancements. However, for scientists wishing to publish obtained images and image-analysis results, there are currently no unified guidelines for best practices. Consequently, microscopy images and image data in publications may be unclear or difficult to interpret. Here, we present community-developed checklists for preparing light microscopy images and describing image analyses for publications. These checklists offer authors, readers and publishers key recommendations for image formatting and annotation, color selection, data availability and reporting image-analysis workflows. The goal of our guidelines is to increase the clarity and reproducibility of image figures and thereby to heighten the quality and explanatory power of microscopy data.

Checklist for image publishing

Image format

-  Focus on relevant image content (e.g., crop, rotate, resize) ☐ Minimal
-  Separate individual images ☐
-  Show example image used for quantifications ☐
-  Indicate position of zoom view/inset in full-view/original image ☐
-  Show images of the range of the described phenotype ☐

Image colors and channels

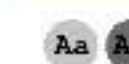
-  Annotation of channels (staining, marker, etc.) visible ☐ Minimal
-  Adjust brightness/contrast, report adjustments, use uniform color scales ☐
-  Image comparison: use the same adjustments ☐
-  Channel colors: high visibility on the background
Best visibility: grayscale ☐
-  Multicolors: provide grayscale for each color channel ☐
-  Multicolor: if channels are merged, make accessible to color-blind individuals ☐
-  Provide intensity scales (calibration bar) for grayscale, color, pseudocolor etc. ☐ Recommended
-  Pseudocolored images: additionally provide grayscale version for comparison ☐ Ideal
-  Gamma adjustments: additionally provide linear-adjusted image for comparison ☐

Image annotation

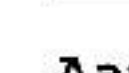
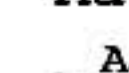
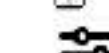
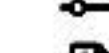
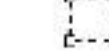
-  Add scale information (scale bar, image length in figure/figure legend) ☐ Minimal
-  Explain all annotations (in figure/figure legend) ☐
-  Annotations should be legible (line width, size/point size, color) ☐
-  Annotations should not obscure key data ☐
-  Annotate imaging details important for interpreting the figure (depending on the main message and imaging technique, this may be, e.g., image pixel size, imaging intervals (time-lapse in movies), exposure time or anatomical section) ☐ Recommended

Image availability

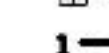
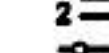
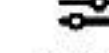
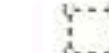
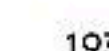
-  Images are shared (lossless compression/microscope images) ☐
-  Image files are freely downloadable (public database) ☐ Recommended
-  Image files are in dedicated image database (added-value database or image archive) ☐ Ideal

Checklists for publication of image-analysis workflows

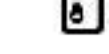
Established workflows

-  Cite workflow and platform ☐ Minimal
-  Key settings ☐
-  Example data ☐
-  Manual ROI ☐
-  193 Exact version ☐
-  All settings ☐ Recommended
-  Public example ☐
-  Document usage (e.g., screen recording or tutorial) ☐ Ideal
-  Cloud hosted or container ☐

New workflows

-  Cite components and platform ☐ Minimal
-  1 2 Describe sequence ☐
-  Key settings ☐
-  Example data and code ☐
-  Manual ROI ☐
-  193 Exact versions ☐
-  All settings ☐ Recommended
-  Public example data and code ☐
-  Rationale ☐
-  Limitations ☐
-  Screen recording or tutorial ☐ Ideal
-  Easy install and usage, container ☐

Machine learning workflows

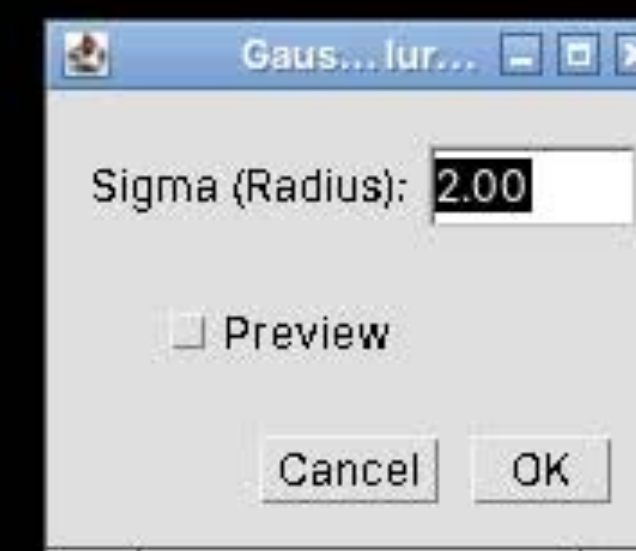
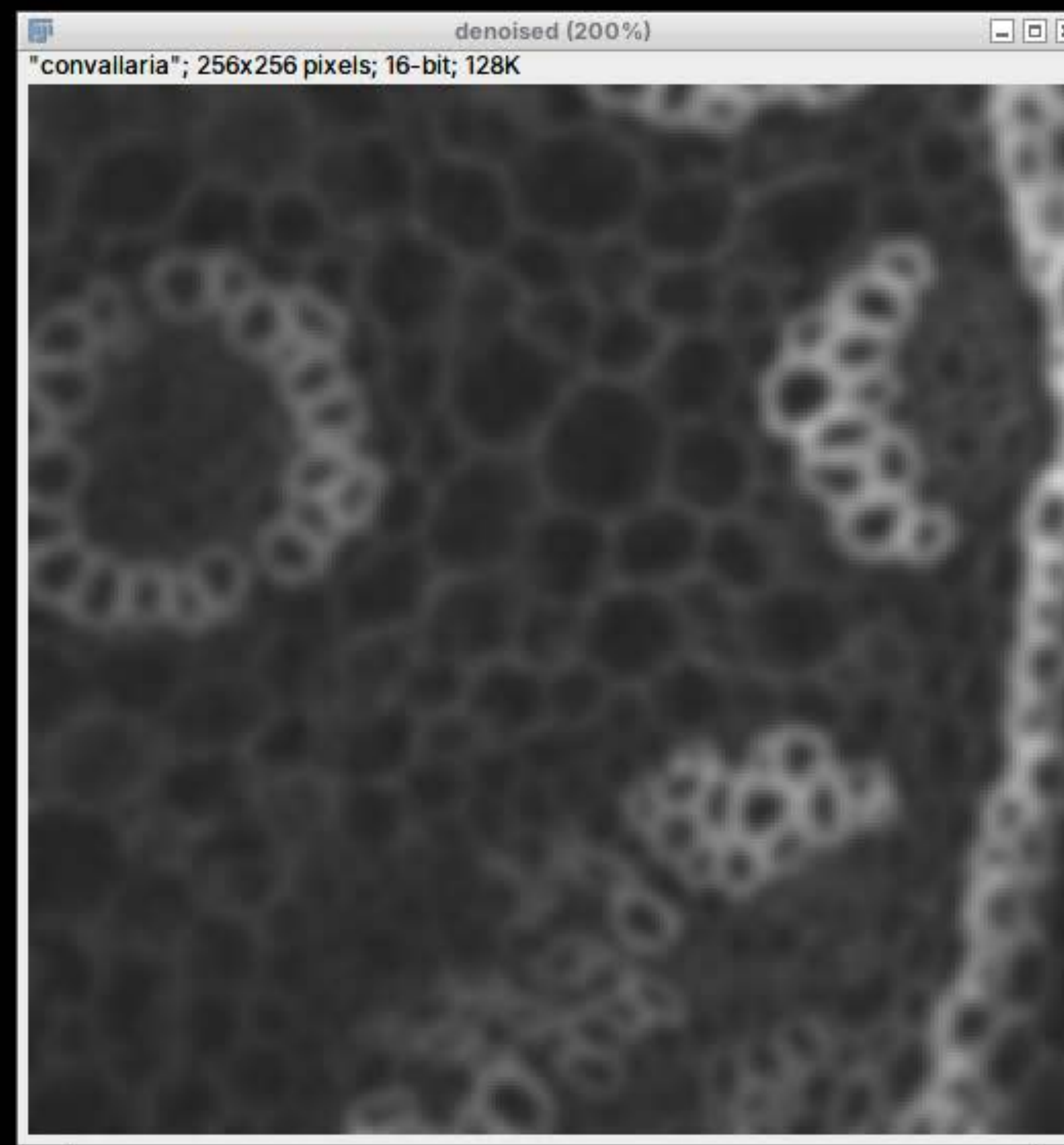
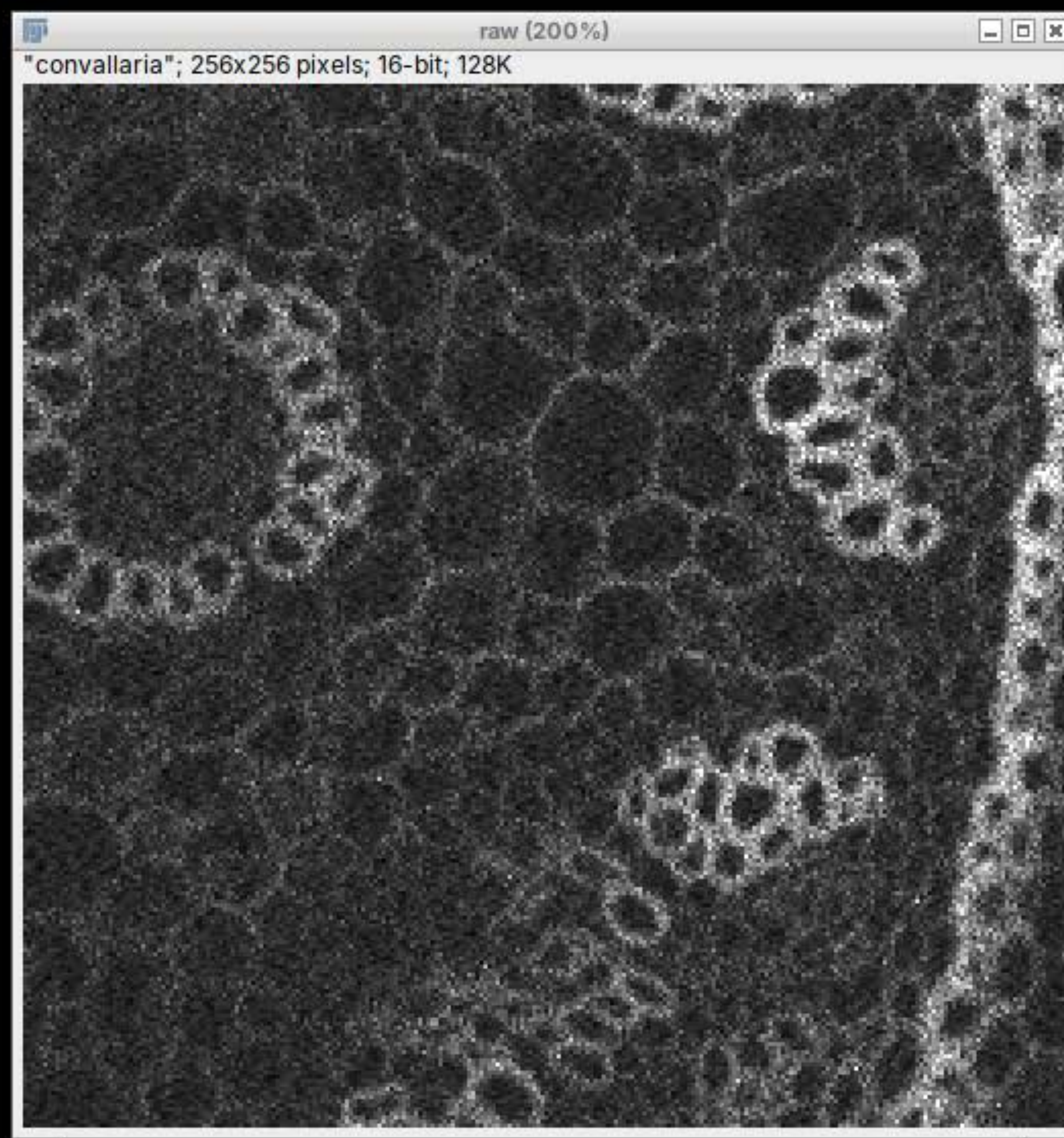
-  Cite original method ☐ Minimal (all models)
-  Access to model ☐
-  Example or validation data ☐
-  Training and testing data and metadata ☐ Recommended (pretrained and new models)
-  Code available ☐
-  Limitations ☐
-  Cloud hosted or container ☐
-  Standardized format ☐ Ideal (new models)



1. WHAT IS AN IMAGE? WHAT IS IMAGE ANALYSIS?
2. CLASSICAL ANALYSIS & MACHINE LEARNING
3. FROM DEEP LEARNING TO AGENTIC SYSTEMS
4. PRACTICAL EXERCISE: GETTING YOUR HANDS DIRTY WITH UNSUPERVISED DENOISING

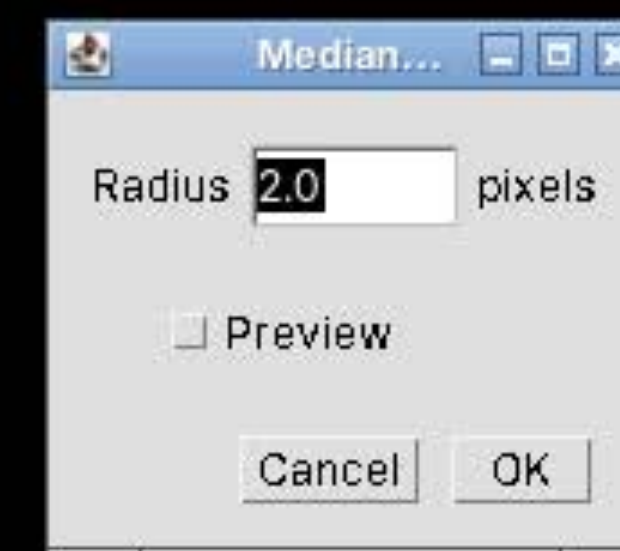
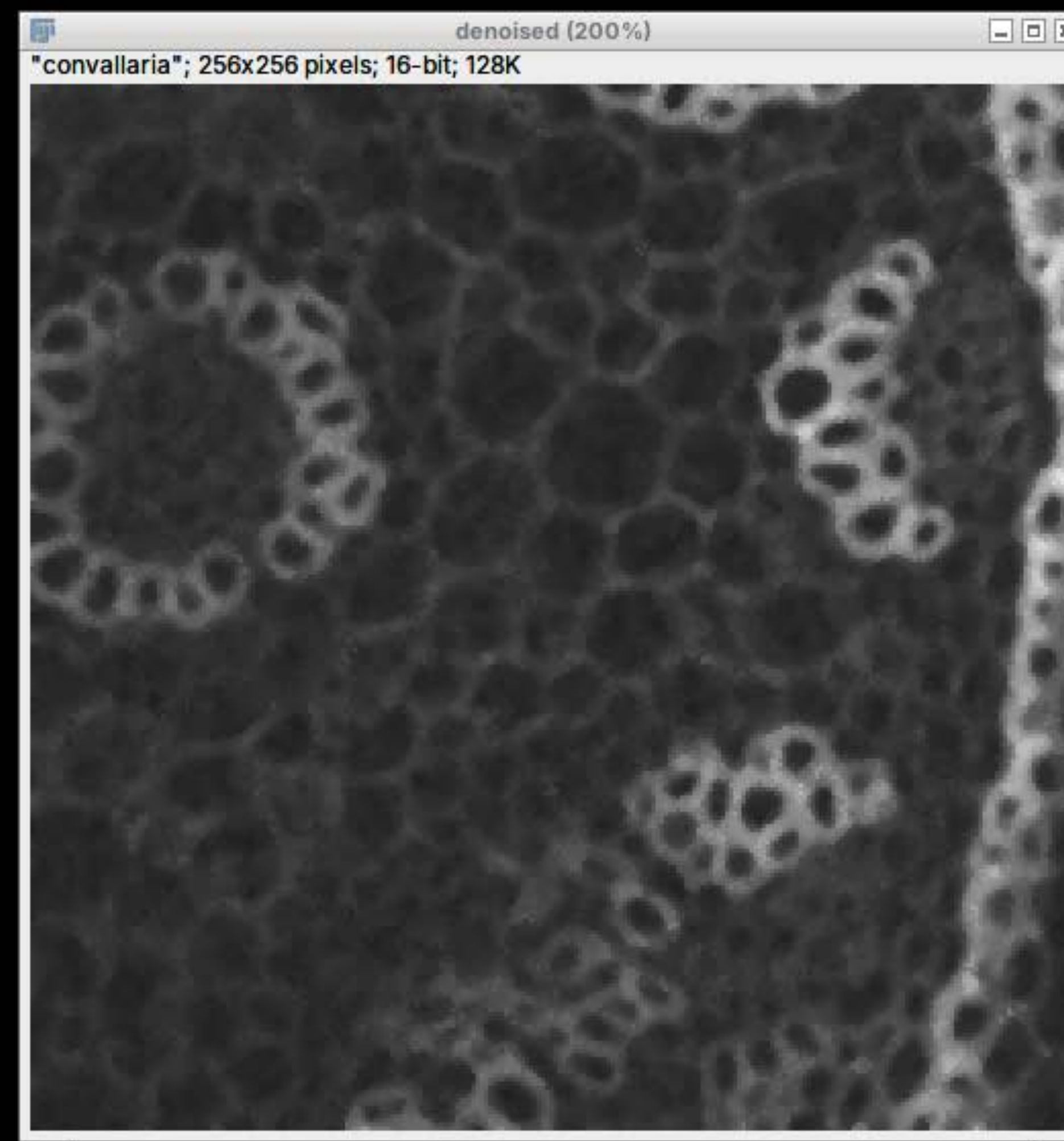
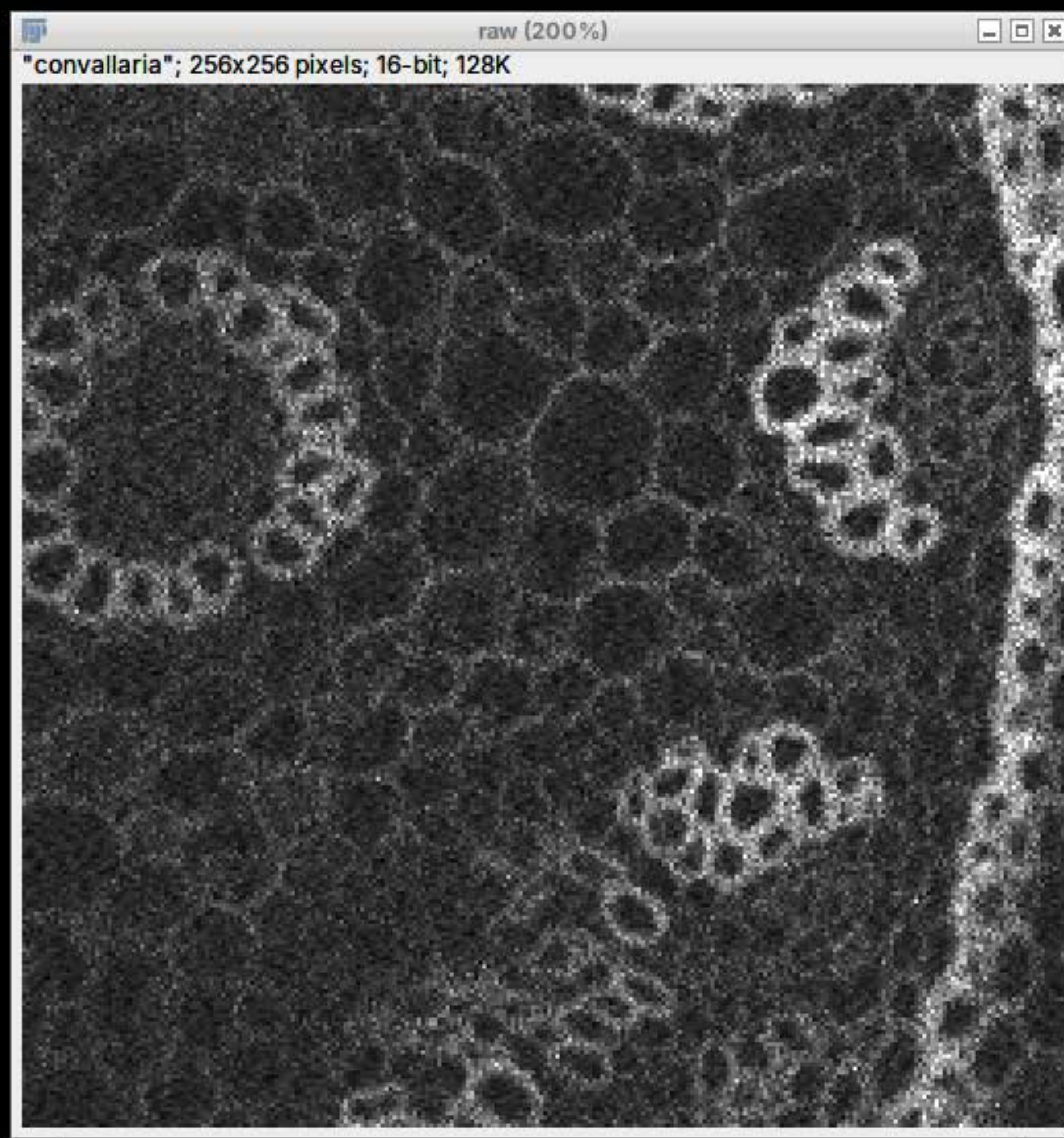
Gaussian blur

Fiji: Process > Filters > Gaussian Blur... (sigma = 2 px)



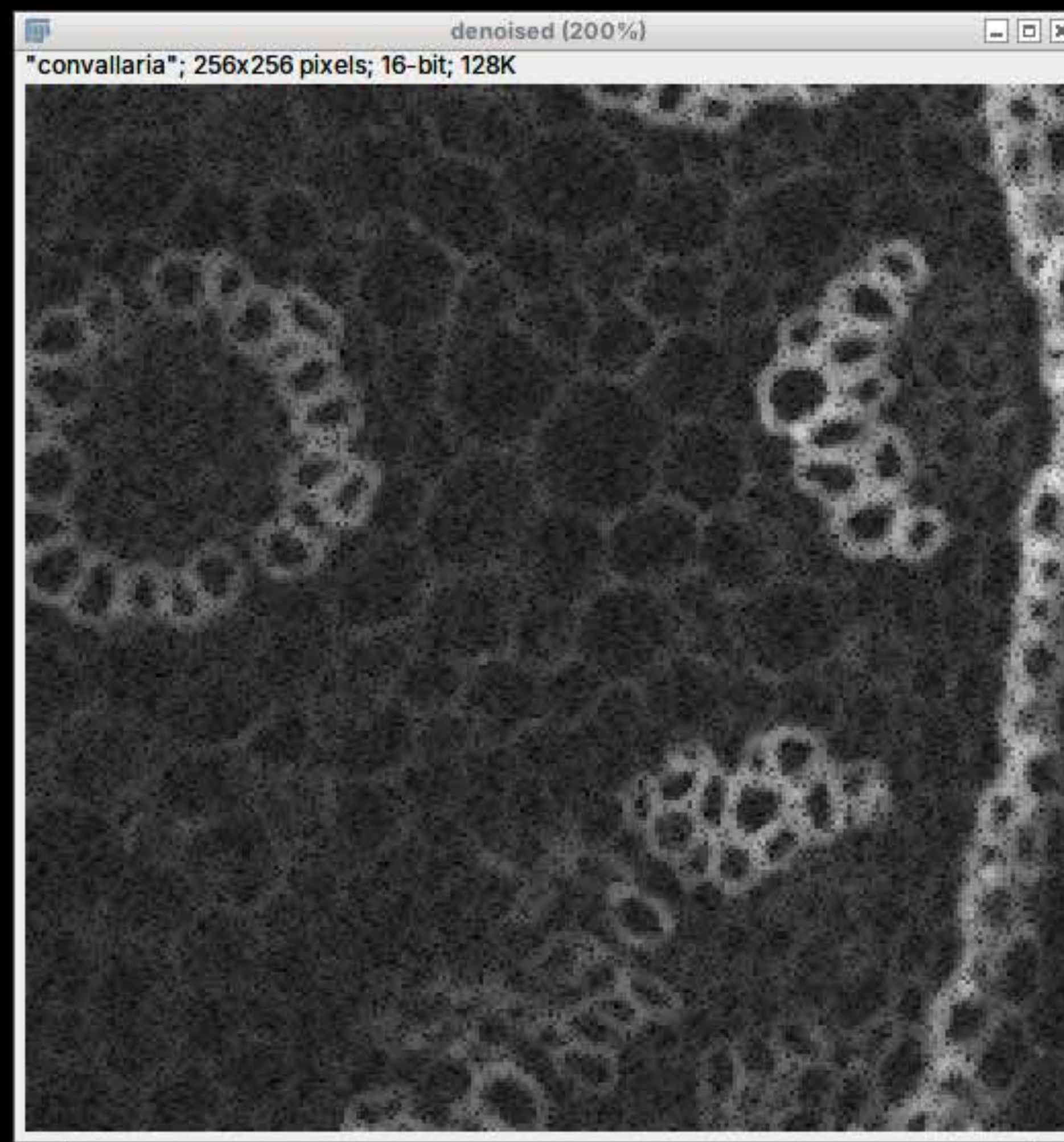
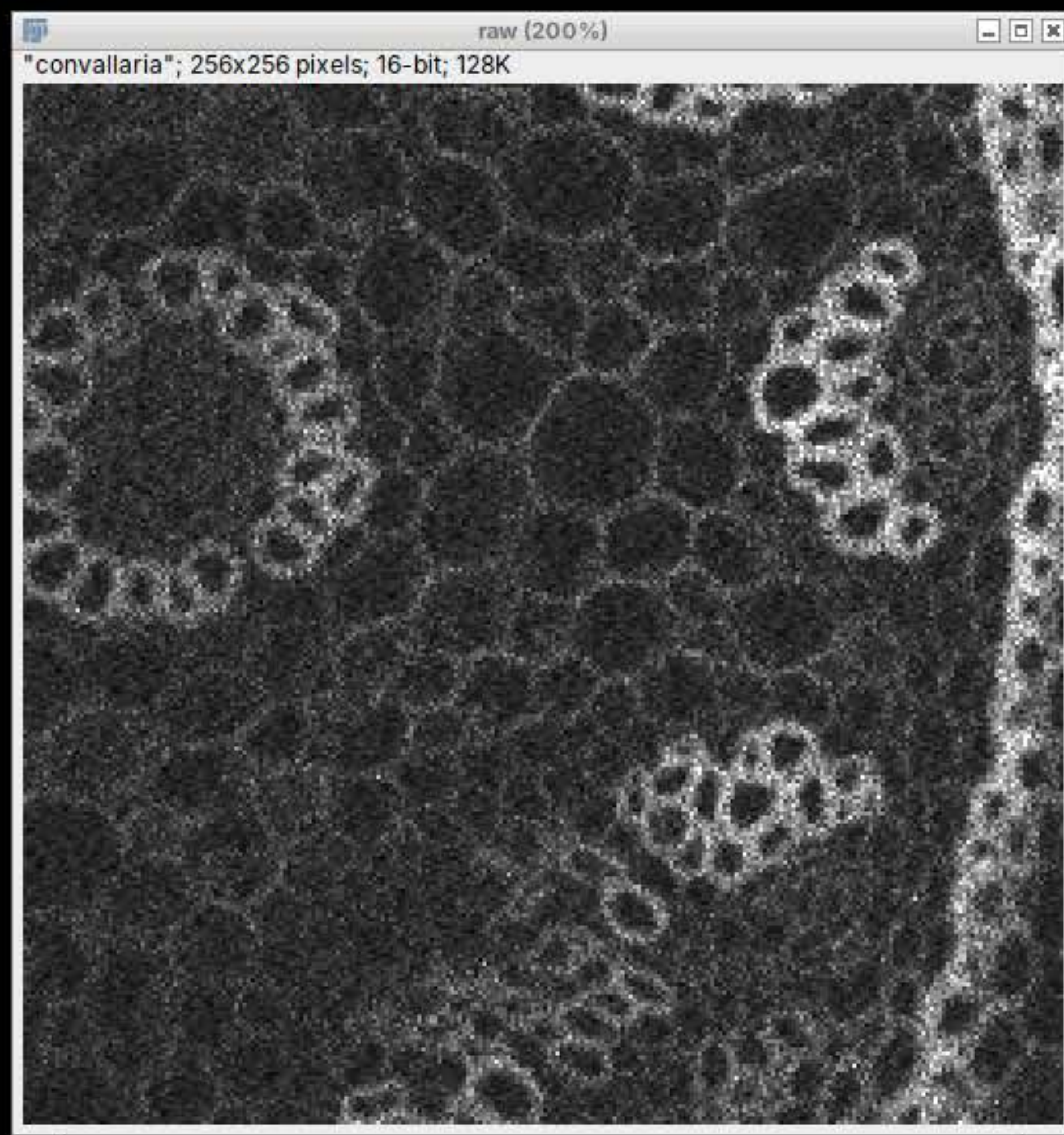
Median filter

Fiji: Process > Filters > Median... (radius = 2 px)



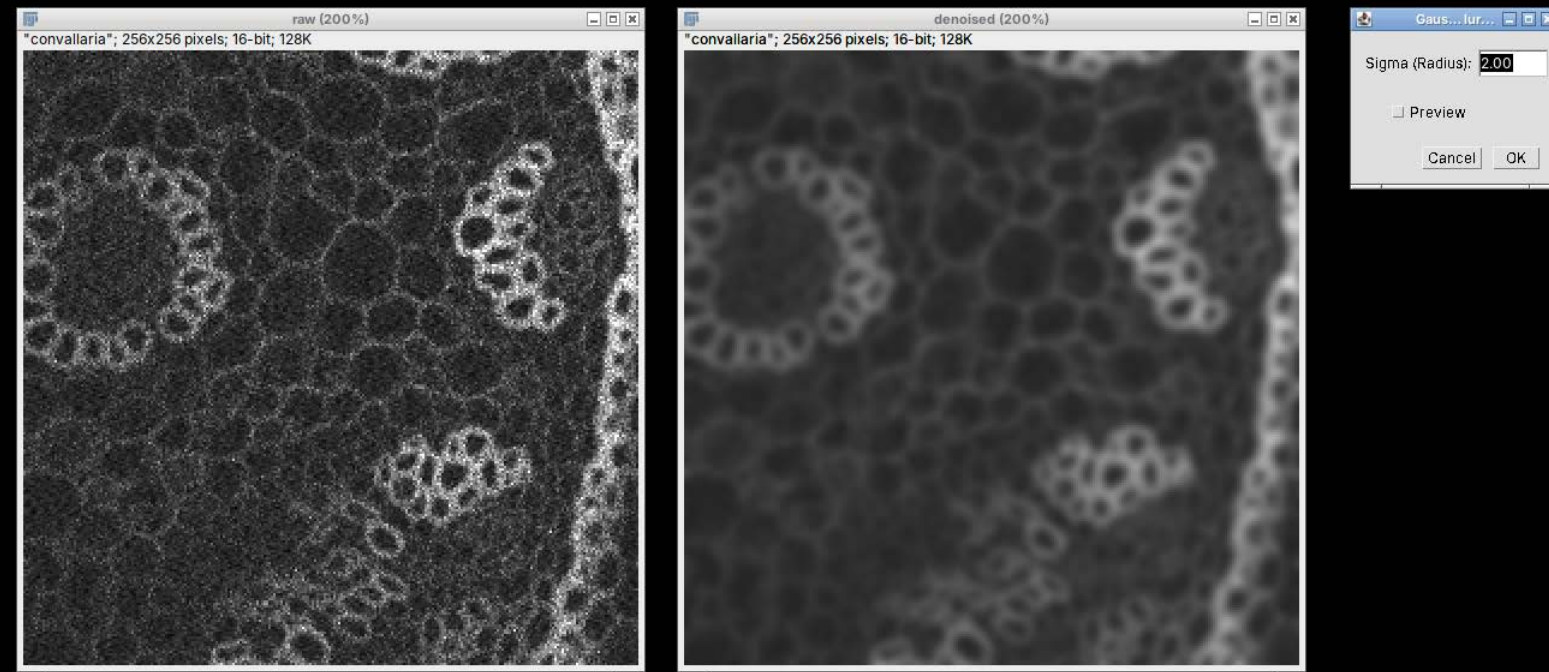
Remove outliers (hot pixels)

Fiji: Process > Noise > Remove Outliers... (radius 2, threshold 50, bright)



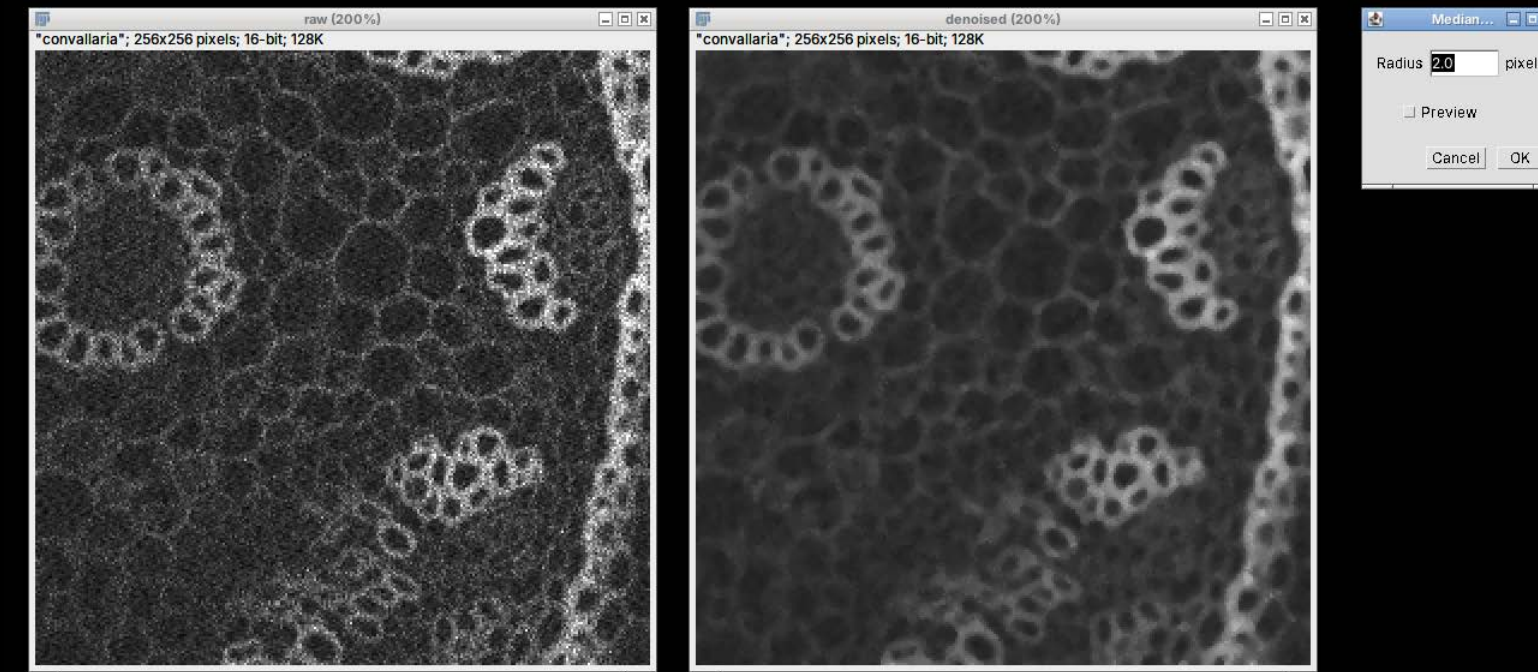
Gaussian blur

Fiji: Process > Filters > Gaussian Blur... (sigma = 2 px)



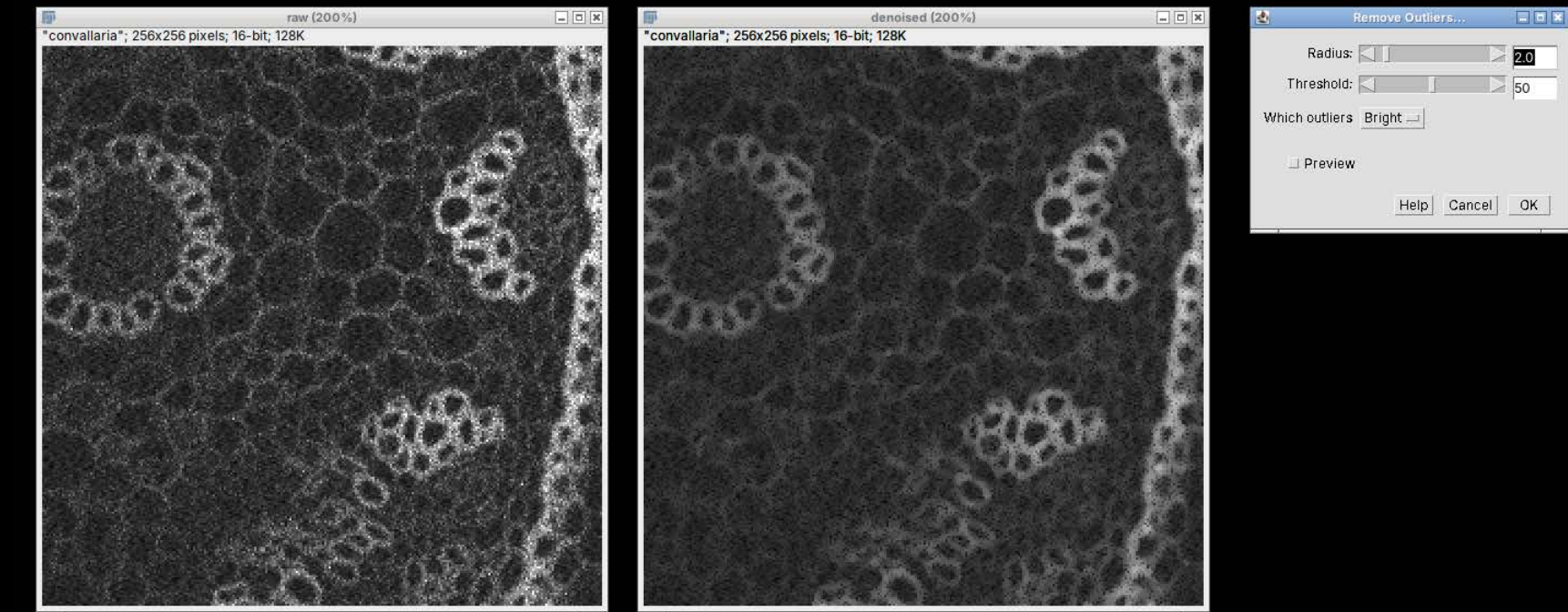
Median filter

Fiji: Process > Filters > Median... (radius = 2 px)



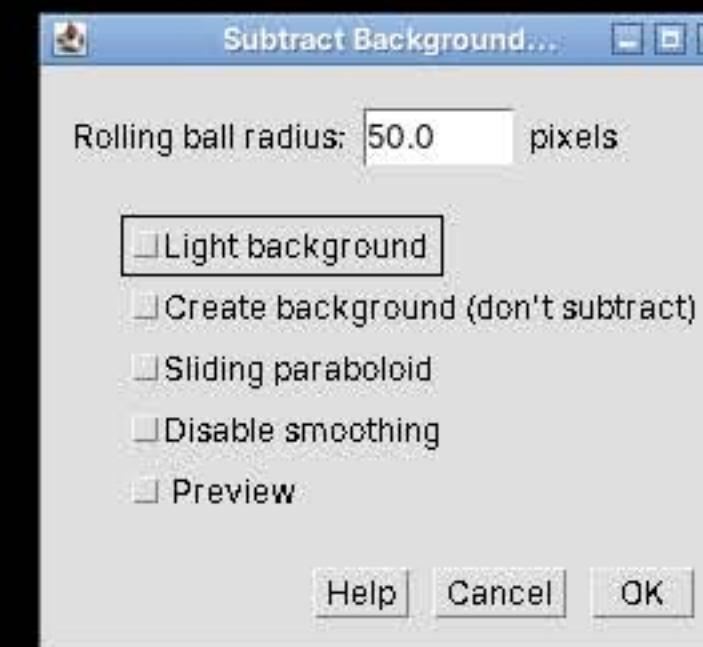
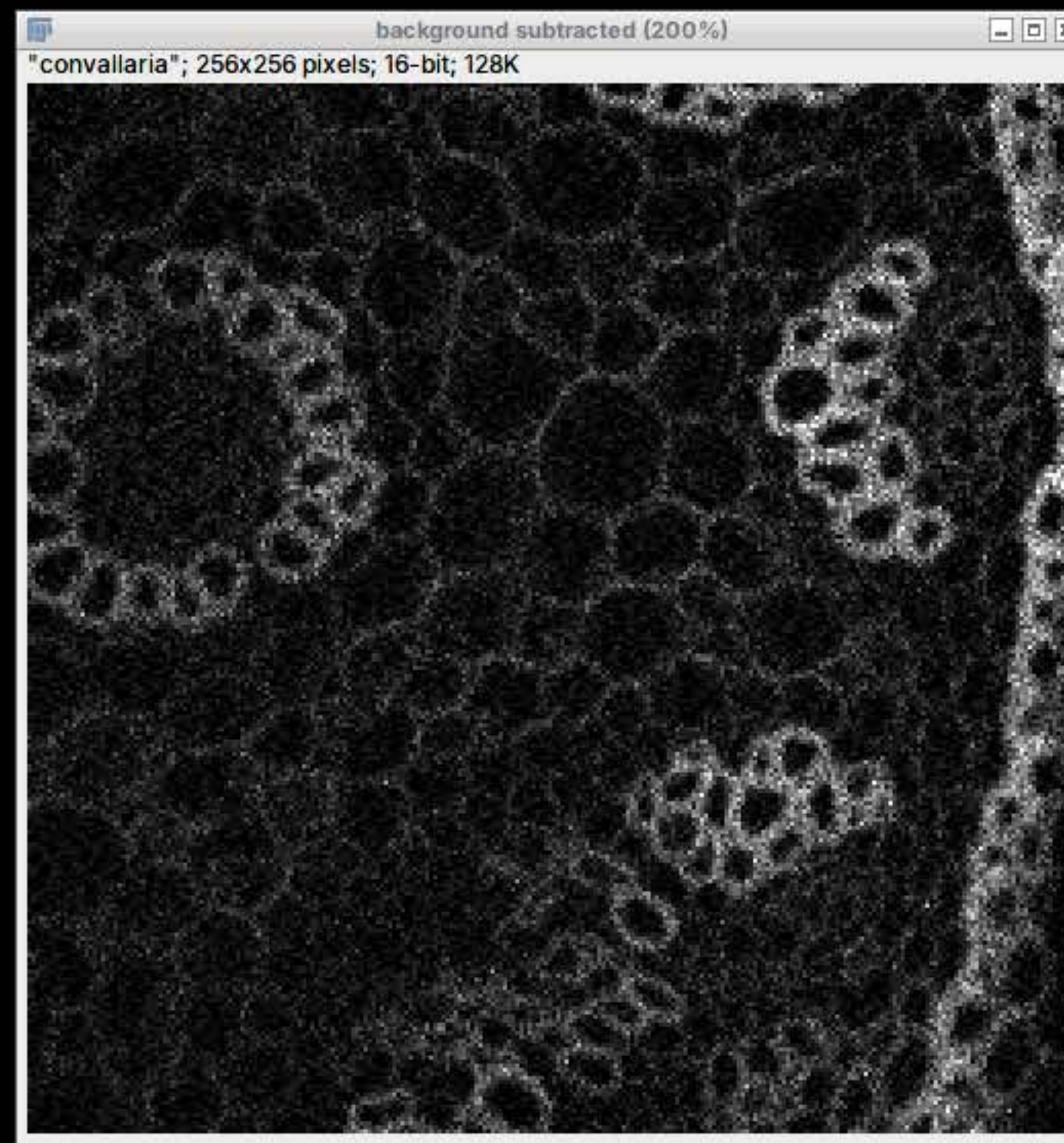
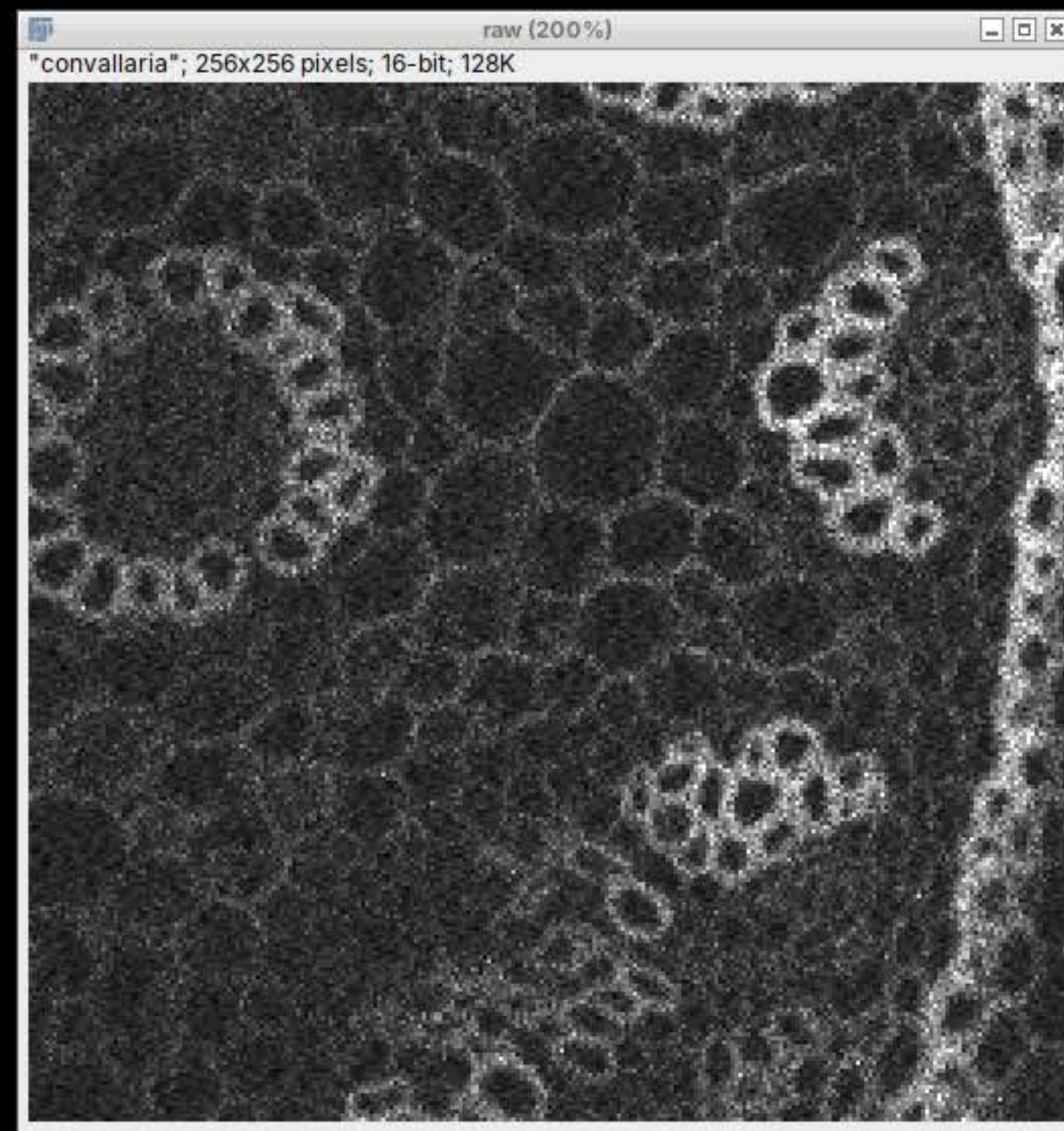
Remove outliers (hot pixels)

Fiji: Process > Noise > Remove Outliers... (radius 2, threshold 50, bright)



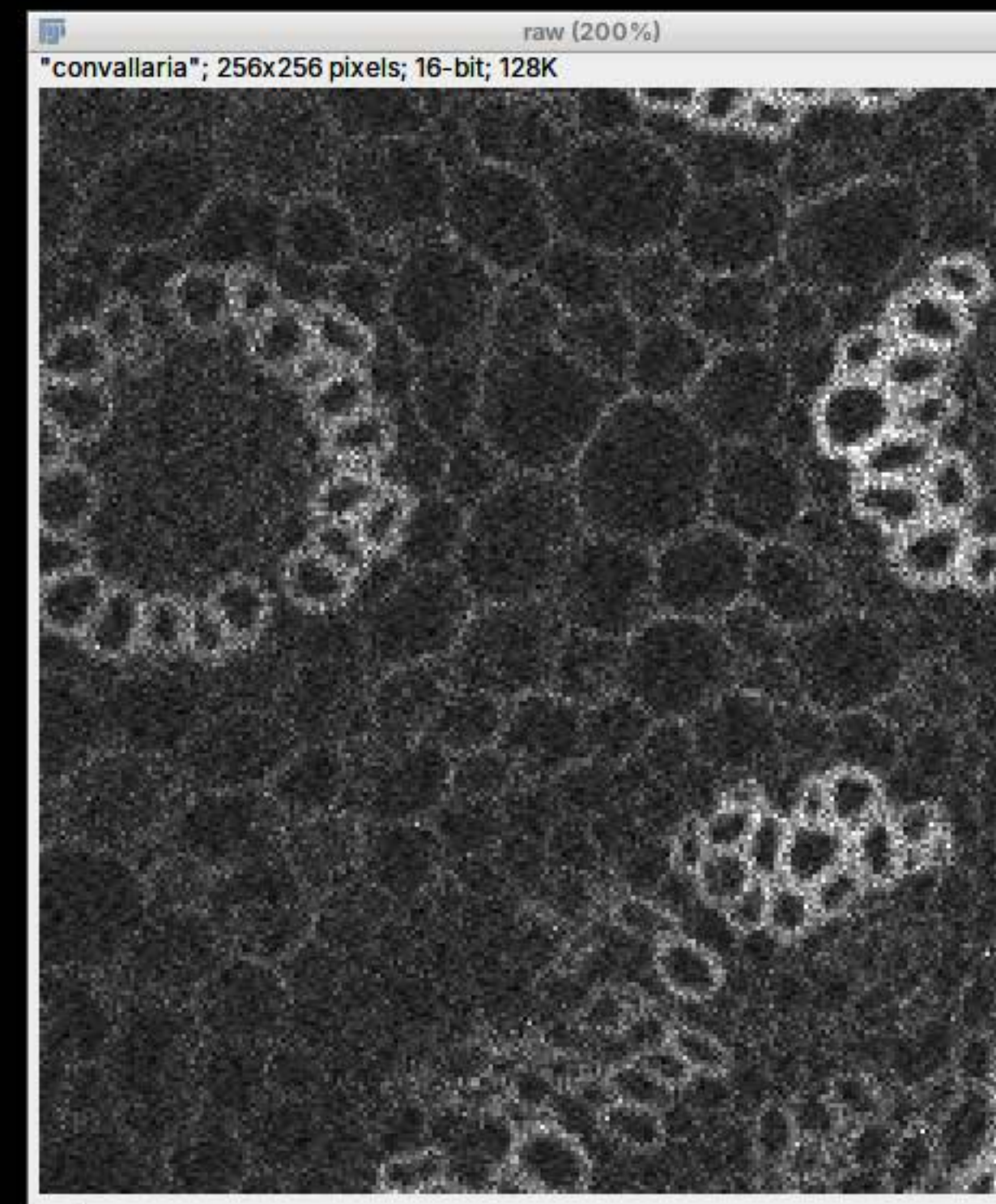
Rolling ball: removes background, NOT noise

Fiji: Process > Subtract Background... (rolling ball radius = 50 px)



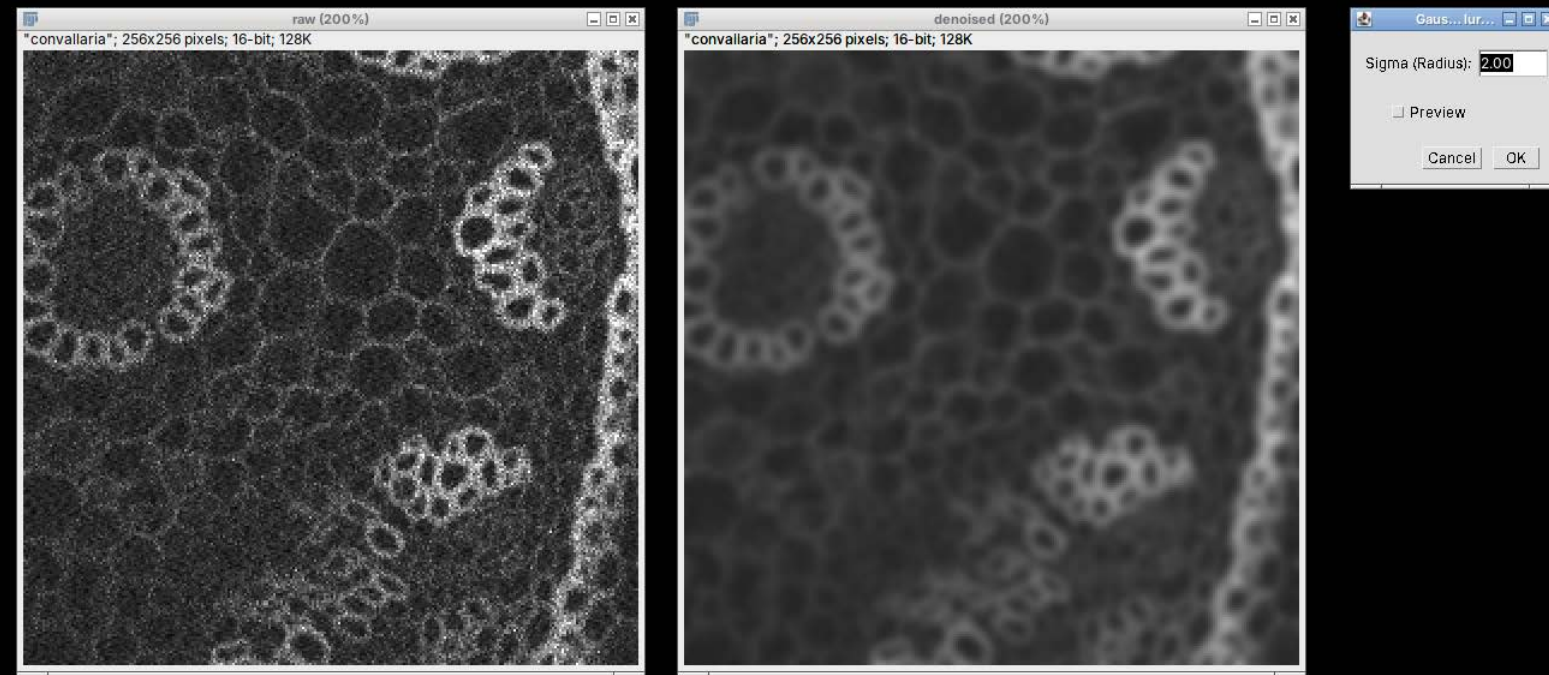
Bilateral filter (edge-preserving)

Fiji: Plugins > Process > Bilateral Filter



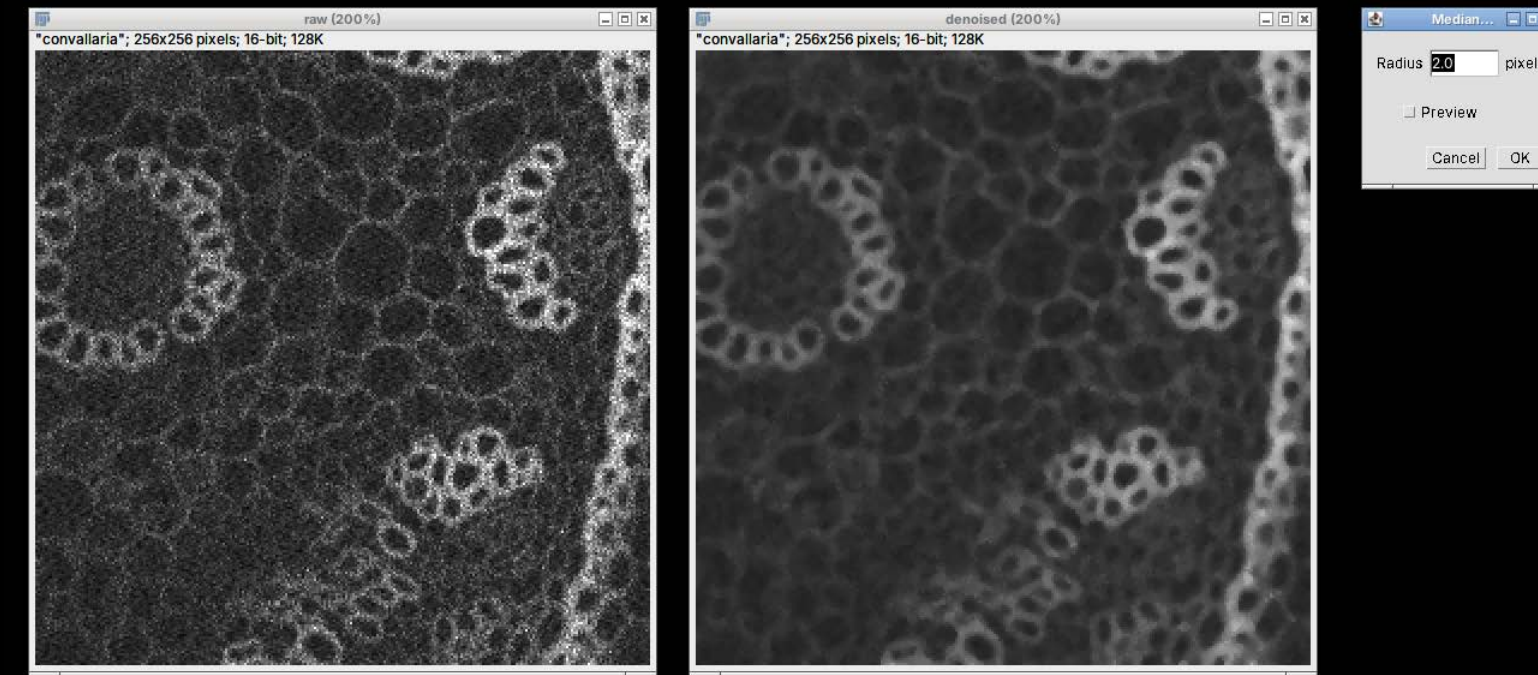
Gaussian blur

Fiji: Process > Filters > Gaussian Blur... (sigma = 2 px)



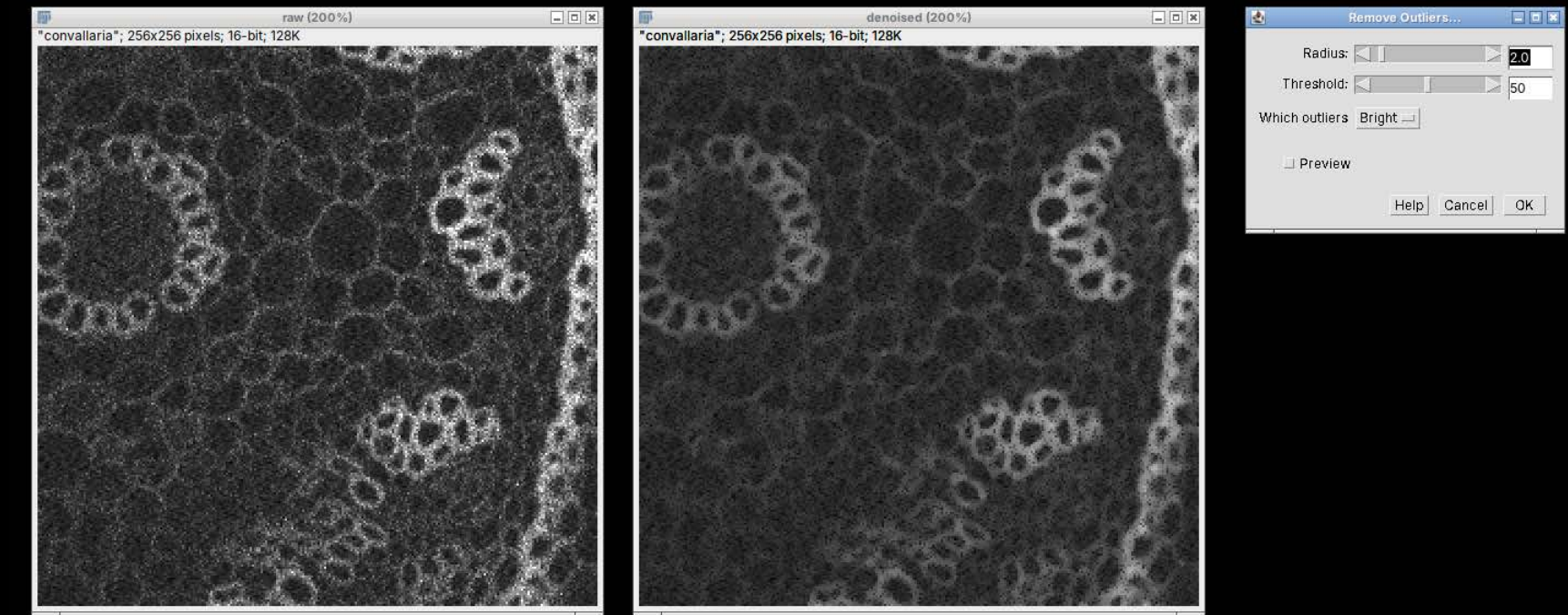
Median filter

Fiji: Process > Filters > Median... (radius = 2 px)



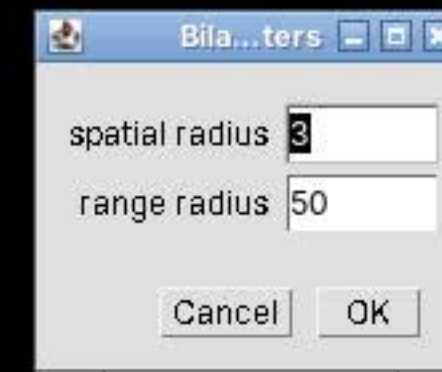
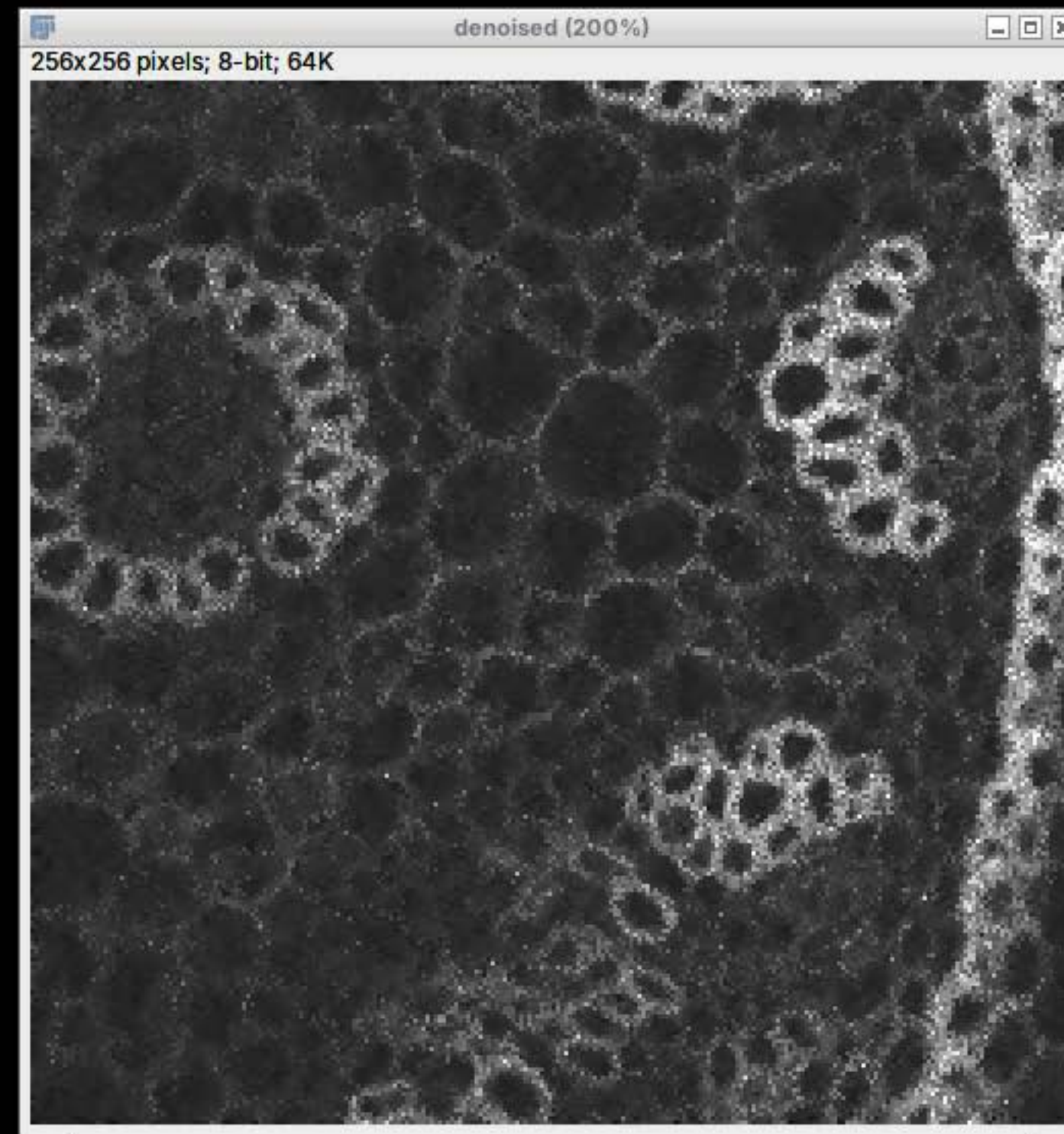
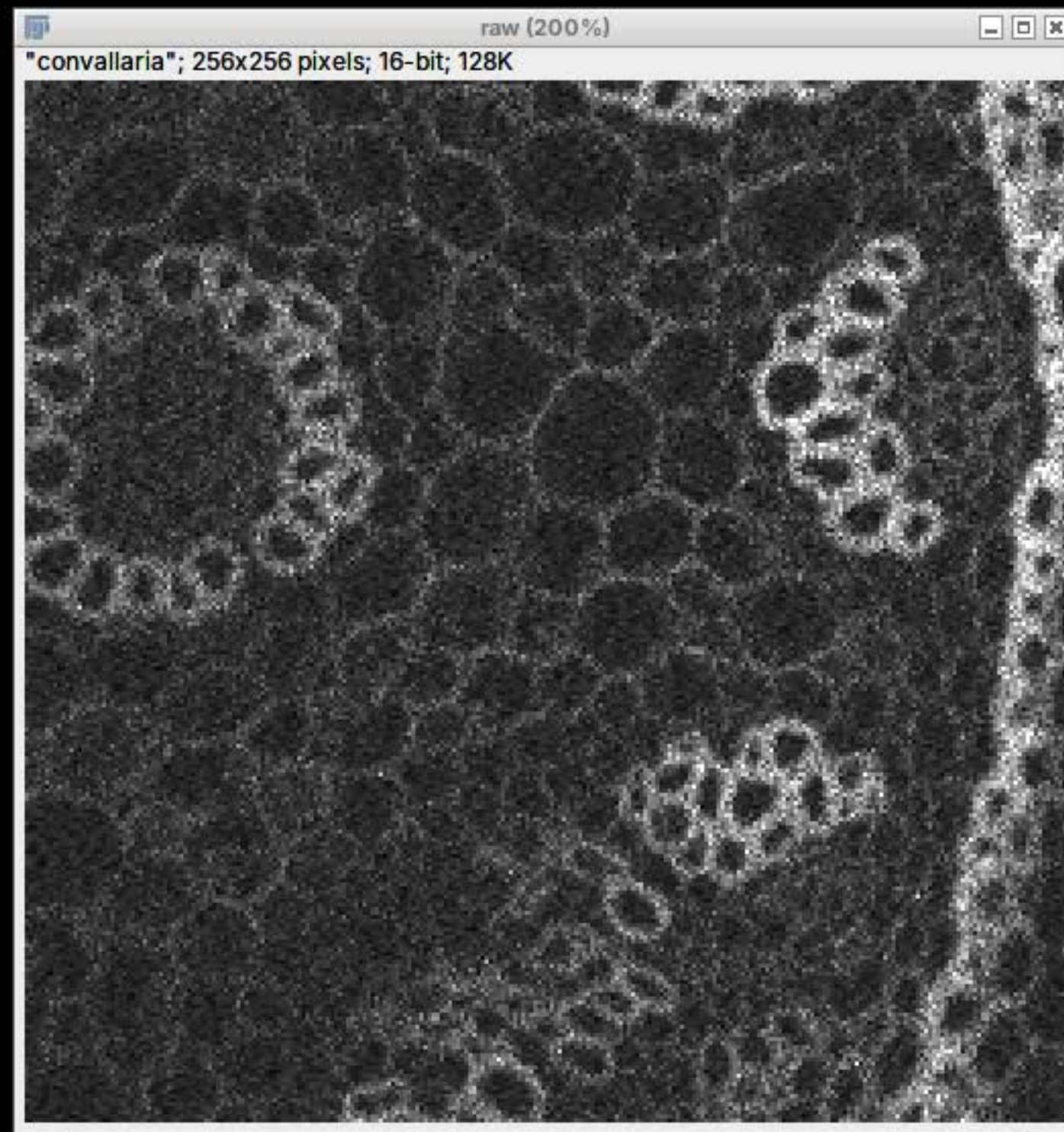
Remove outliers (hot pixels)

Fiji: Process > Noise > Remove Outliers... (radius 2, threshold 50, bright)



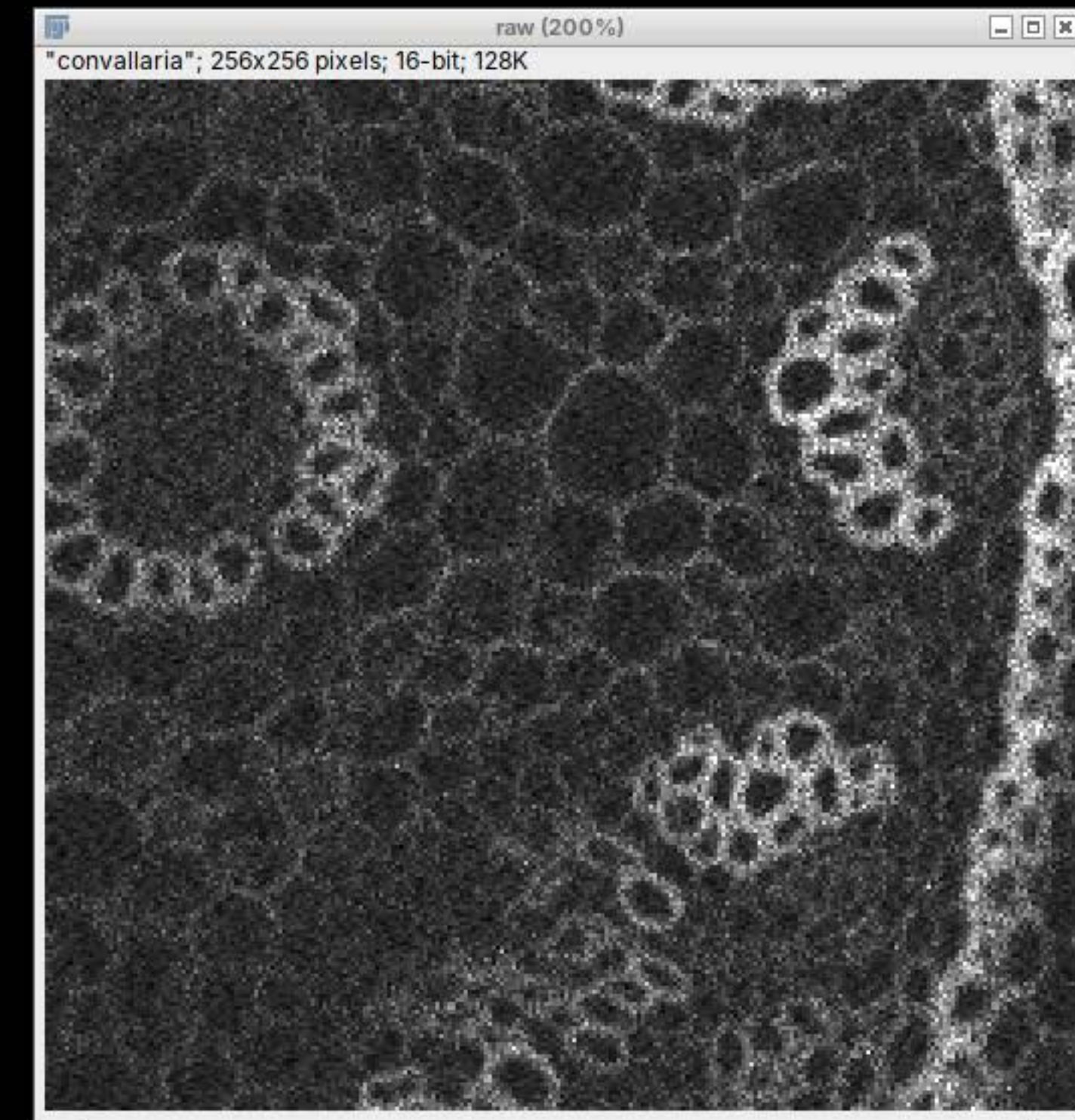
Bilateral filter (edge-preserving)

Fiji: Plugins > Process > Bilateral Filter (spatial 3, range 50; 8-bit only)



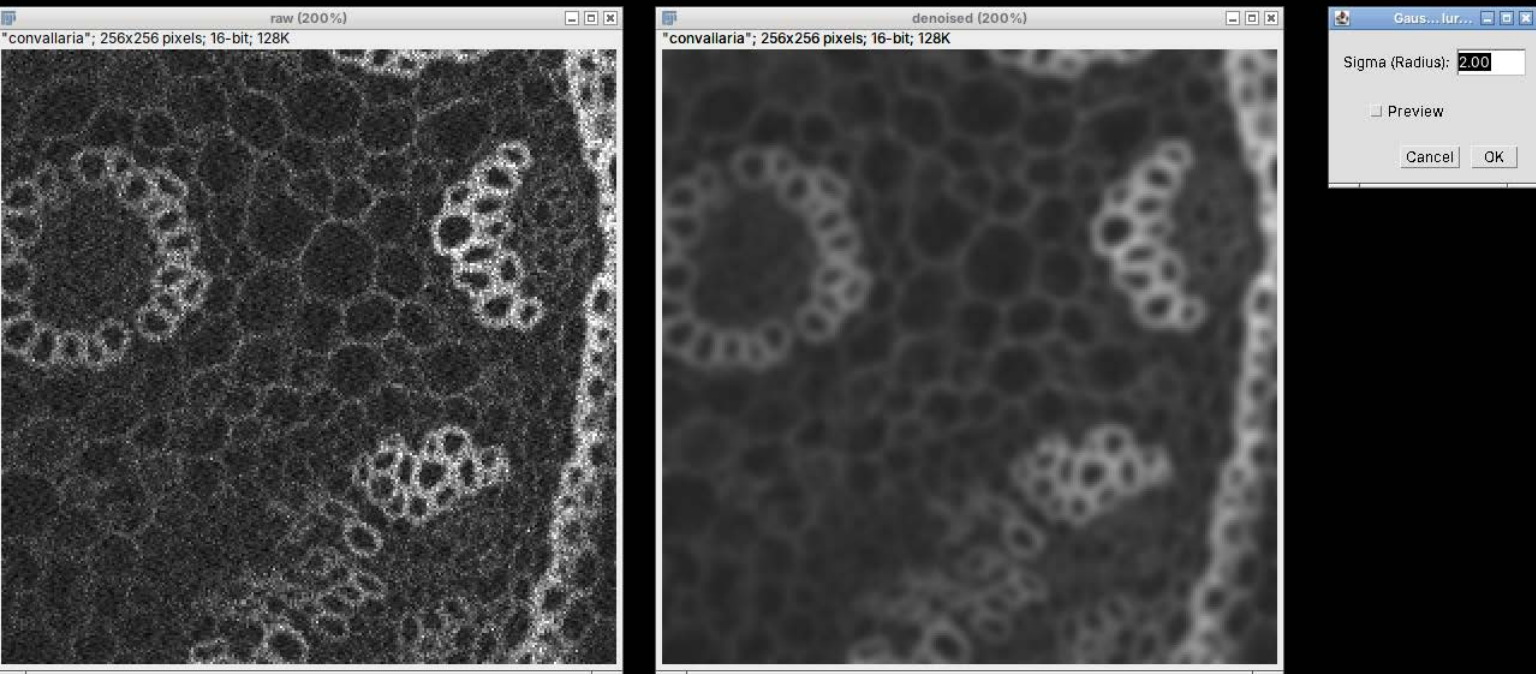
Total-variation (ROF) denoising

Fiji: Process > Noise > ROF Denoise (theta = 2)



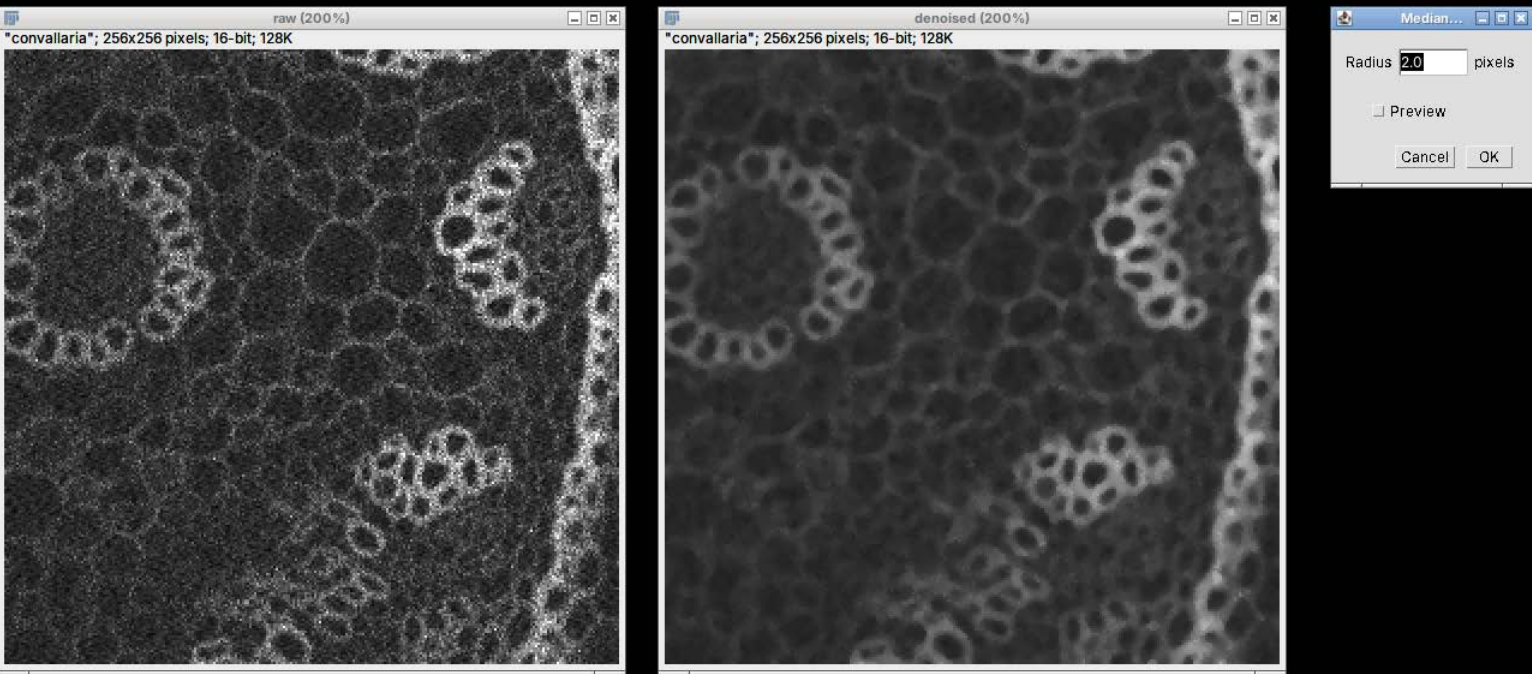
Gaussian blur

Fiji: Process > Filters > Gaussian Blur... (sigma = 2 px)



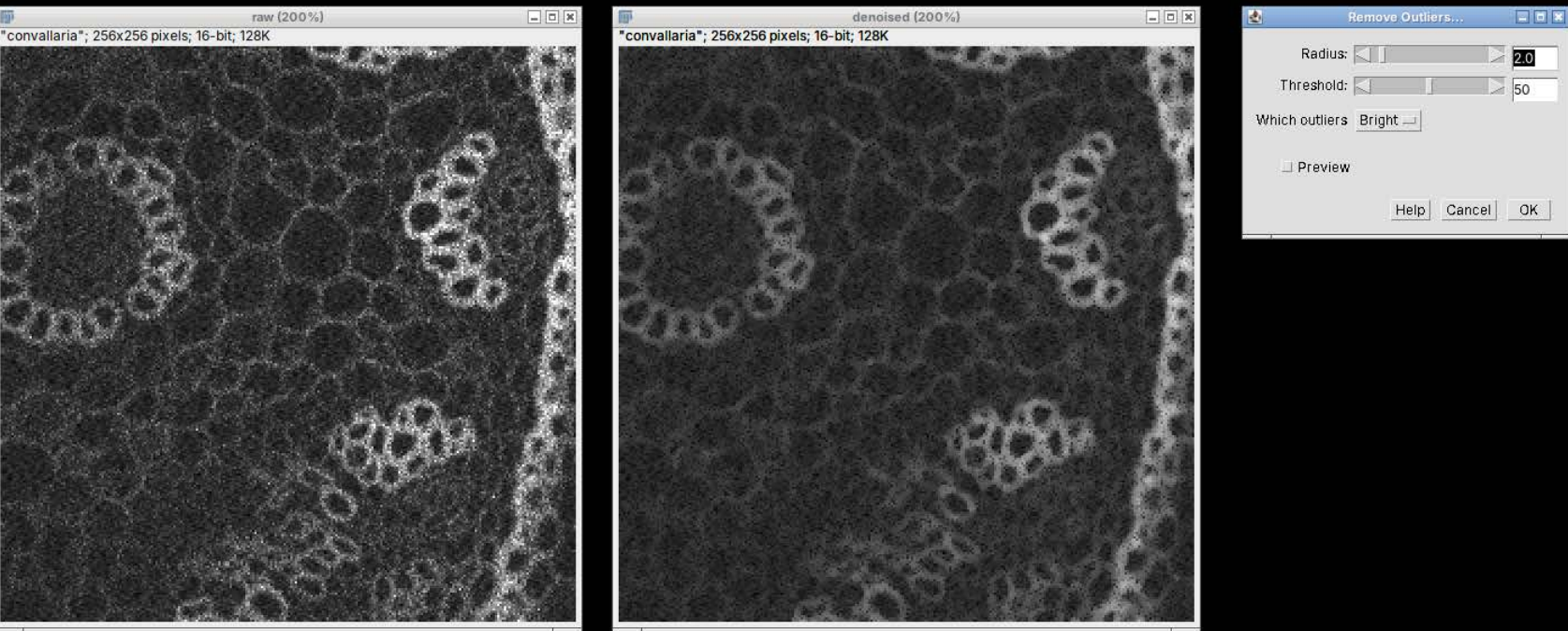
Median filter

Fiji: Process > Filters > Median... (radius = 2 px)



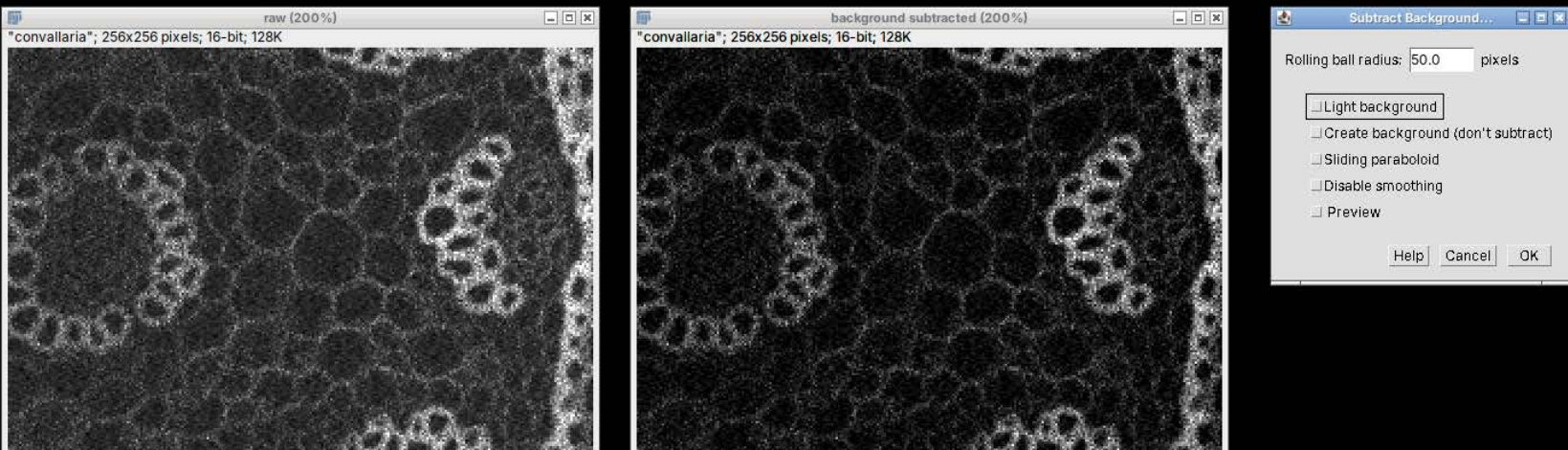
Remove outliers (hot pixels)

Fiji: Process > Noise > Remove Outliers... (radius 2, threshold 50, bright)



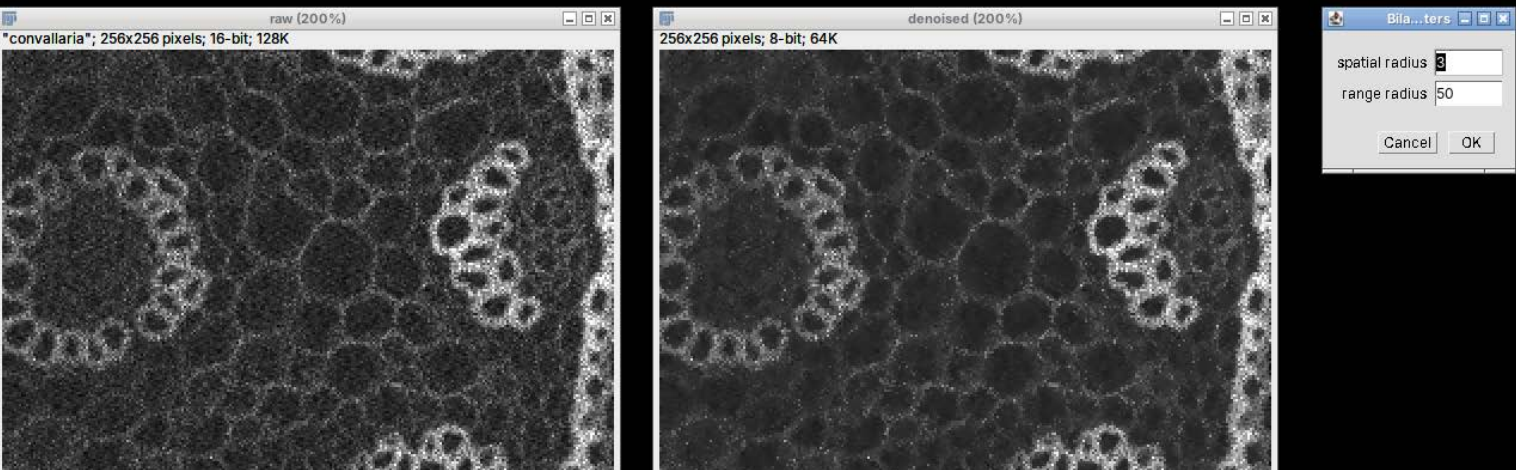
Rolling ball: removes background, NOT noise

Fiji: Process > Subtract Background... (rolling ball radius = 50 px)



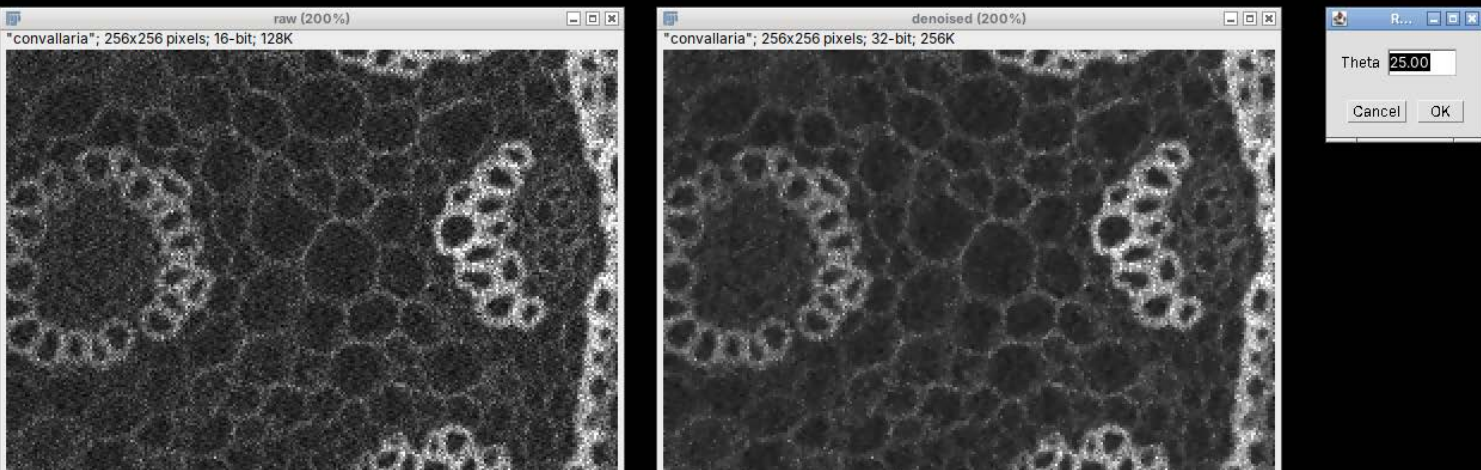
Bilateral filter (edge-preserving)

Fiji: Plugins > Process > Bilateral Filter (spatial 3, range 50; 8-bit only)



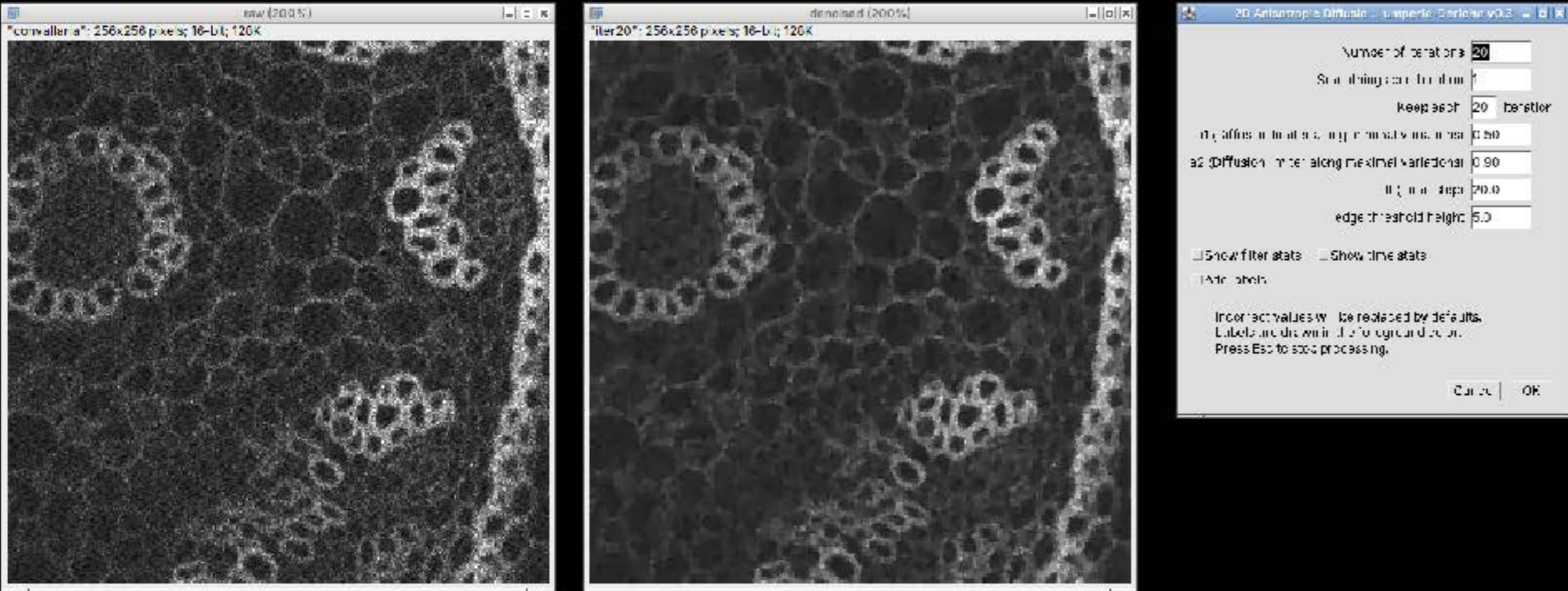
Total-variation (ROF) denoising

Fiji: Process > Noise > ROF Denoise (theta = 25; 32-bit only)



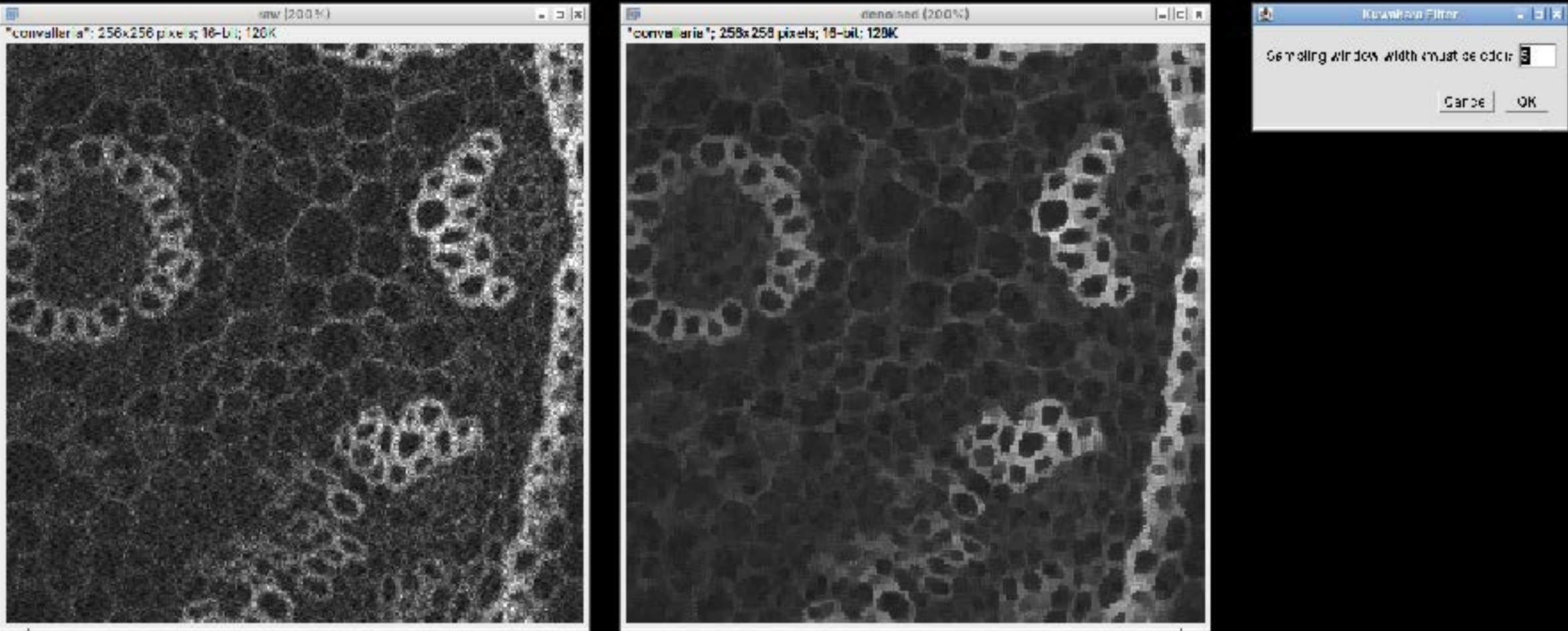
Anisotropic diffusion (edge-preserving)

Fiji: Plugins > Process > Anisotropic Diffusion 2D (default settings)



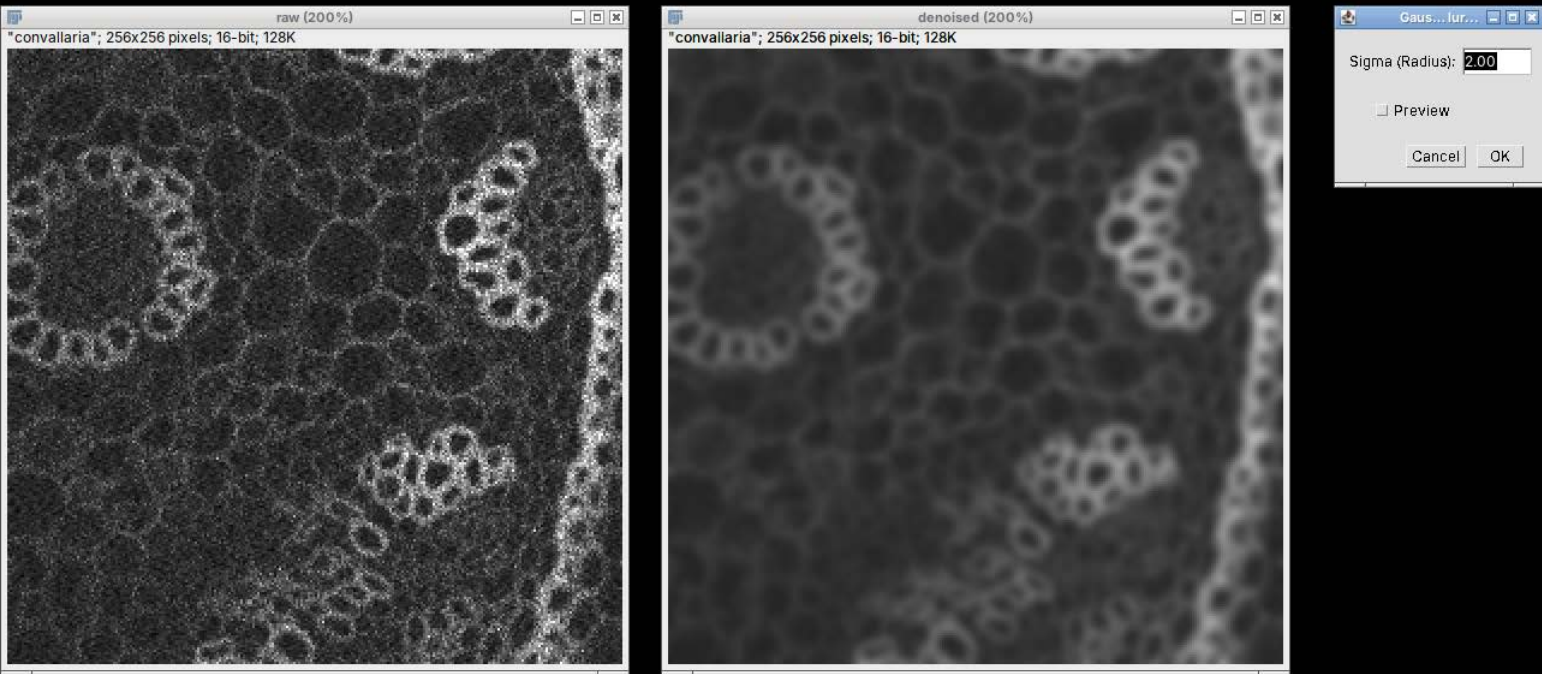
Kuwahara filter (edge-preserving)

Fiji: Process > Filters > Kuwahara Filter (window = 5)



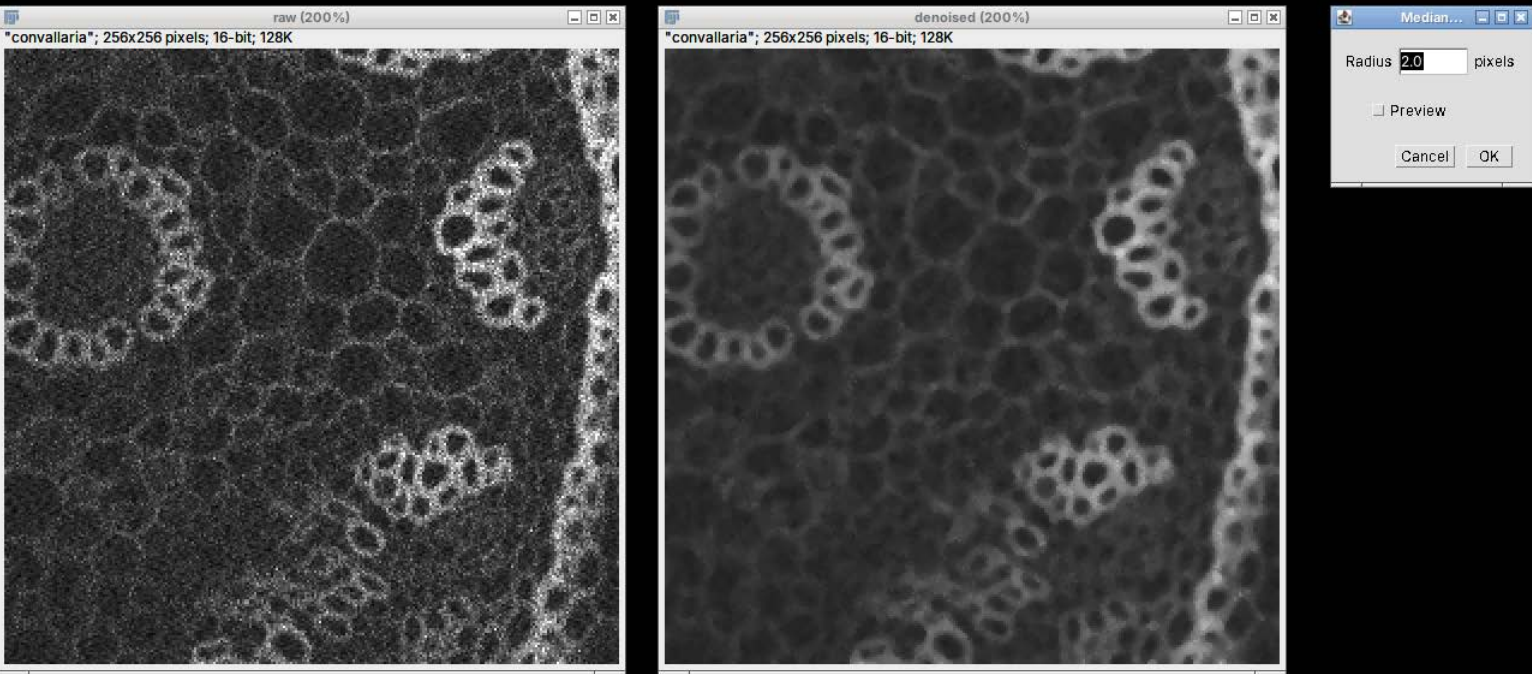
Gaussian blur

Fiji: Process > Filters > Gaussian Blur... (sigma = 2 px)



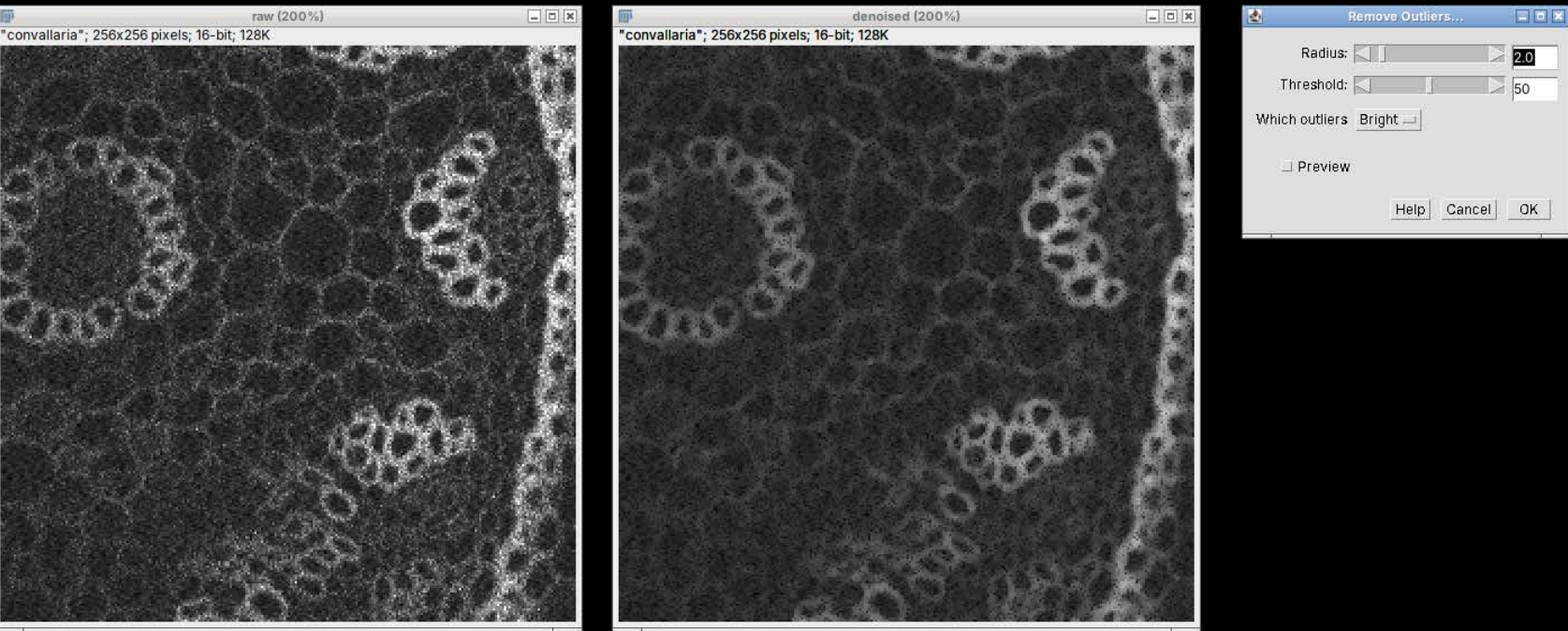
Median filter

Fiji: Process > Filters > Median... (radius = 2 px)



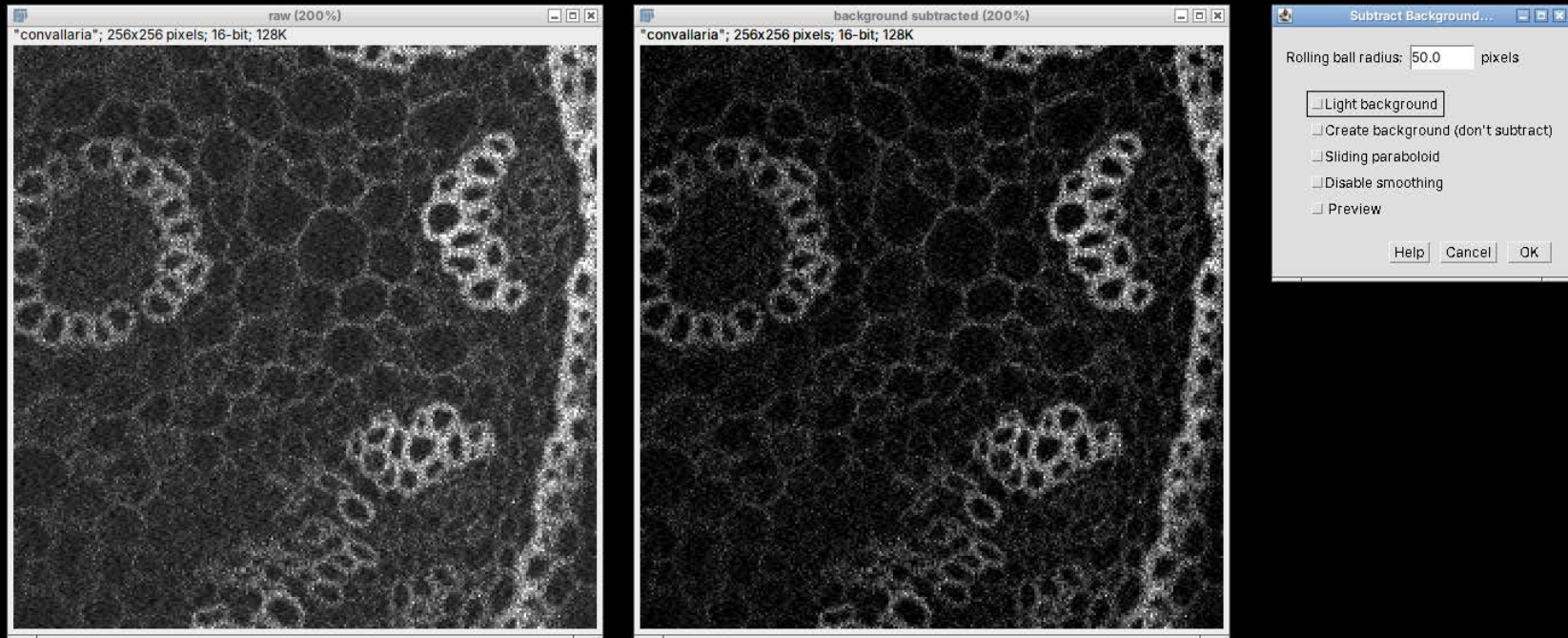
Remove outliers (hot pixels)

Fiji: Process > Noise > Remove Outliers... (radius 2, threshold 50, bright)



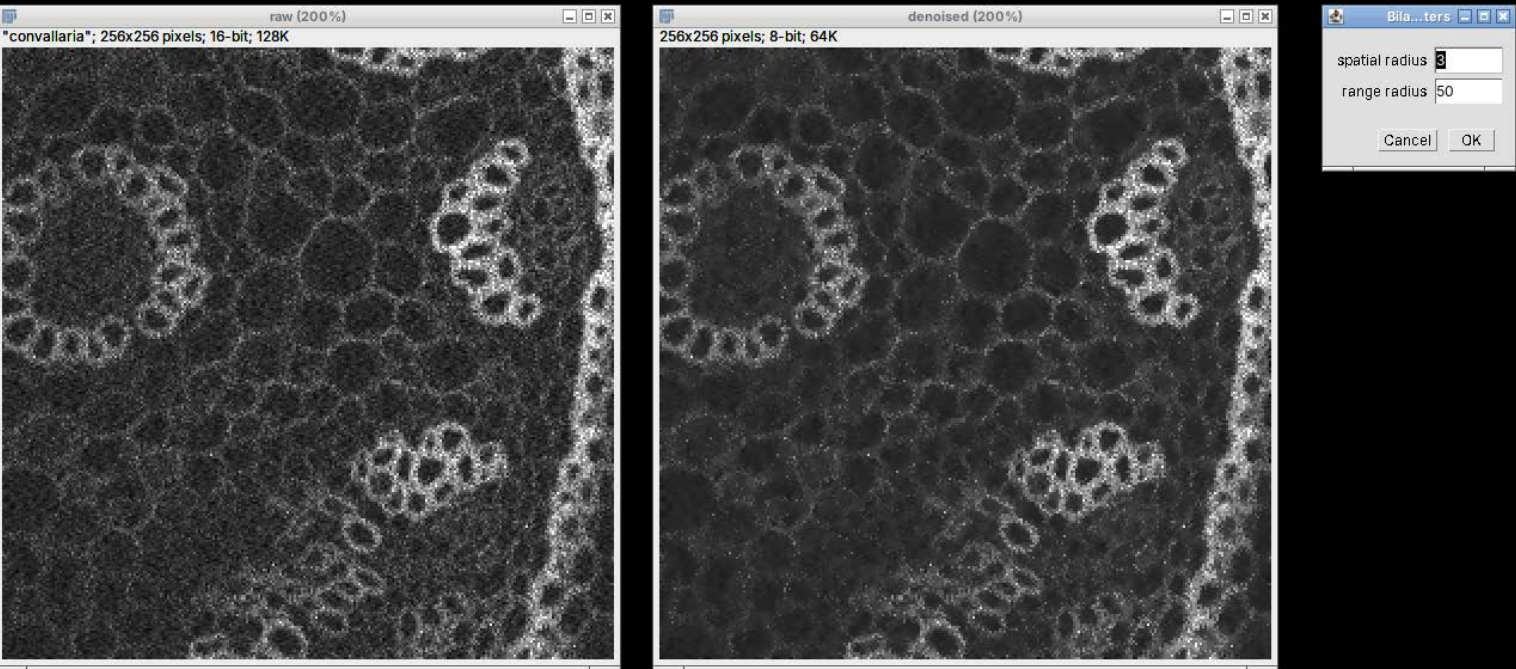
Rolling ball: removes background, NOT noise

Fiji: Process > Subtract Background... (rolling ball radius = 50 px)



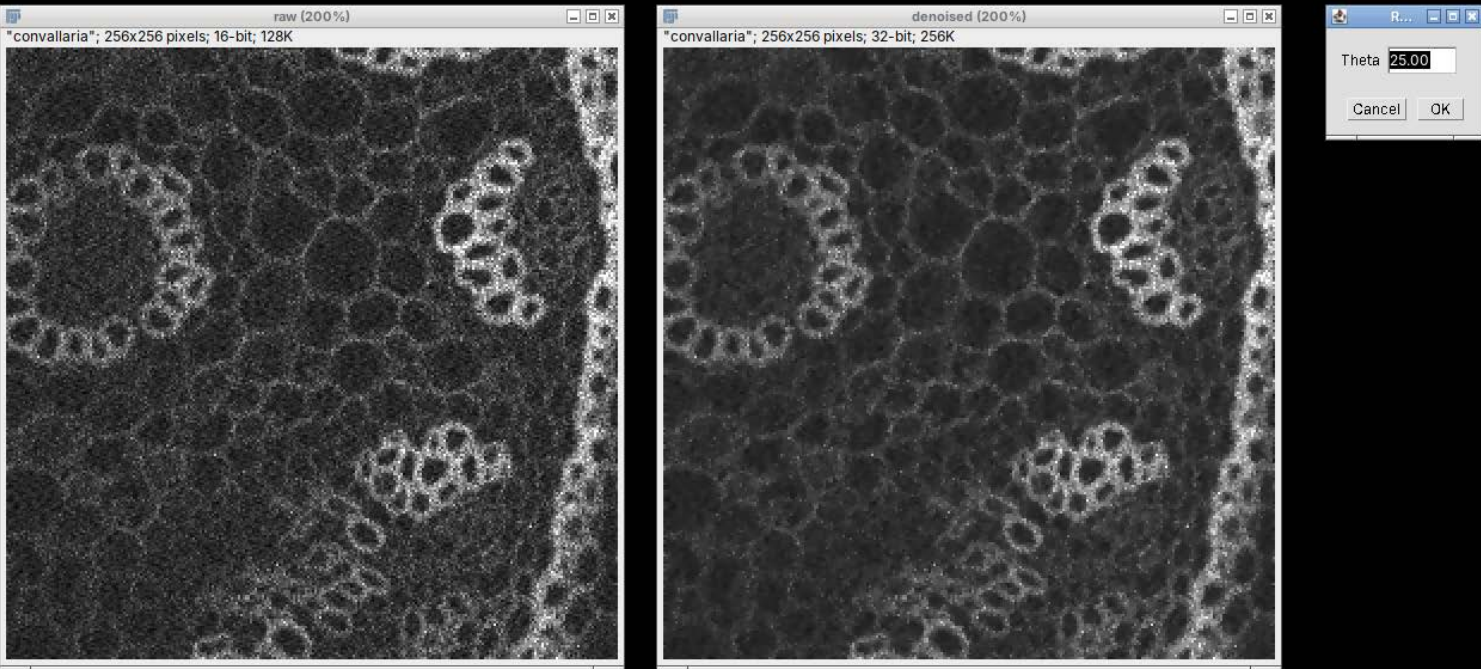
Bilateral filter (edge-preserving)

Fiji: Plugins > Process > Bilateral Filter (spatial 3, range 50; 8-bit only)



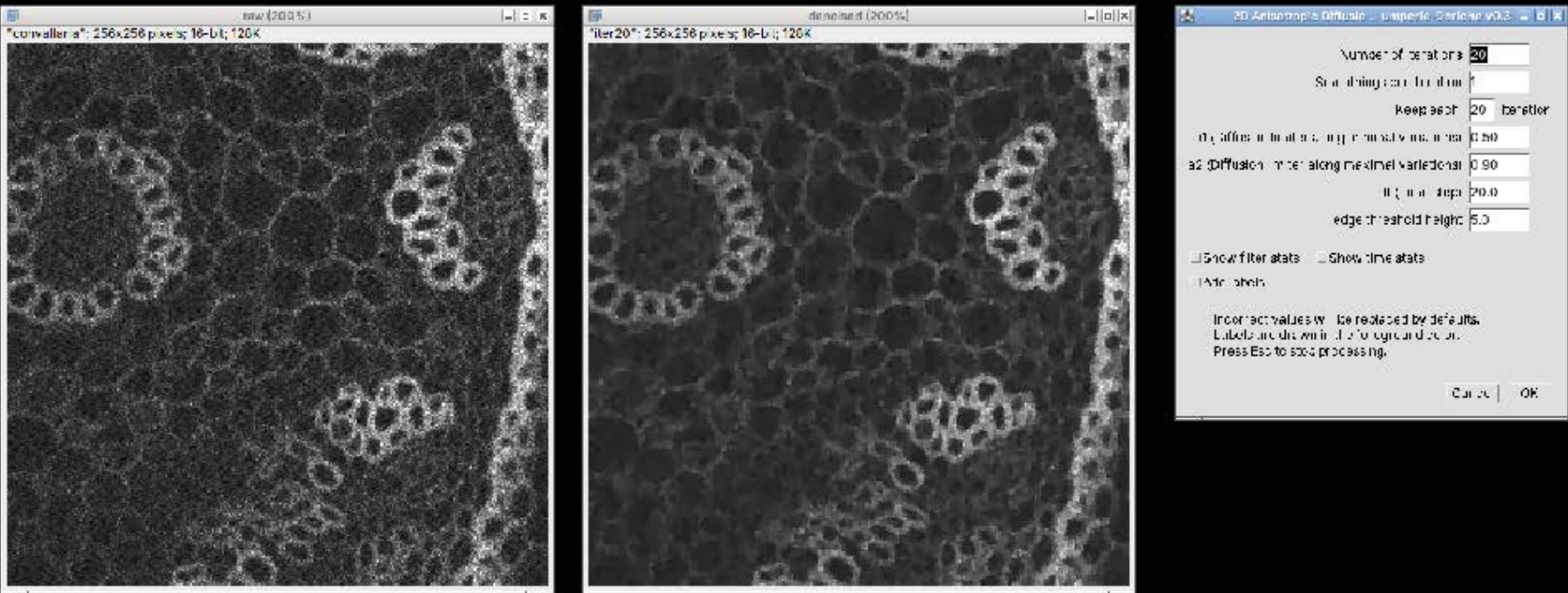
Total-variation (ROF) denoising

Fiji: Process > Noise > ROF Denoise (theta = 25; 32-bit only)



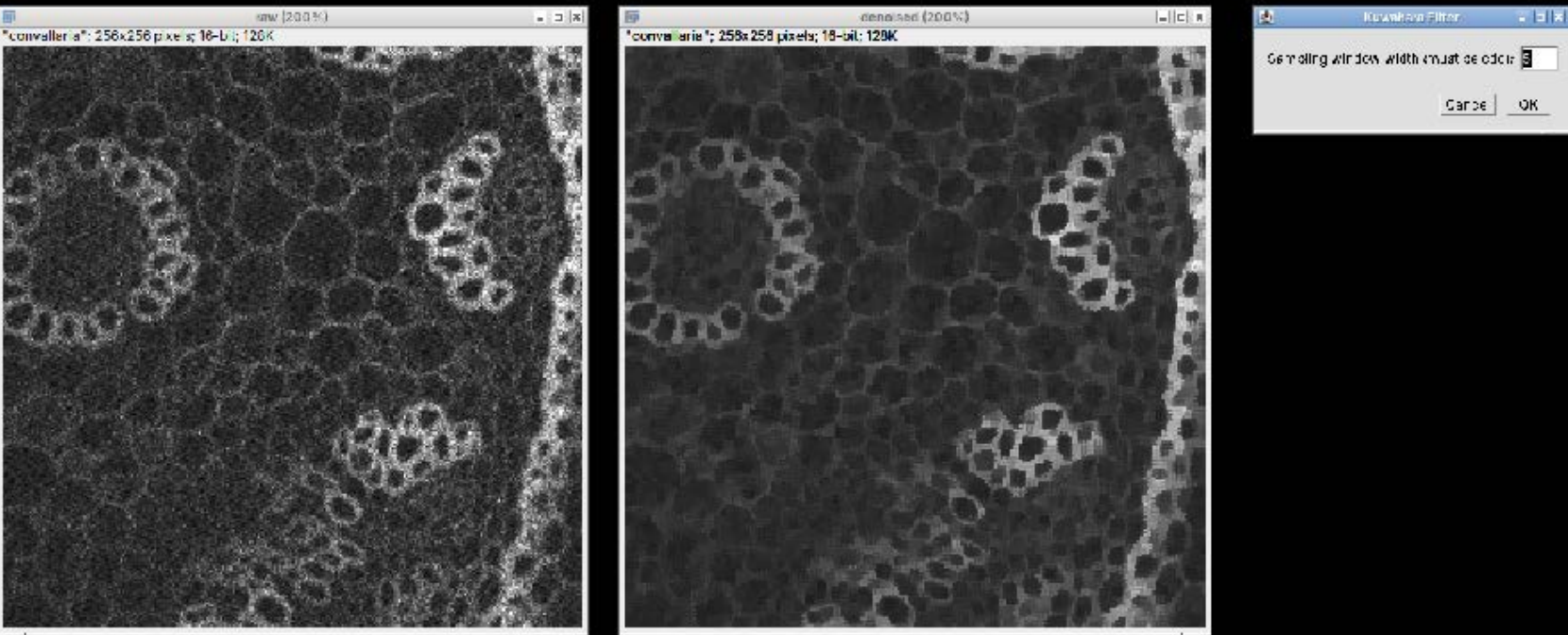
Anisotropic diffusion (edge-preserving)

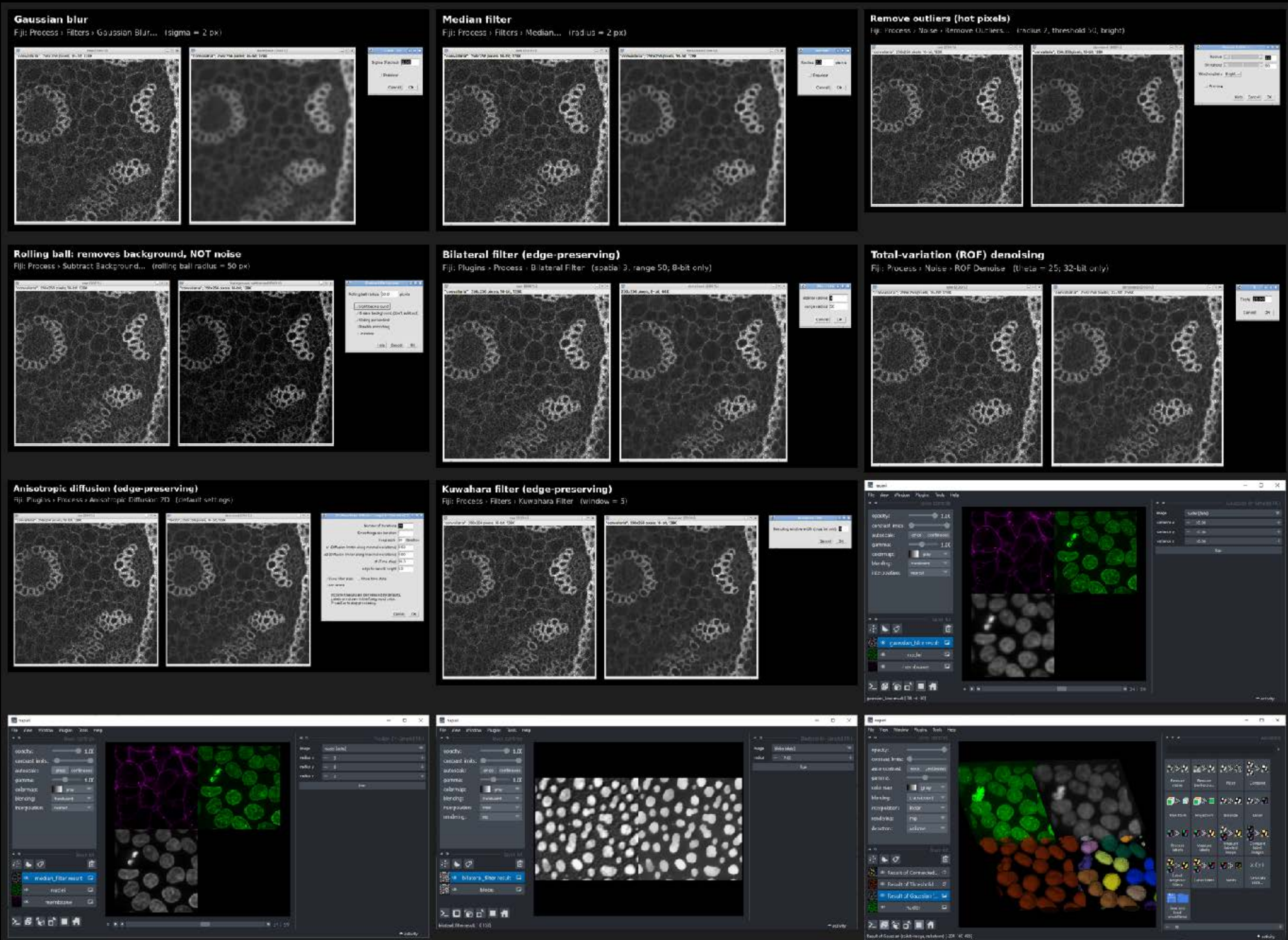
Fiji: Plugins > Process > Anisotropic Diffusion 2D (default settings)

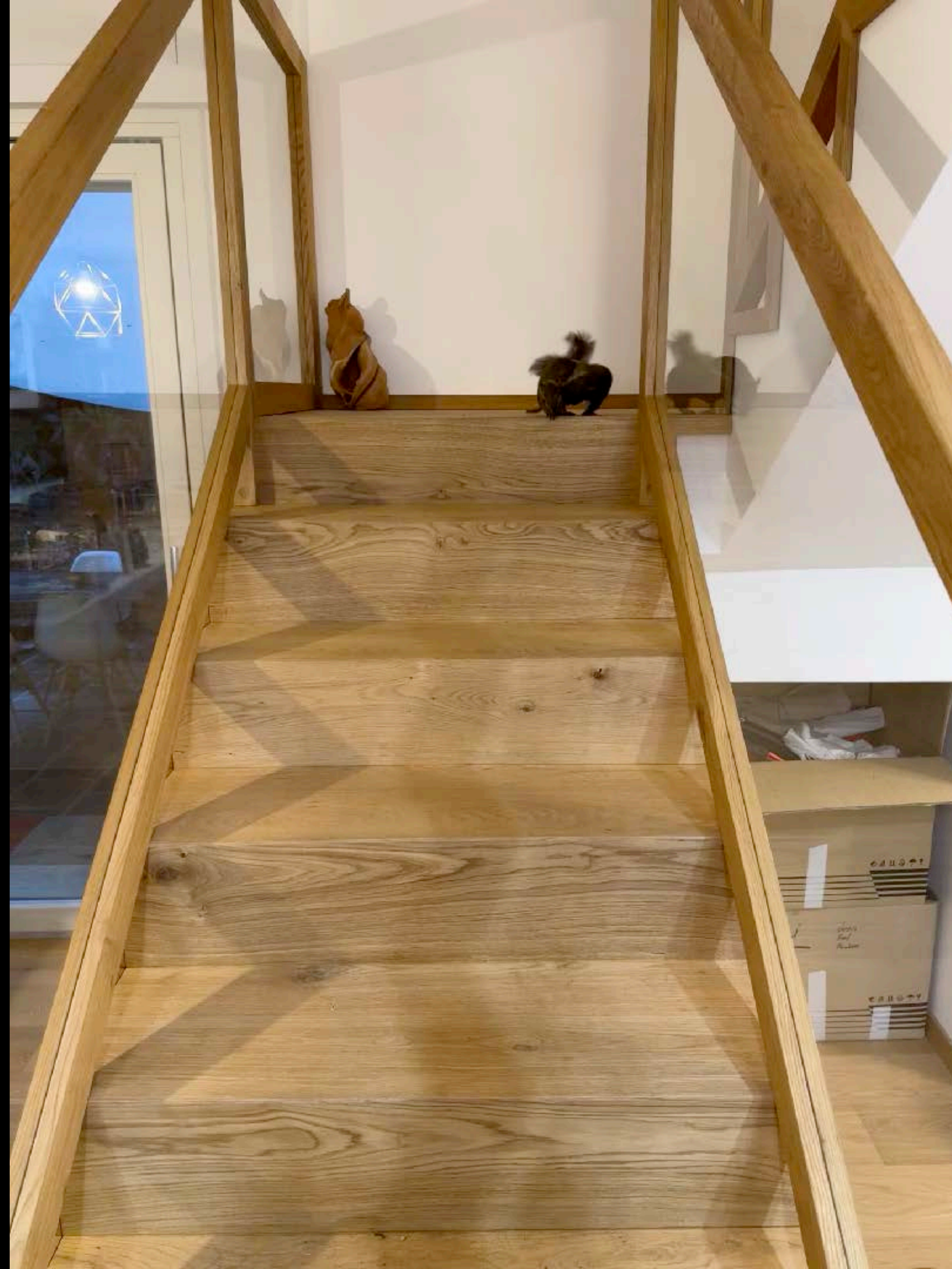


Kuwahara filter (edge-preserving)

Fiji: Process > Filters > Kuwahara Filter (window = 5)







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 fjug and claude	README: keep the program only up to Thursday's recap	941add · 2 hours ago	8 Commits
data	Add denoising exercise (BM3D and Noise2Void)	20 hours ago	
denoising	README: let students choose between Colab and local track	3 hours ago	
.gitignore	Ignore instructor-only material and Claude app outputs	19 hours ago	
README.md	README: keep the program only up to Thursday's recap	2 hours ago	
check_setup.py	Add setup check script for students	19 hours ago	

README

<https://github.com/juglab/CiliaAI-minicourse>

CiliaAI mini-course: hands-on exercises

Material for the hands-on part of the Cilia-AI / CiliaConnect mini-course on image analysis and AI.

Before the course, please choose how you want to work (in the browser or on your own laptop) and get ready as described in [Before the course: get ready.](#)

Program

About

Hands-on exercises for the Cilia-AI / CiliaConnect mini-course on image analysis and AI (denoising with BM3D and Noise2Void)

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About

Hands-on exercises for the Cilia-AI / CiliaConnect mini-course on image analysis and AI (denoising with BM3D and Noise2Void)

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Denoising, Part 1: solve it with AI

In this part you get **no recipe**. You get an image, a goal and a short list of what to deliver. How you get there is up to you, and you are explicitly invited to use AI to help you:

- chat assistants such as ChatGPT, Claude or Gemini (ask, copy the code, run it yourself), or
- coding agents such as Claude Code, Codex or Cursor (they write and run code for you).

Work in pairs. Tutors will help you with installation problems, but not with choosing or tuning a method.

Please do not open the other files in the `denoising/` folder yet: they belong to Part 2. If you use a coding agent, start it in a new, empty folder of its own, not inside the course repository.

The image

`data/convollaria.tif` : a fluorescence microscopy image of a section through a *Convallaria* (lily of the valley) rhizome.

- 2D, 16-bit, 1024 × 1024 pixels
- intensities roughly 330 to 1850, most of the signal between 400 and 1200
- the image is noisy, and **no clean (noise-free) version of it exists**

Open it in Fiji first and zoom in. What do you see?

The goal

Produce a denoised version of this image.

A good result is not simply the smoothest-looking image. A good denoiser removes noise and keeps the real biological structure. A good denoiser must not remove real structure, must not invent structure that is not there, and must not shift intensities. Imagine that some day you want to measure membrane intensities and to make a figure for a paper.

What to deliver

Put everything in one folder, for example `denoising-<your names>/`.

https://github.com/juglab/CiliaAI-minicourse/blob/main/denoising/denoising_challenge.md



30 min... go! 🤗



5 MINUTES, ON PAPER (?), OR AT LEAST ON YOUR OWN...

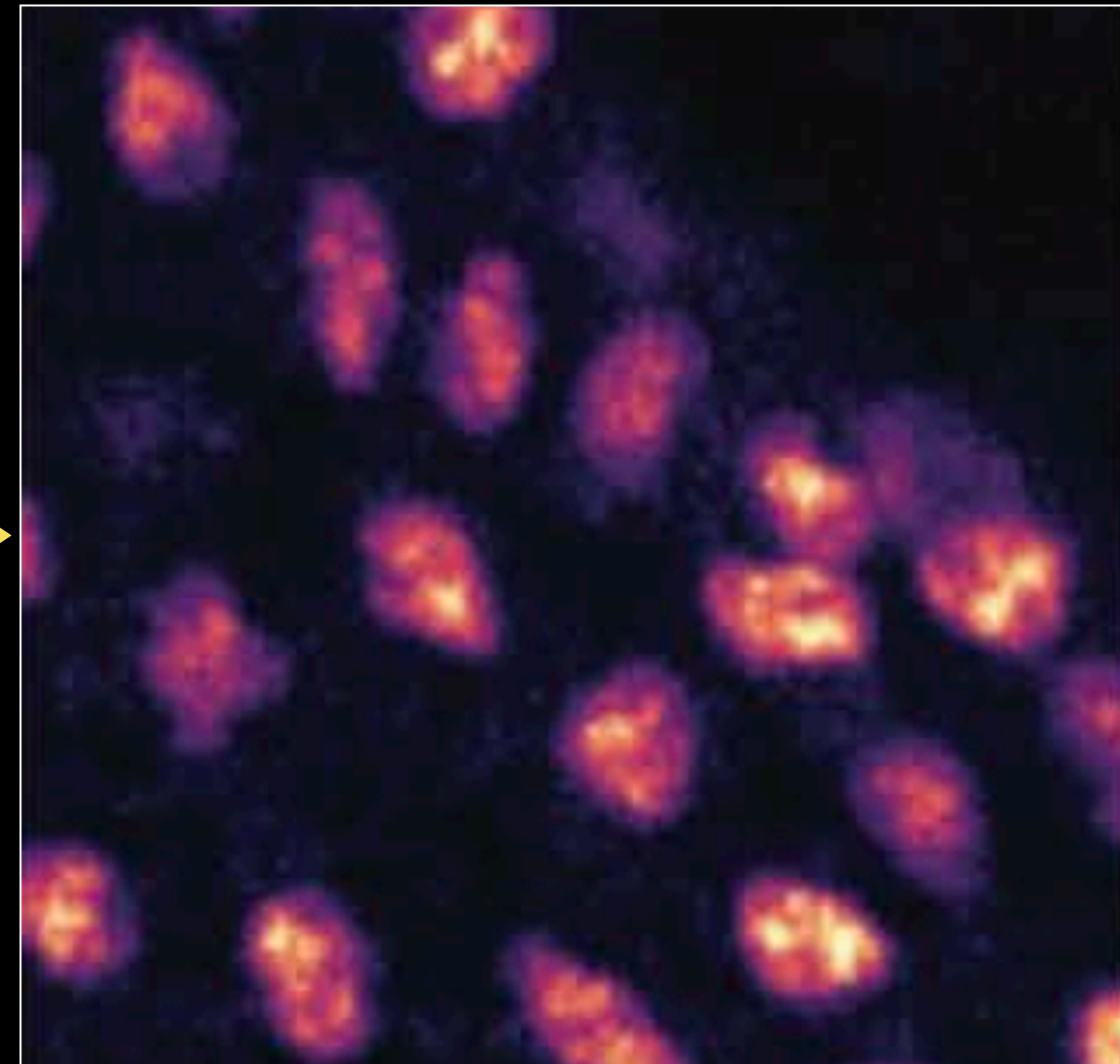
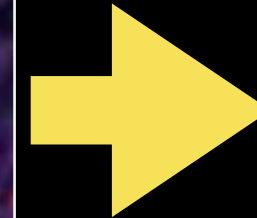
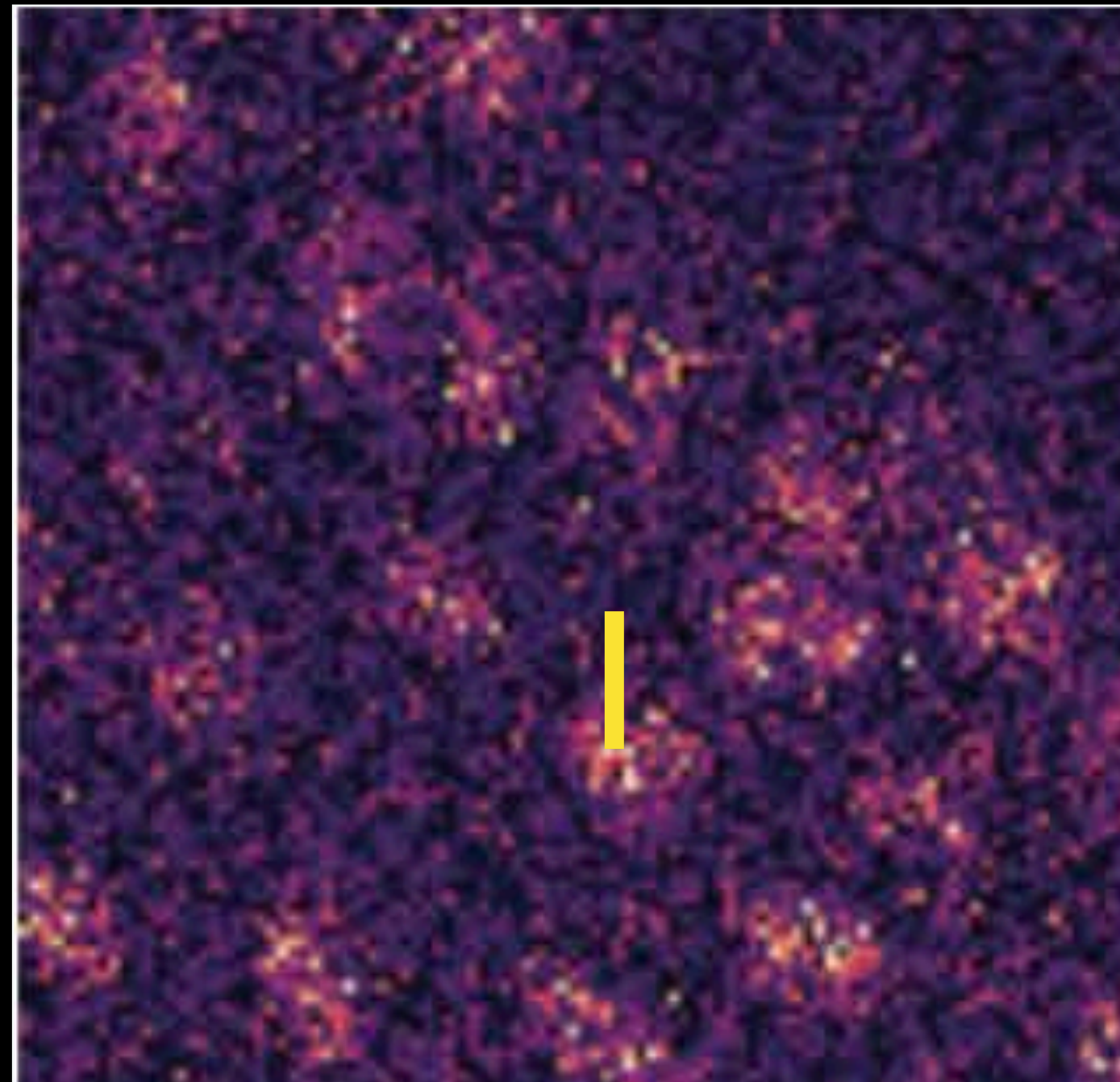
1. In one sentence: what does your method do?
2. What does it assume about the noise? Does that assumption hold for this image?
3. How do you know your result did not remove real structure, or invent structure that isn't there?
4. Could anyone rerun your solution tomorrow and get the same image? How?
5. From 1 to 5: how much of your solution could you explain to a reviewer?

Keep your sheet. We'll come back to it at the end of the day, and tomorrow morning.

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CARE TRAINING IN A NUTSHELL

>> Weigert et al. 2018
>> data, Rink Lab, MPI-CBG



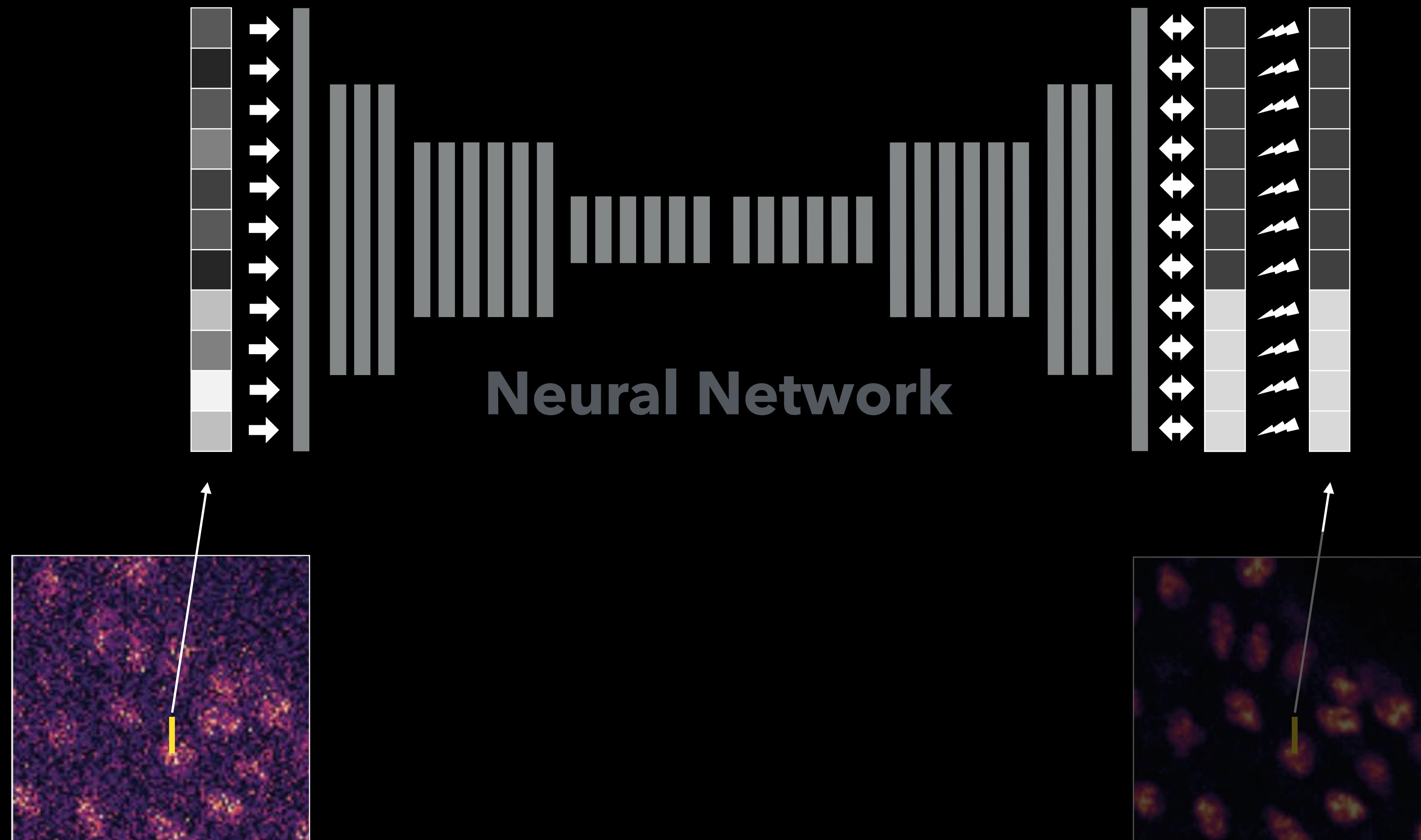
Martin
Weigert



Uwe
Schmidt

CARE TRAINING IN A NUTSHELL

>> Weigert et al. 2018
>> data, Rink Lab, MPI-CBG



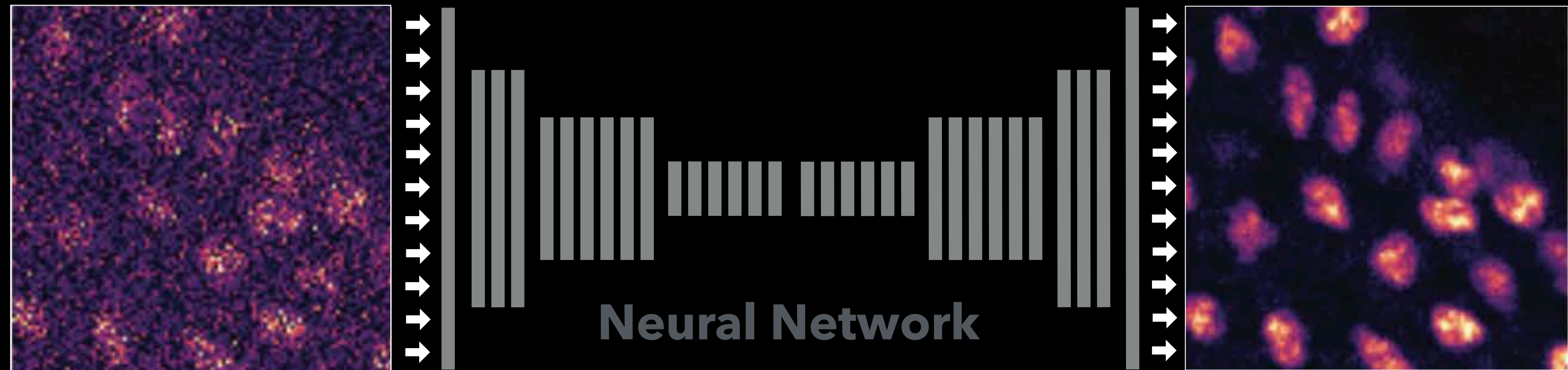
Martin Weigert



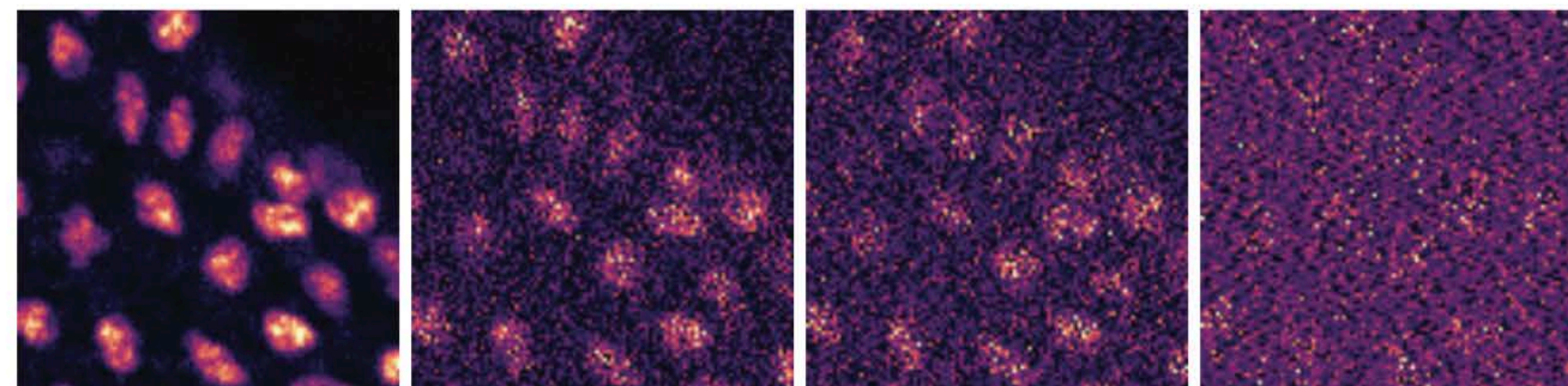
Uwe Schmidt

CARE TRAINING IN A NUTSHELL

>> Weigert et al. 2018
>> data, Rink Lab, MPI-CBG



Imaging conditions

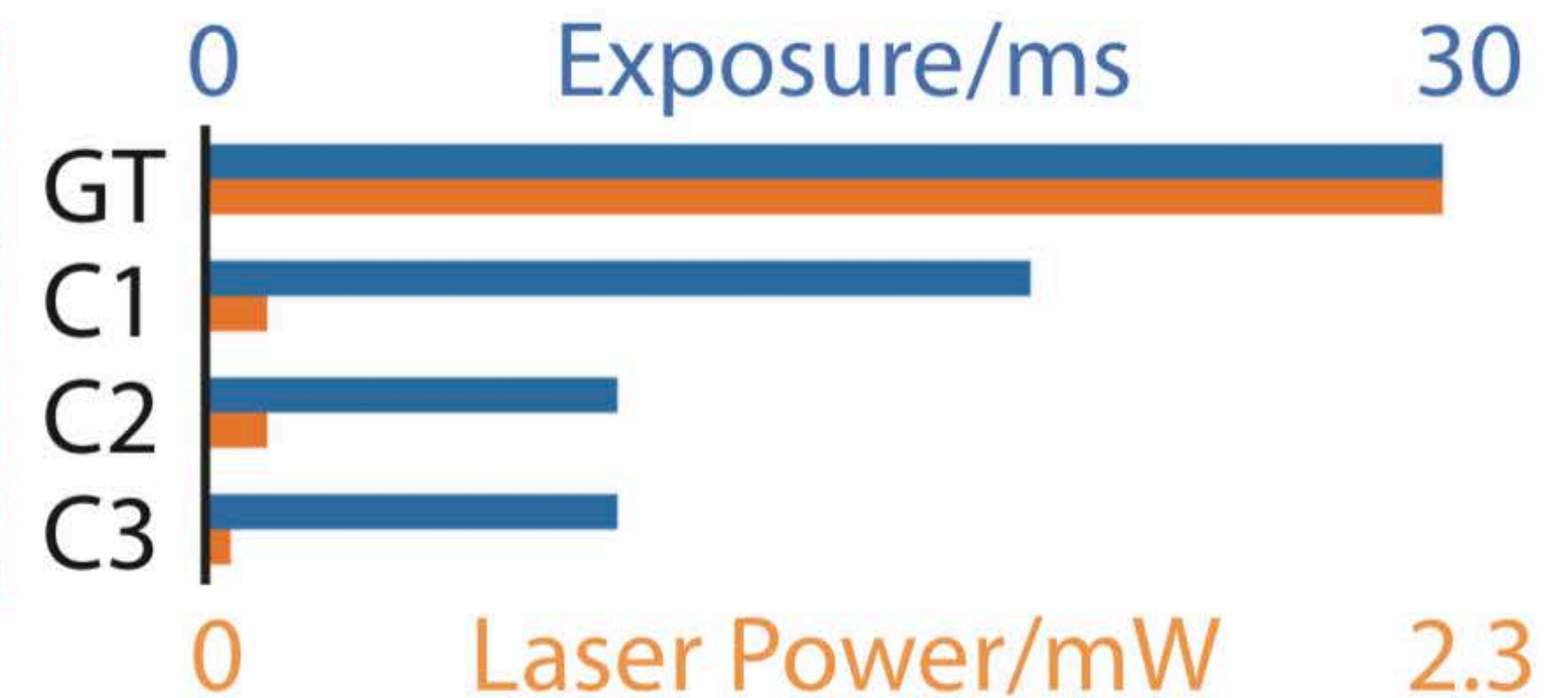


GT/high

C1/medium

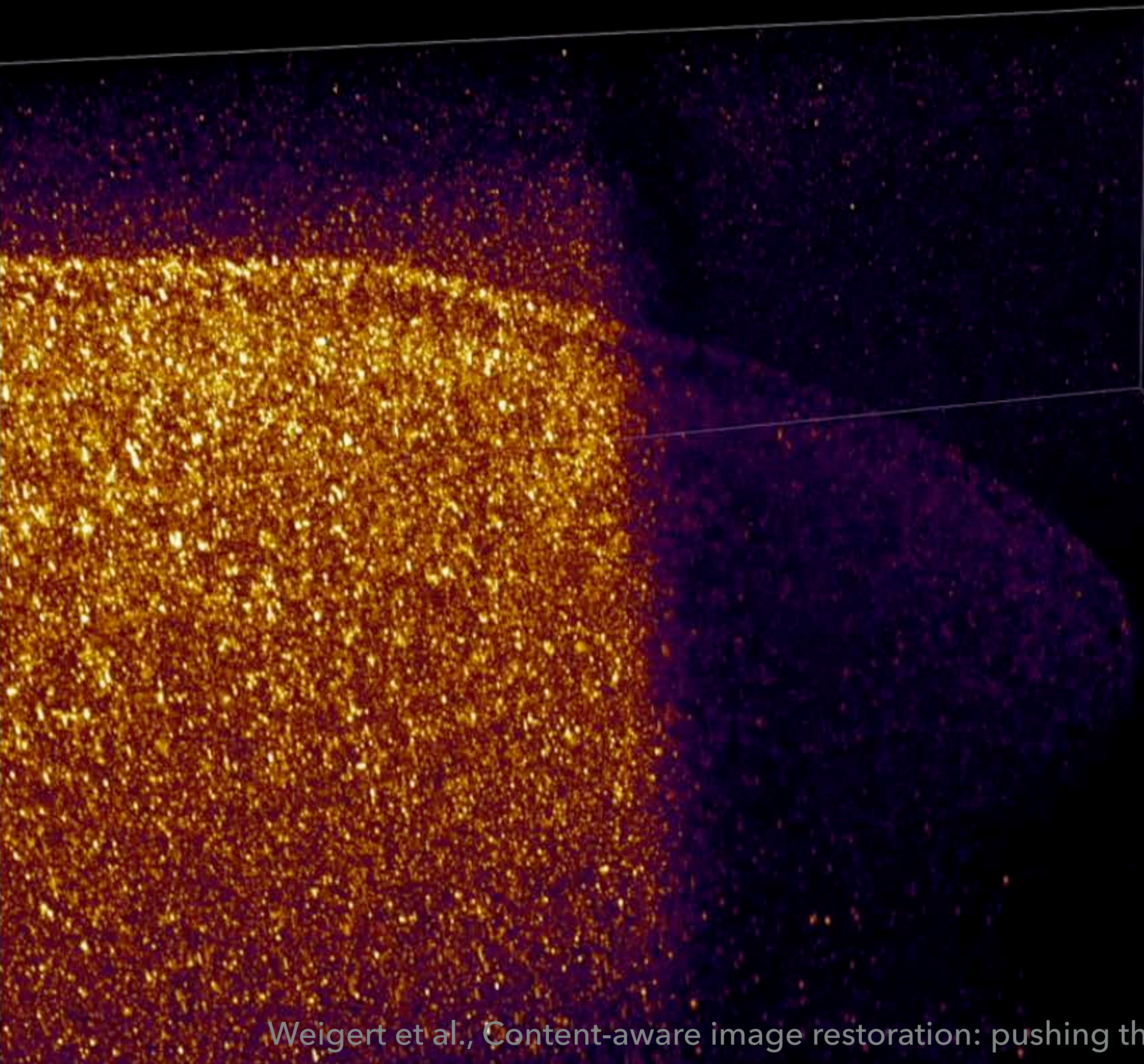
C2/low

C3/very low

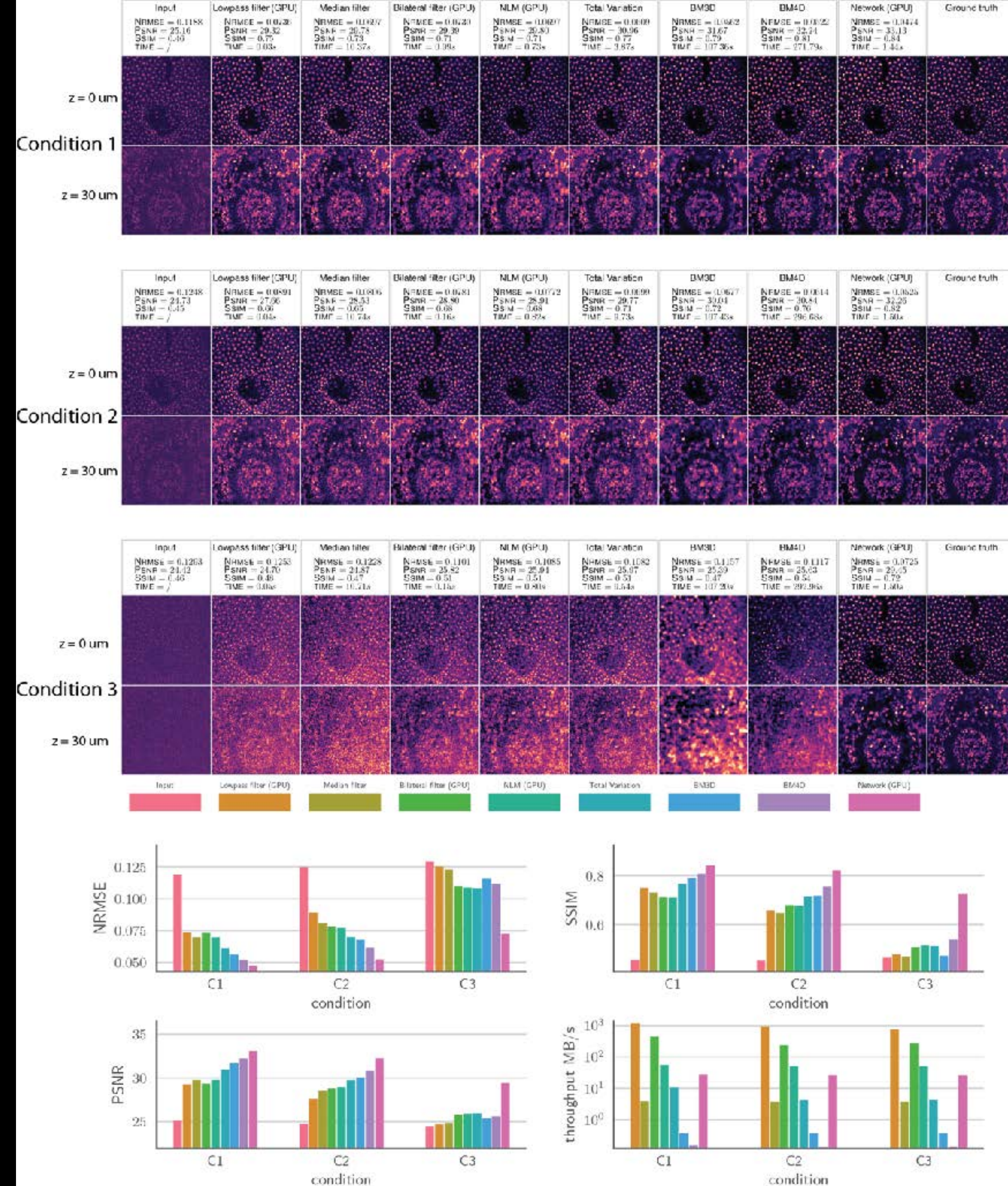


CARE TRAINING IN A NUTSHELL

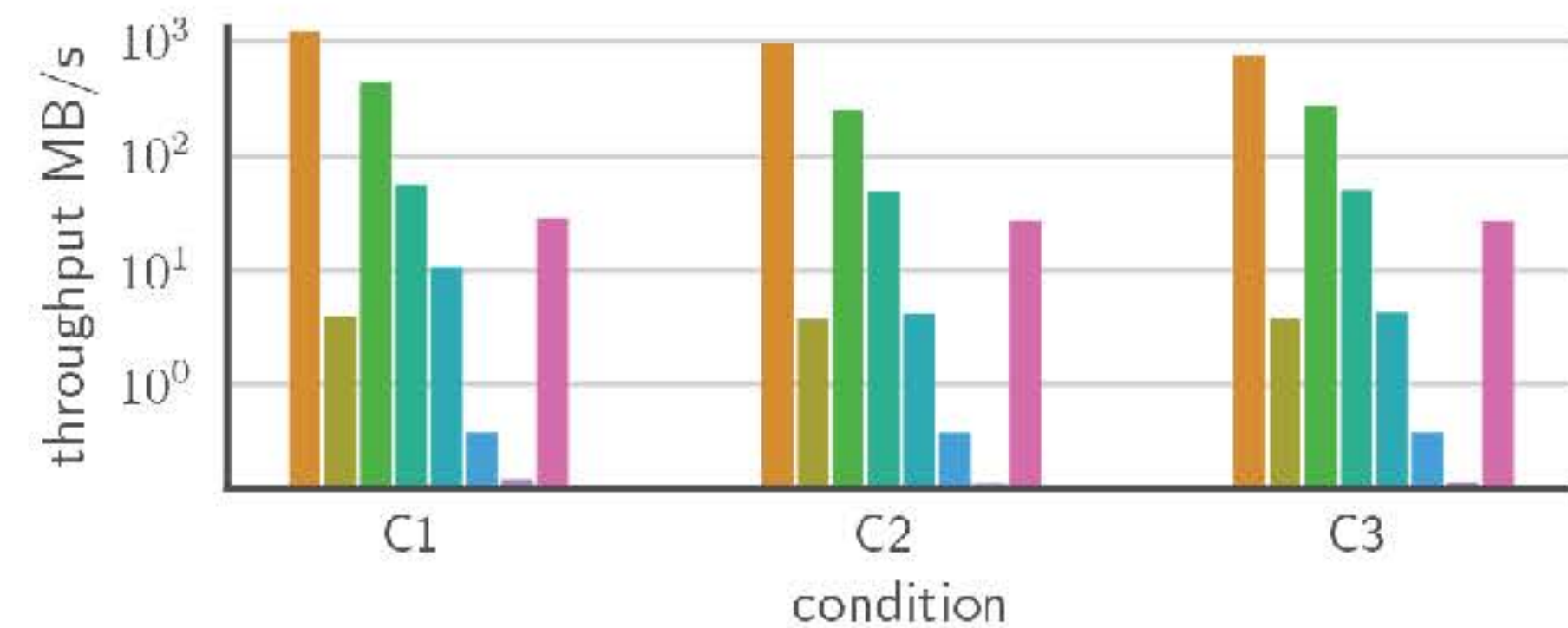
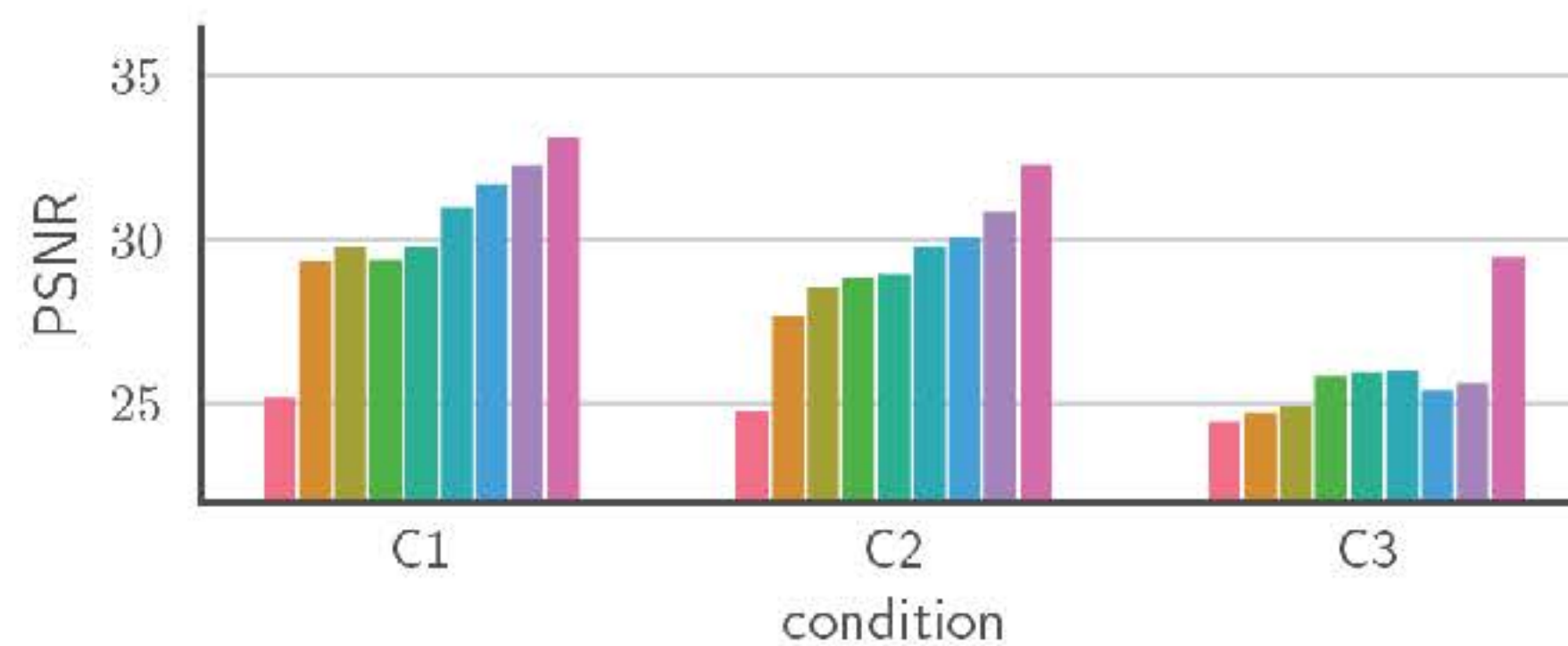
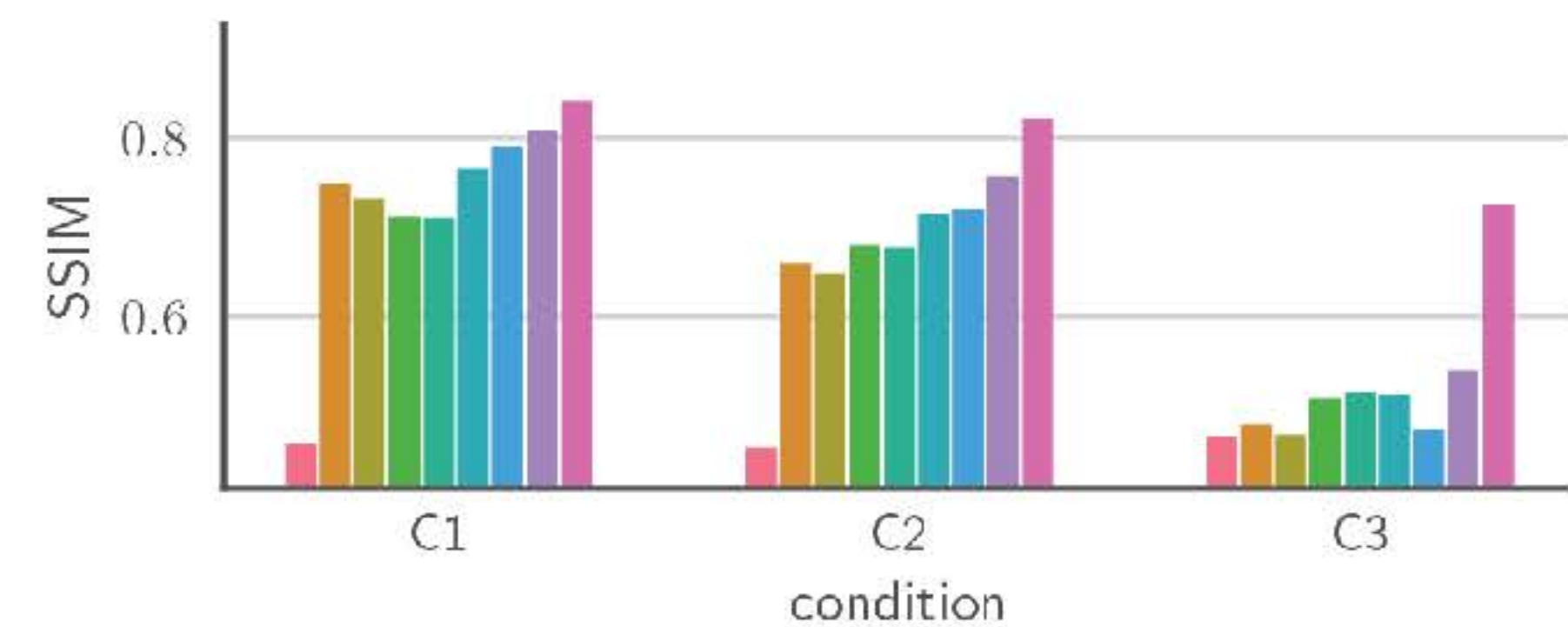
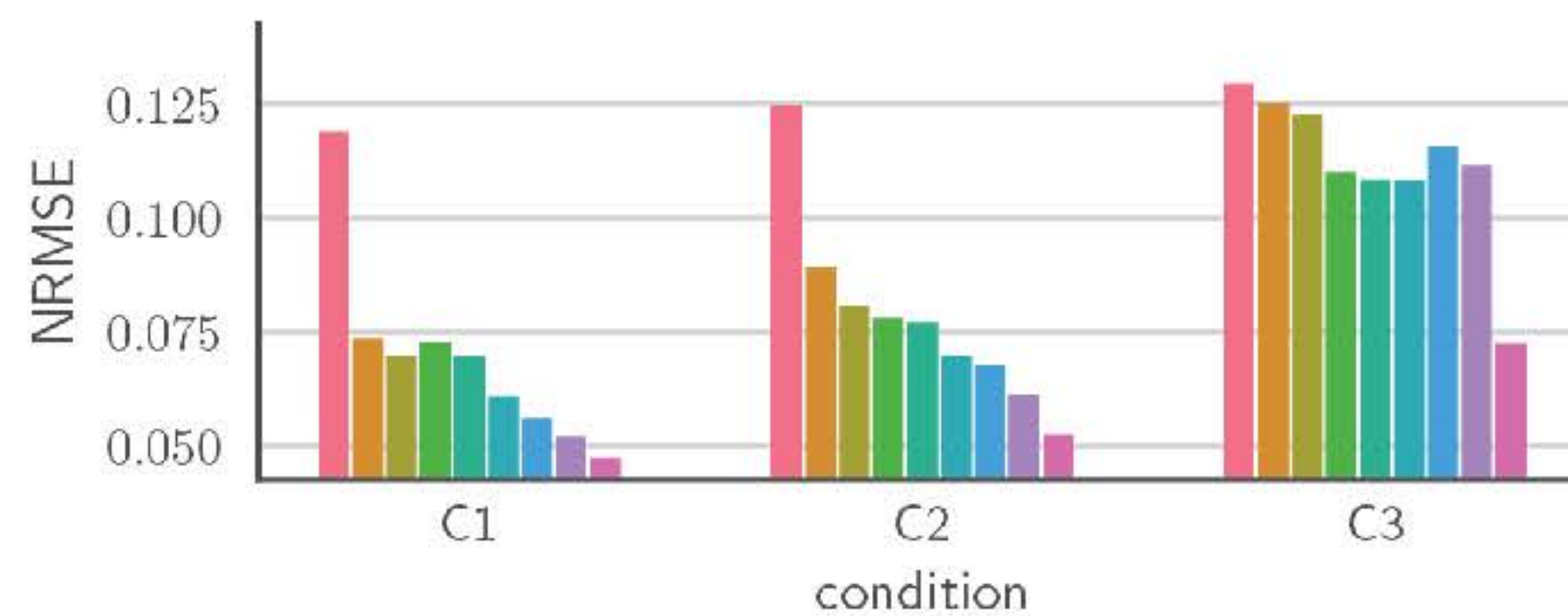
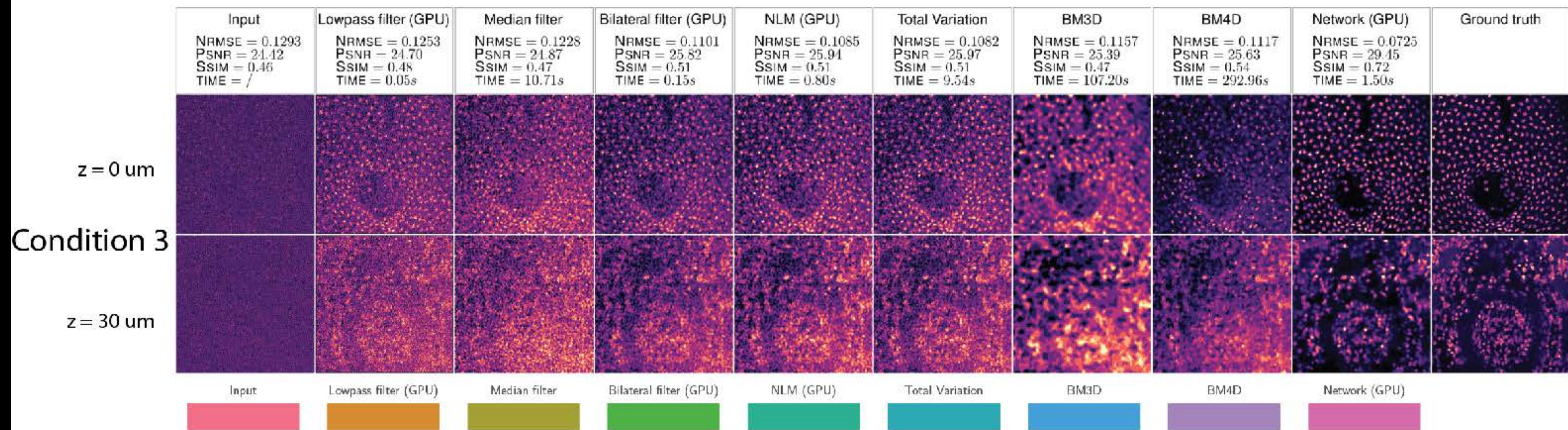
>>data, **Rink Lab**, MPI-CBG



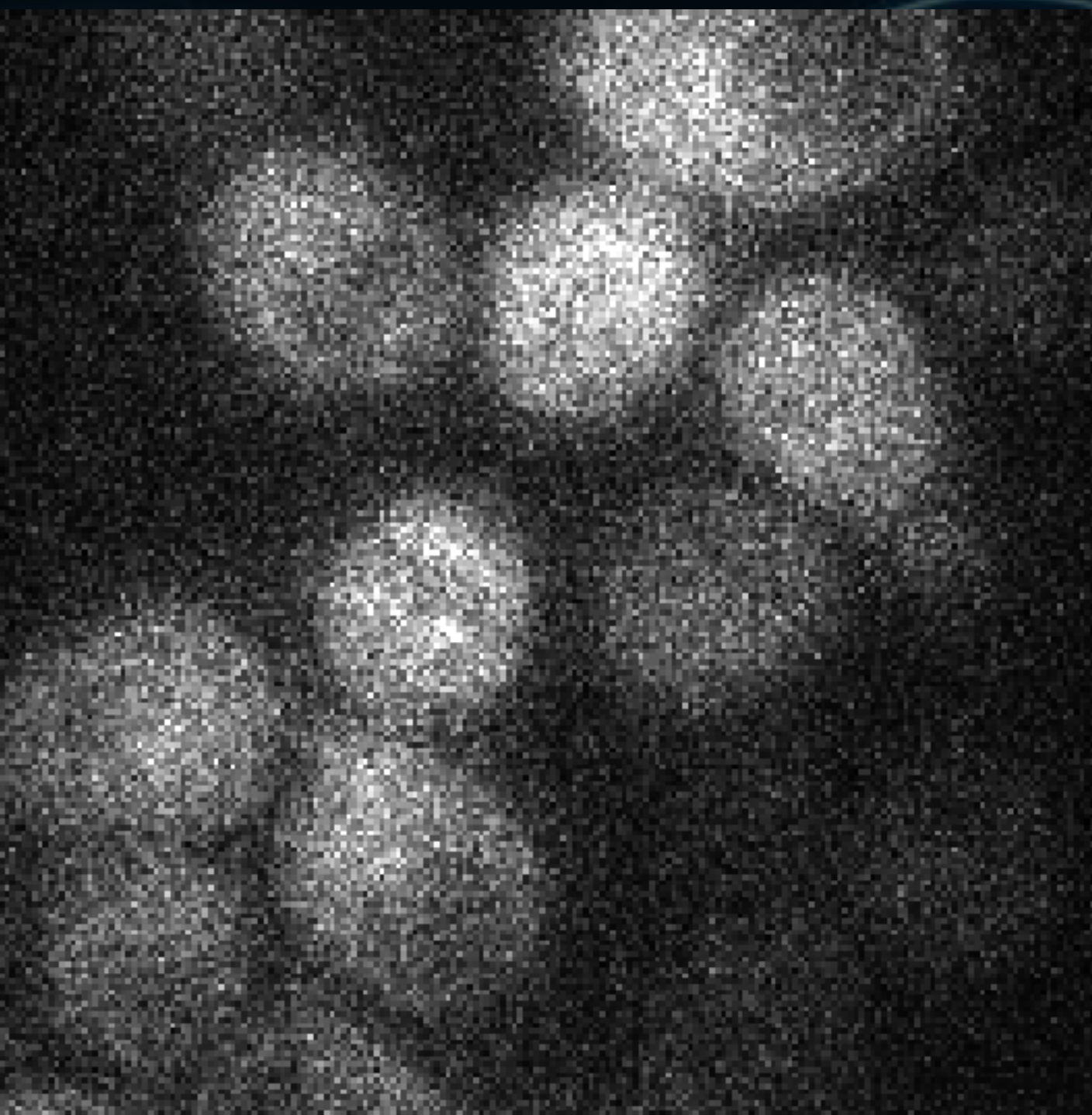
Weigert et al., Content-aware image restoration: pushing the limits of fluorescence microscopy. Nat Methods 15, 1090-1097 (2018).



Weigert et al., Content-aware image restoration: pushing the limits of fluorescence microscopy. Nat Methods 15, 1090-1097 (2018).

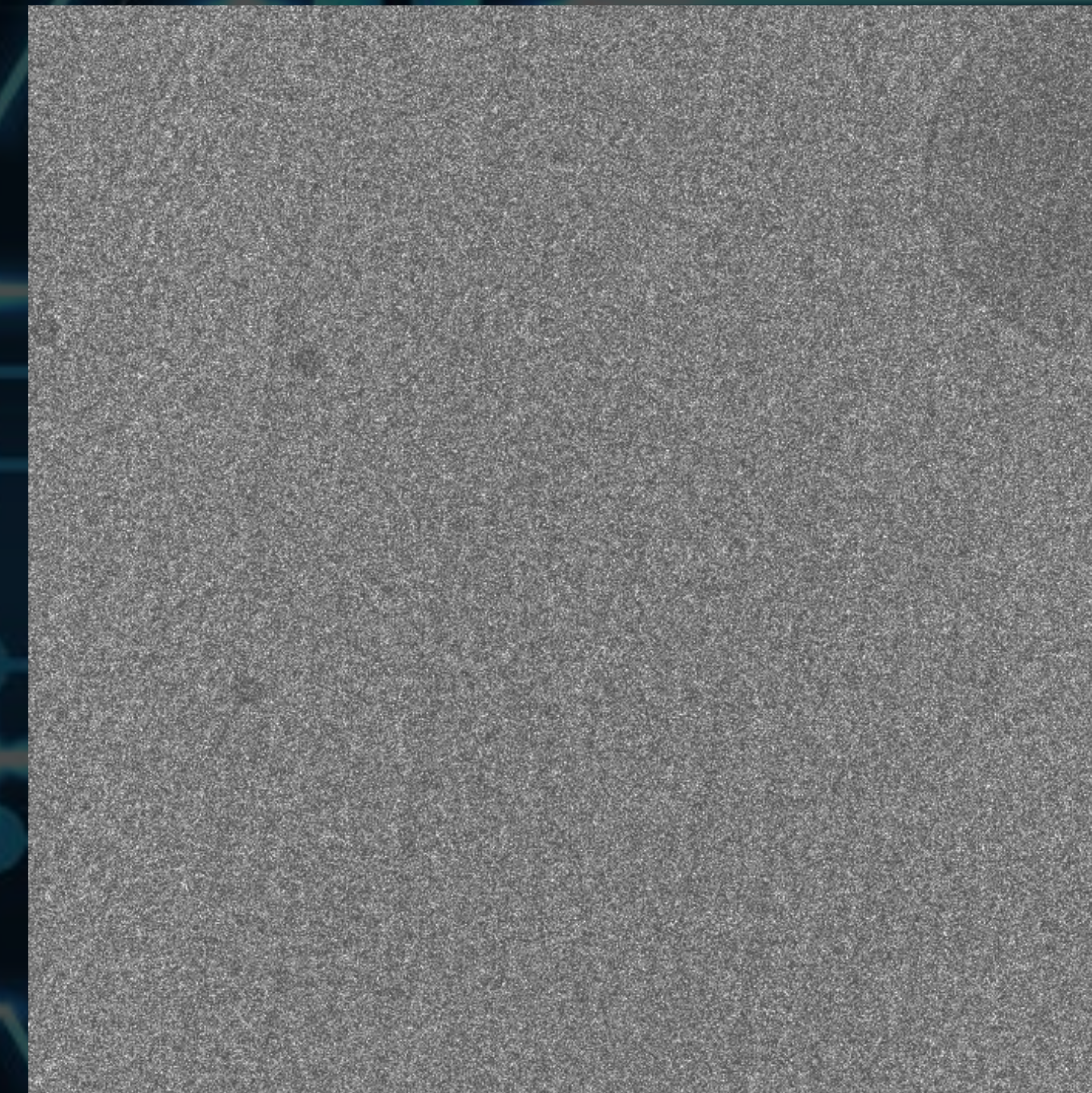


**WHAT IF SUFFICIENT
QUALITY TRAINING DATA
CAN'T BE IMAGED?**



FAST PROCESSES

WHAT IF SUFFICIENT
QUALITY TRAINING DATA
CAN'T BE IMAGED?



CRYO-TOMOGRAPHY



Alex
Krull

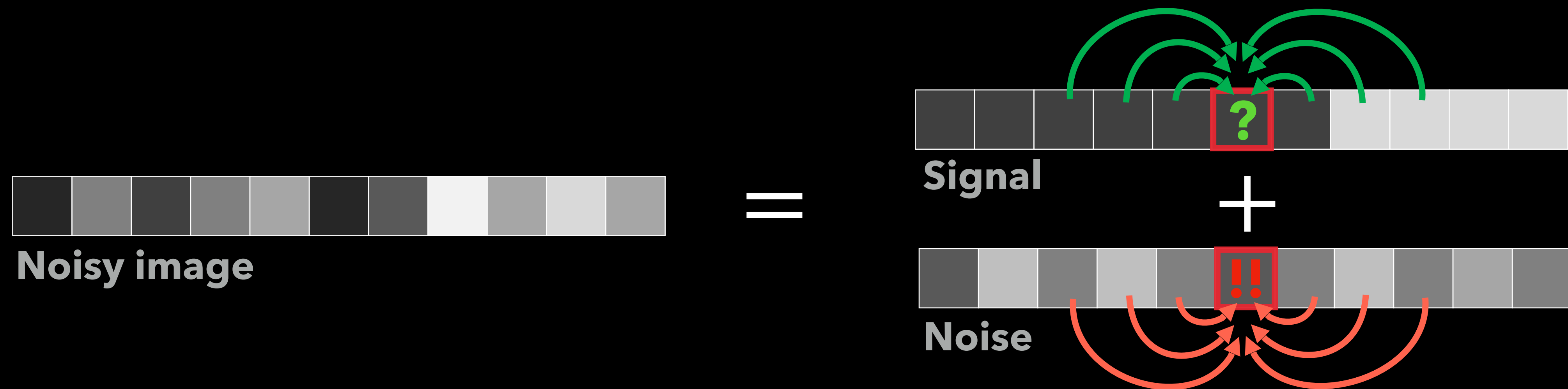


Tim-Oliver
Buchholz

Another new idea: Noise2Void



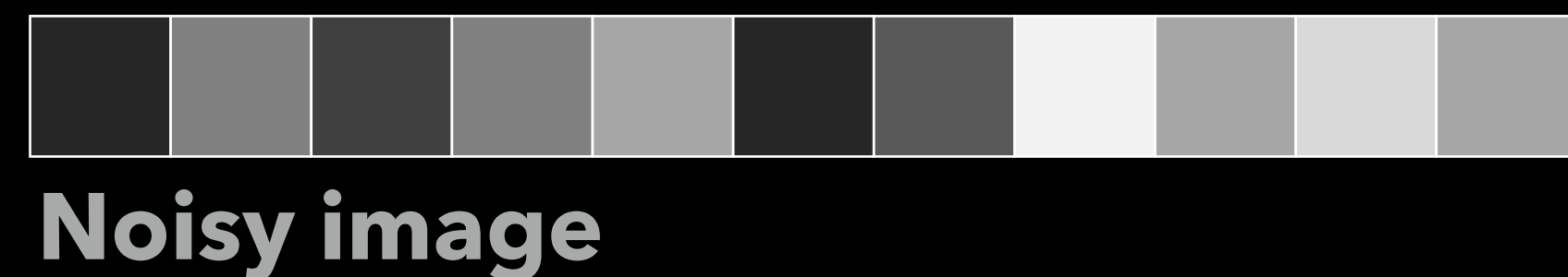
What if you'd only have one set of noisy images?



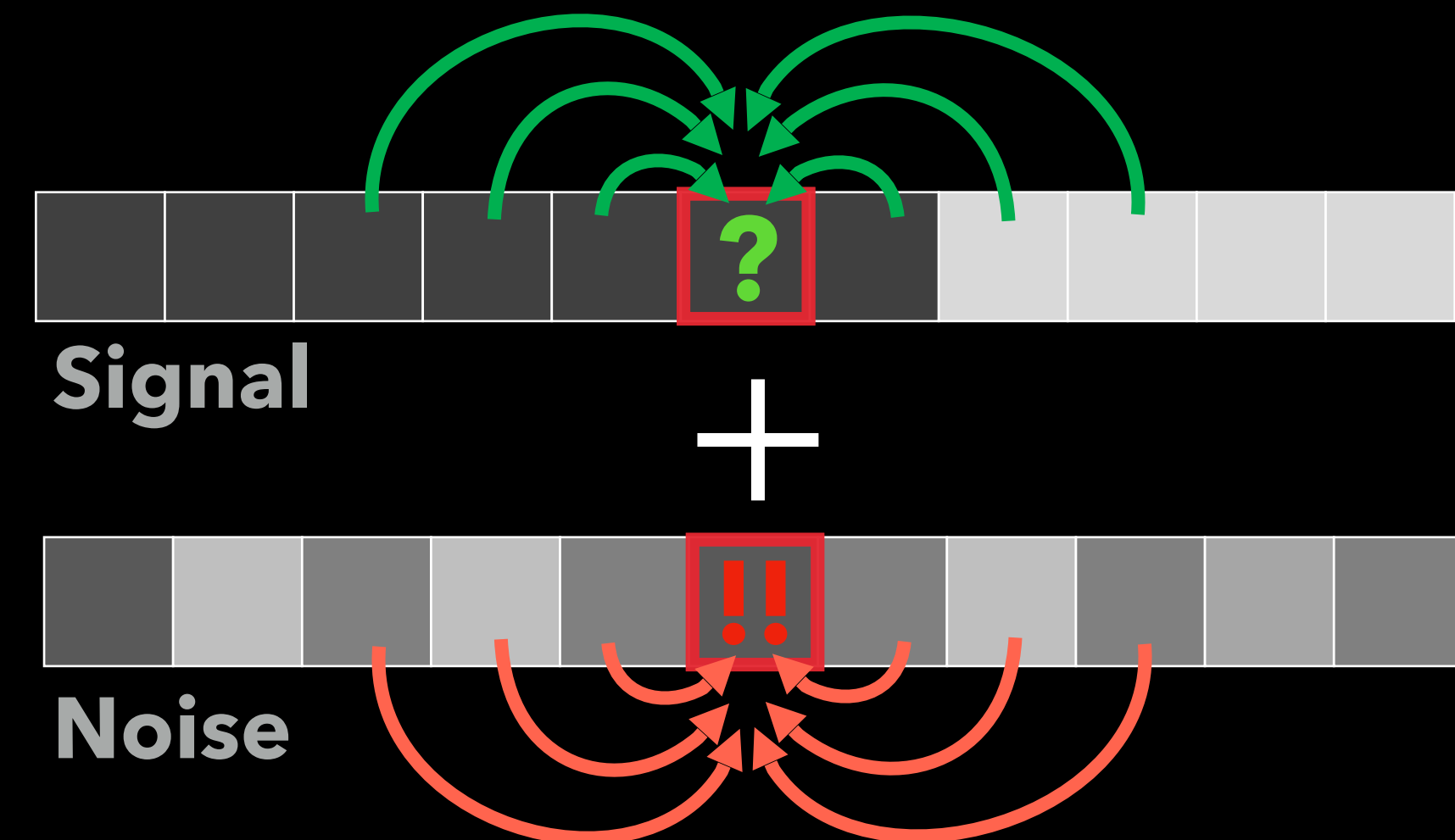
Another new idea:
Noise2Void



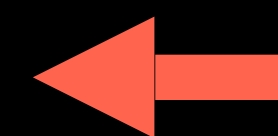
What if you'd only have one set of noisy images?



=

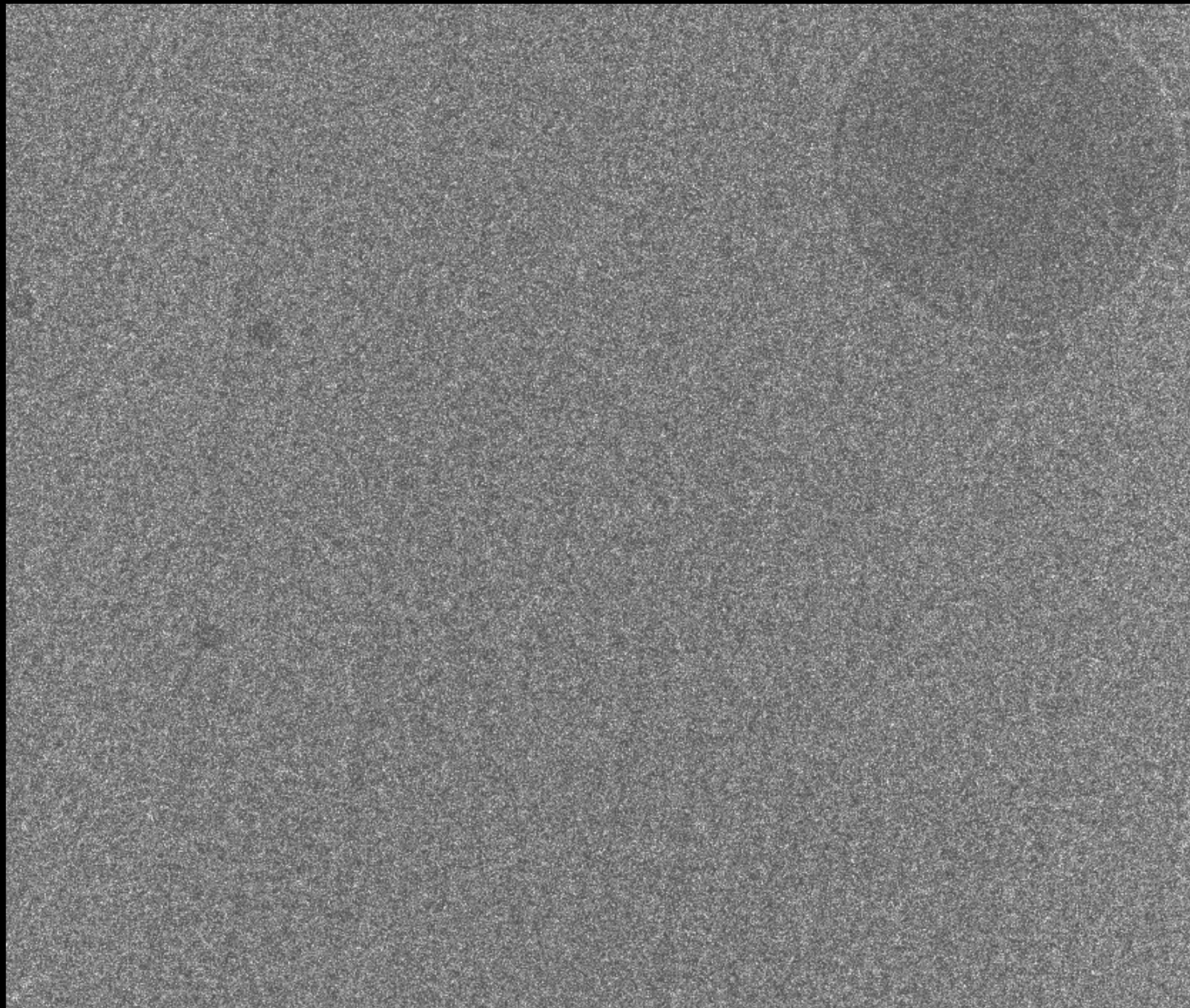


noise component of a pixel
cannot possibly be predicted!

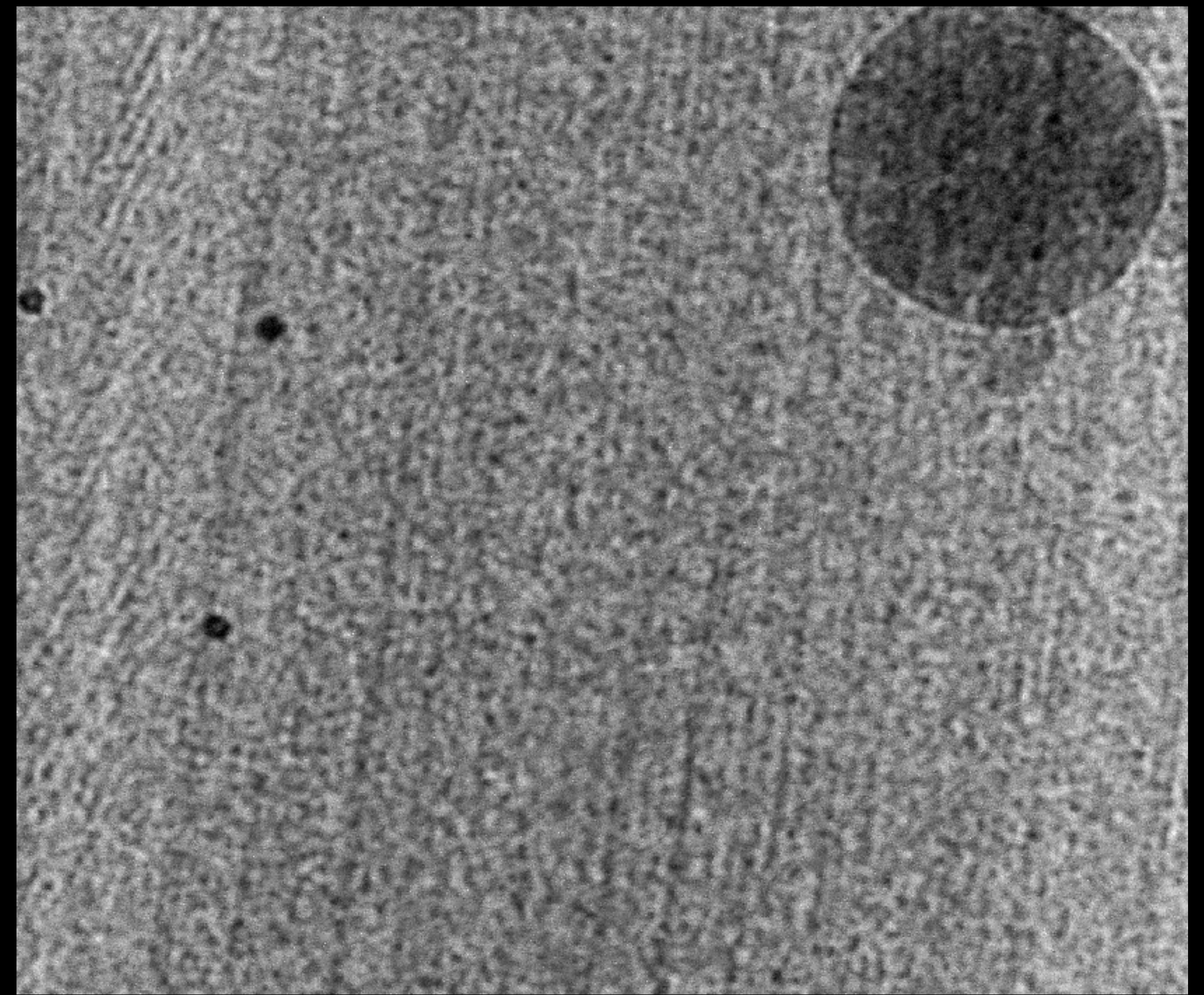


key assumption:
noise is independent per pixel
(given the signal)



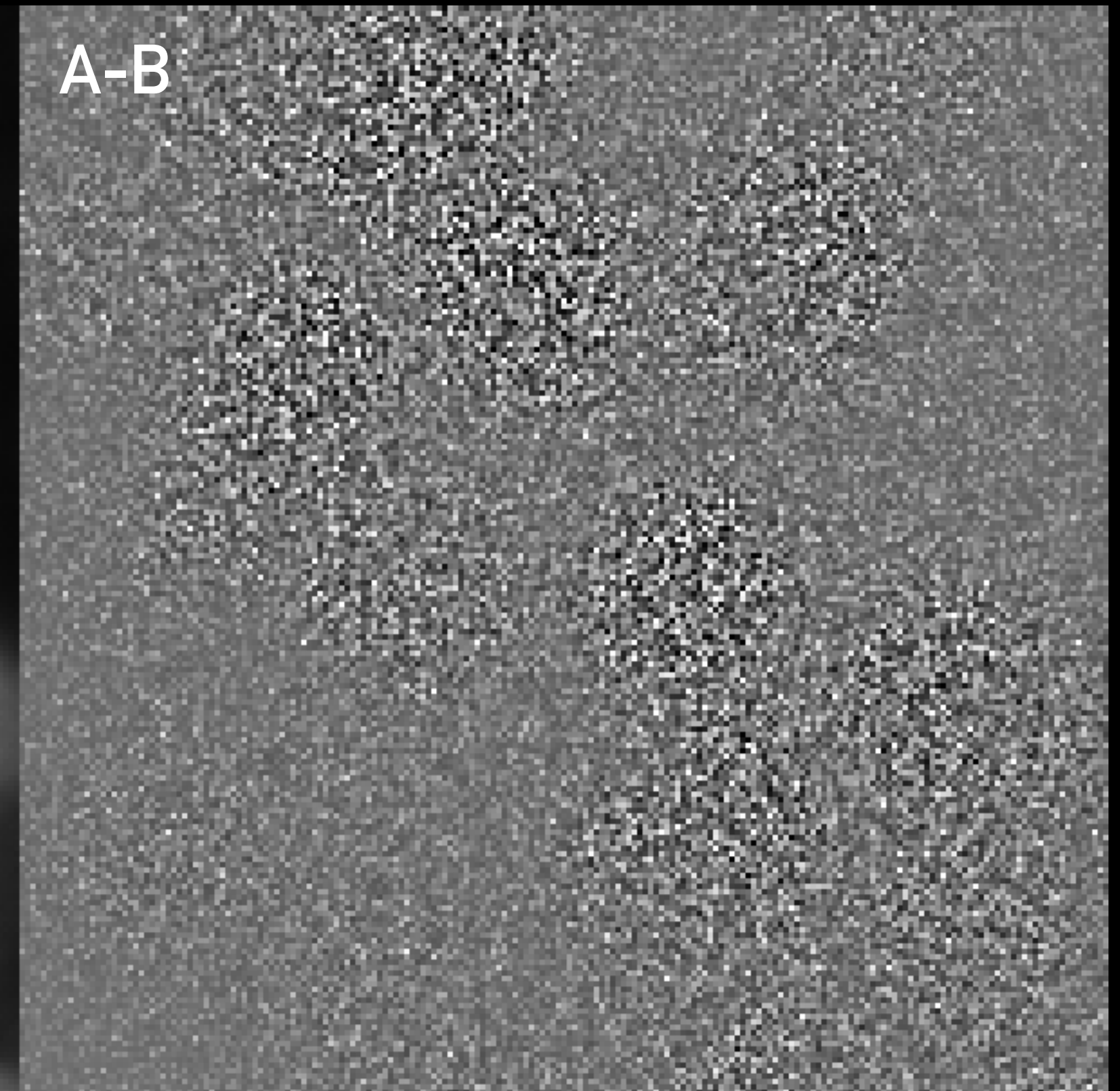
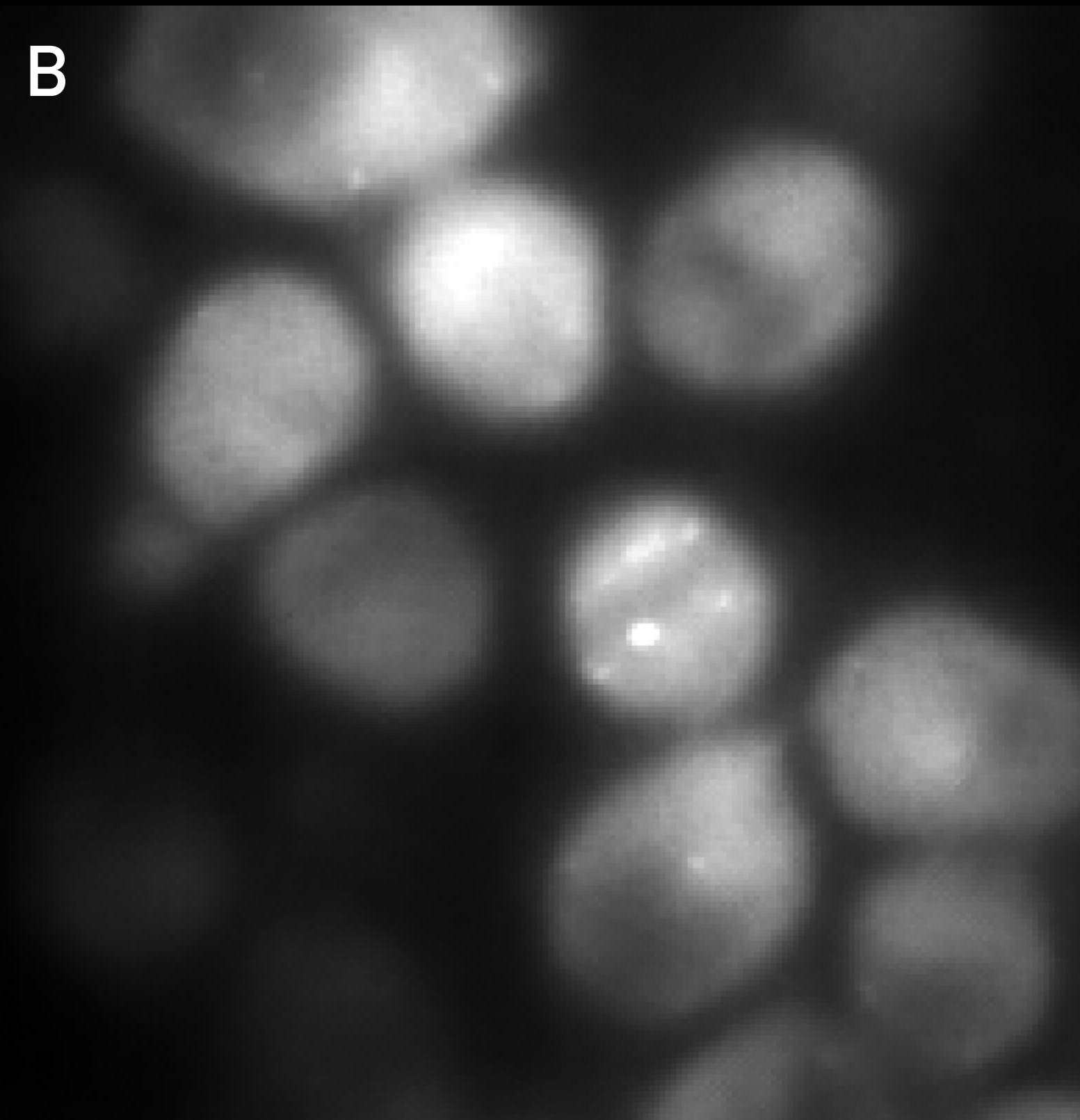
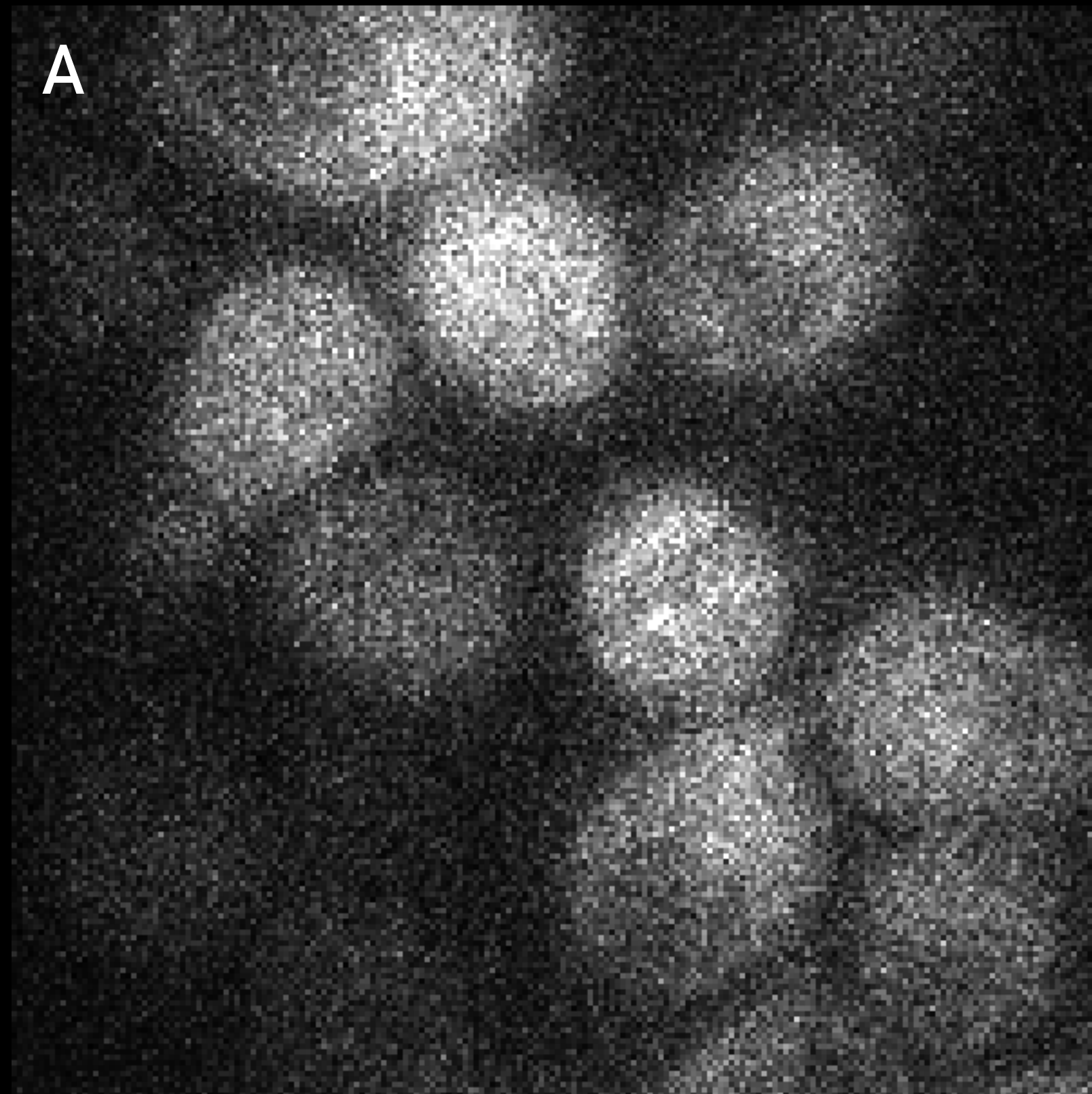


INPUT



NOISE2VOID

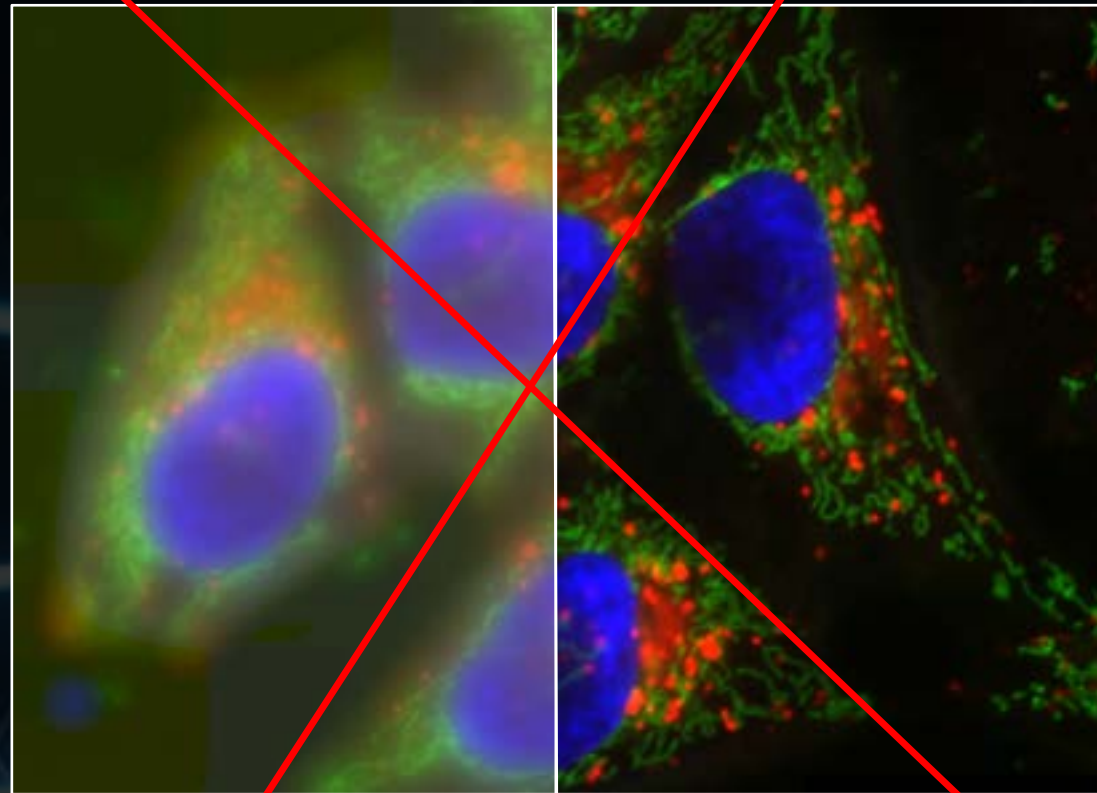
>> Data: **Stephanie Heinrich**, ETH Zurich



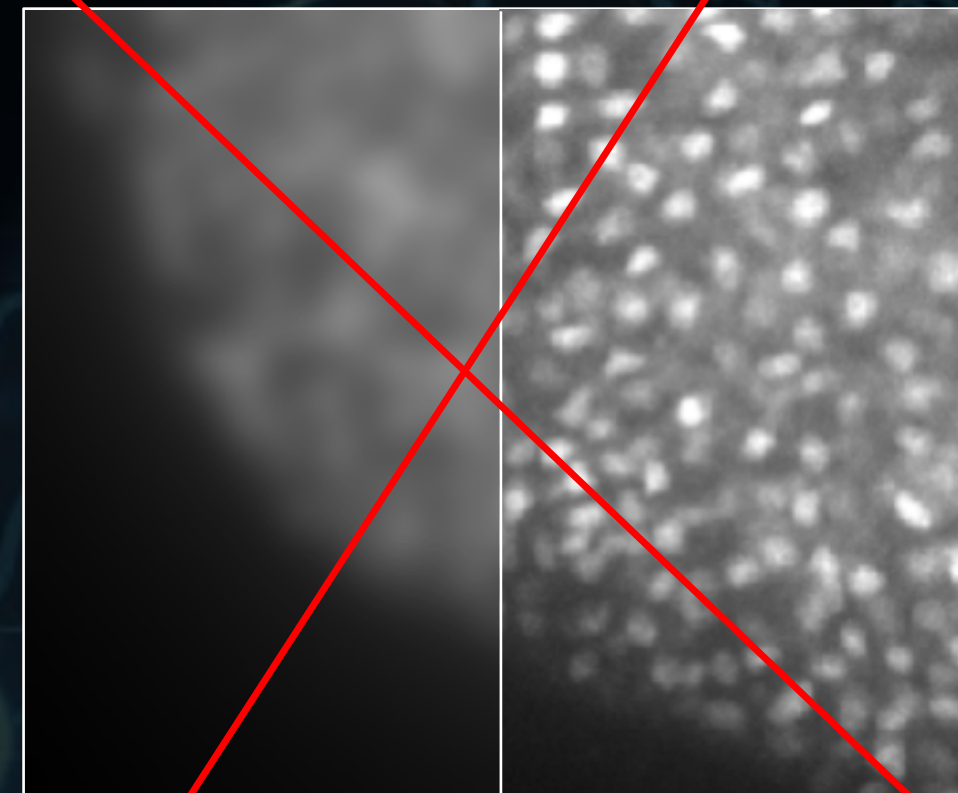
NOISE2VOID
OF COURSE ALSO USEFUL FOR LM DATA!

Krull et al., Noise2Void - Learning Denoising from Single Noisy Images, CVPR'19

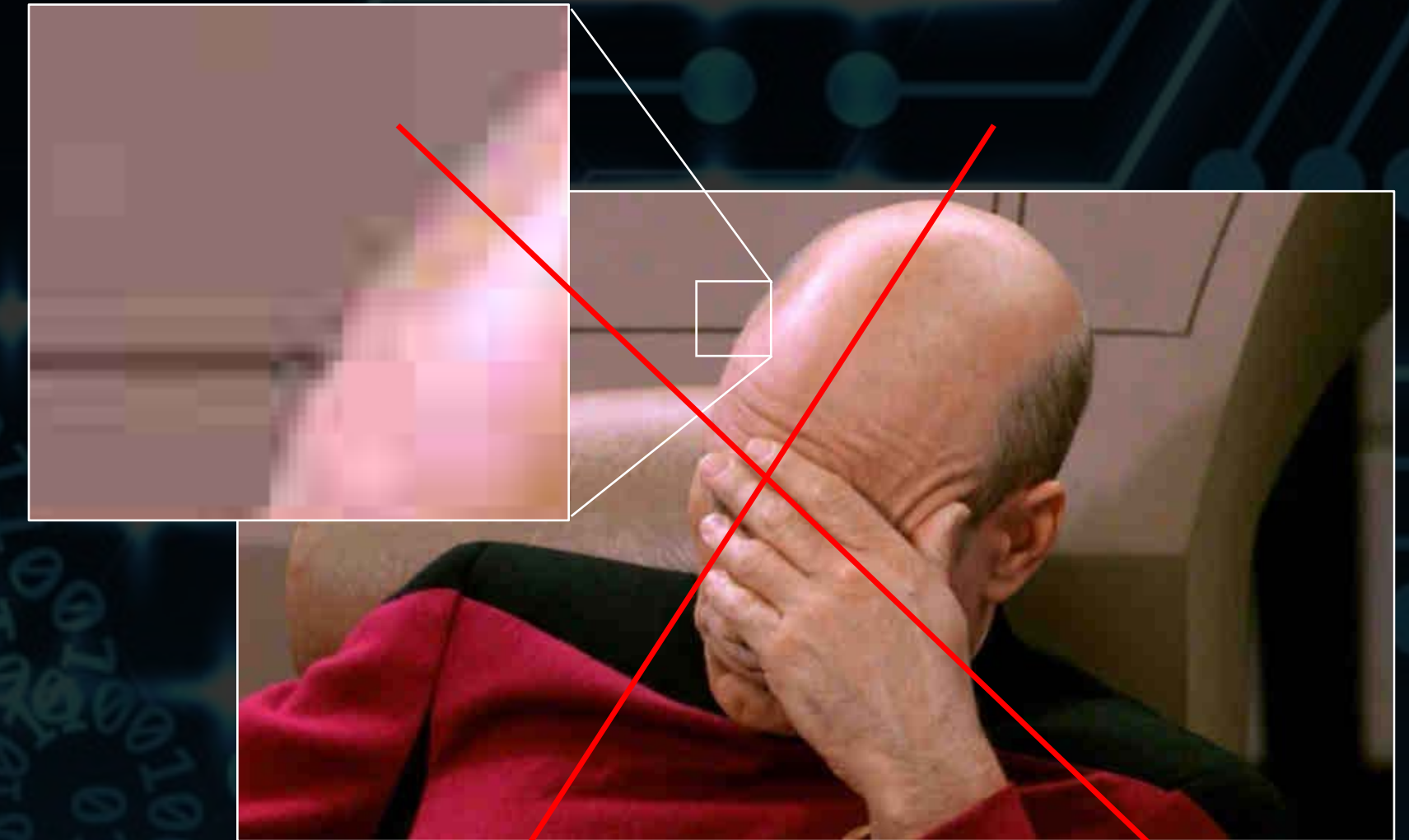




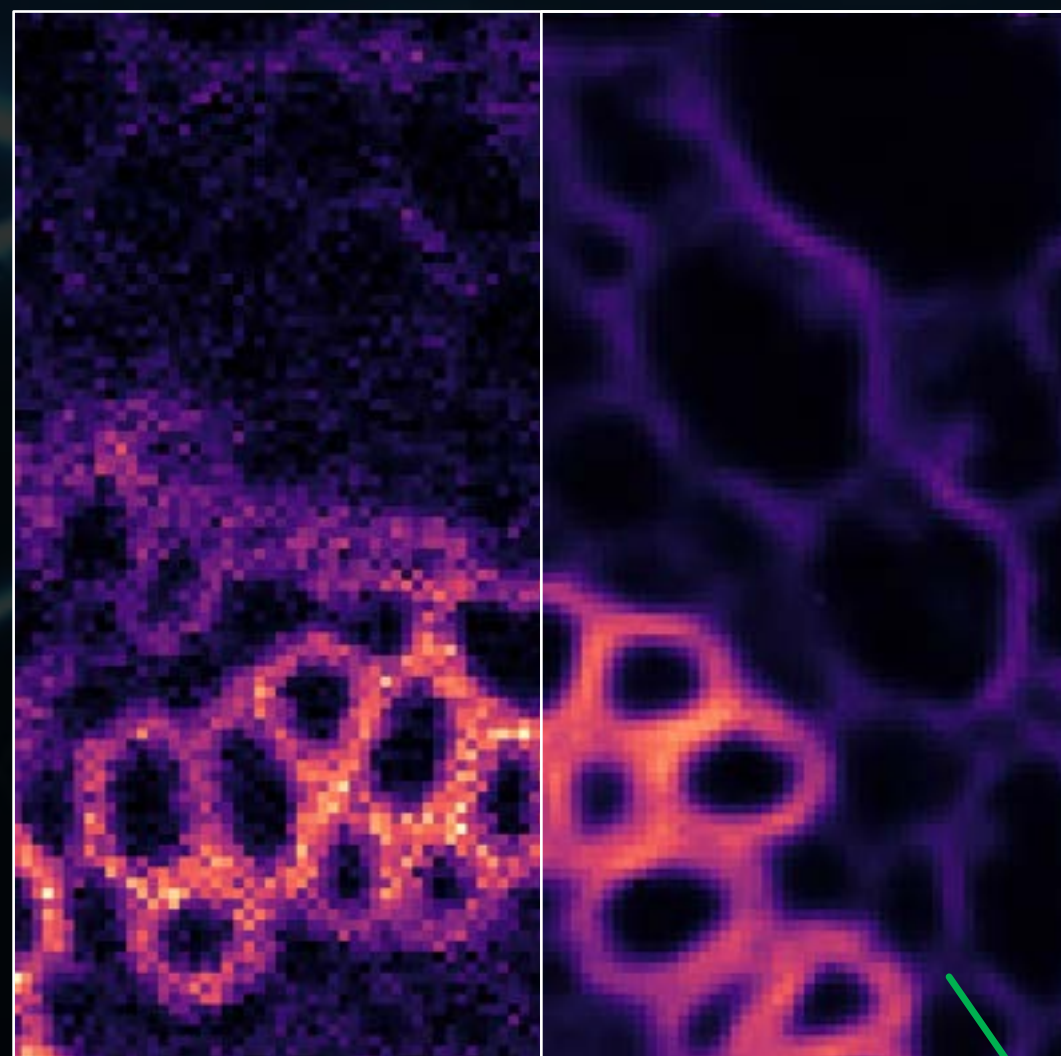
Computational clearing



Deconvolution



JPEG compression artifacts



per-pixel noise

- Low light images
- Shot noise
- Readout noise
- Non-structured noise

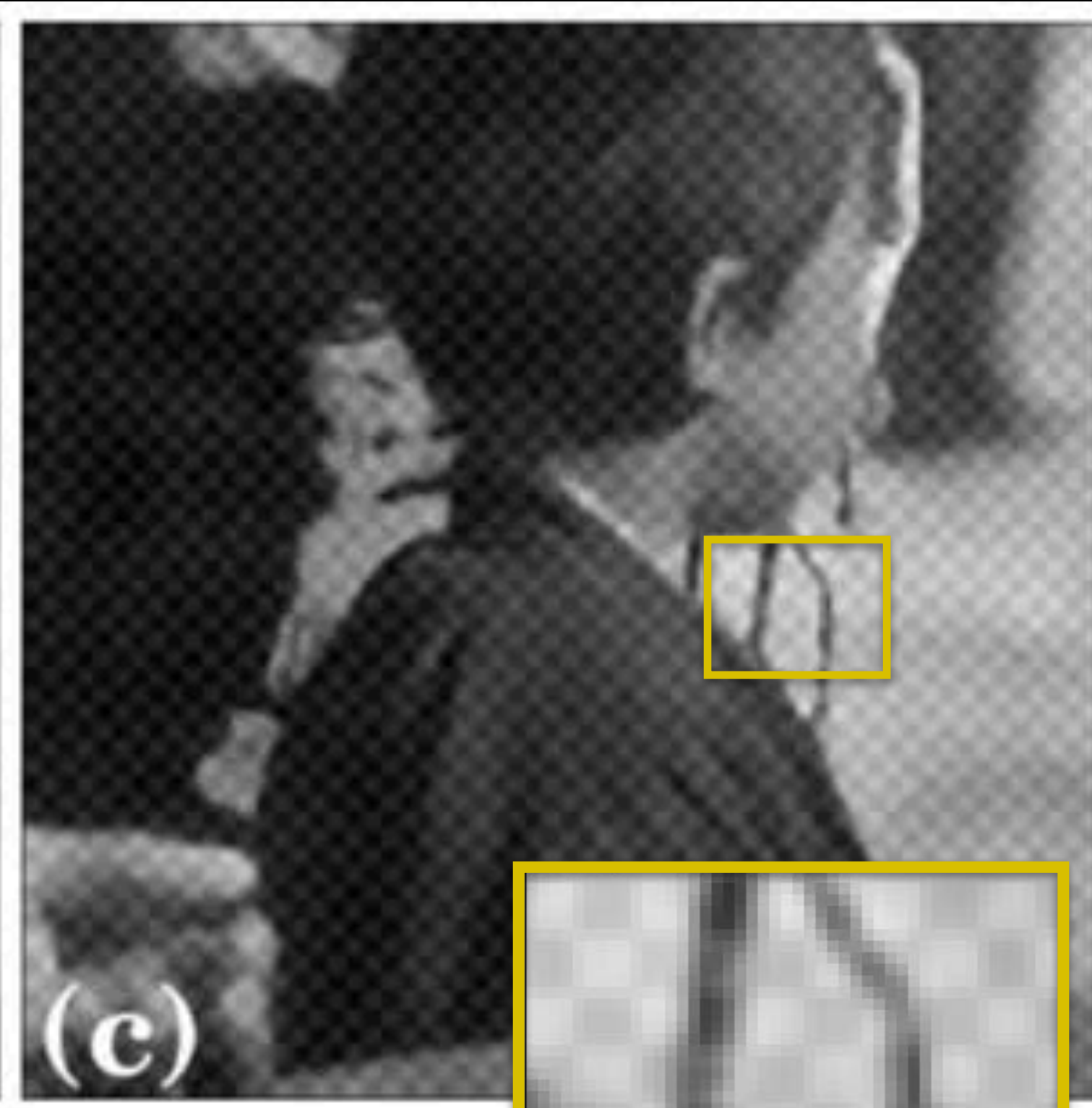
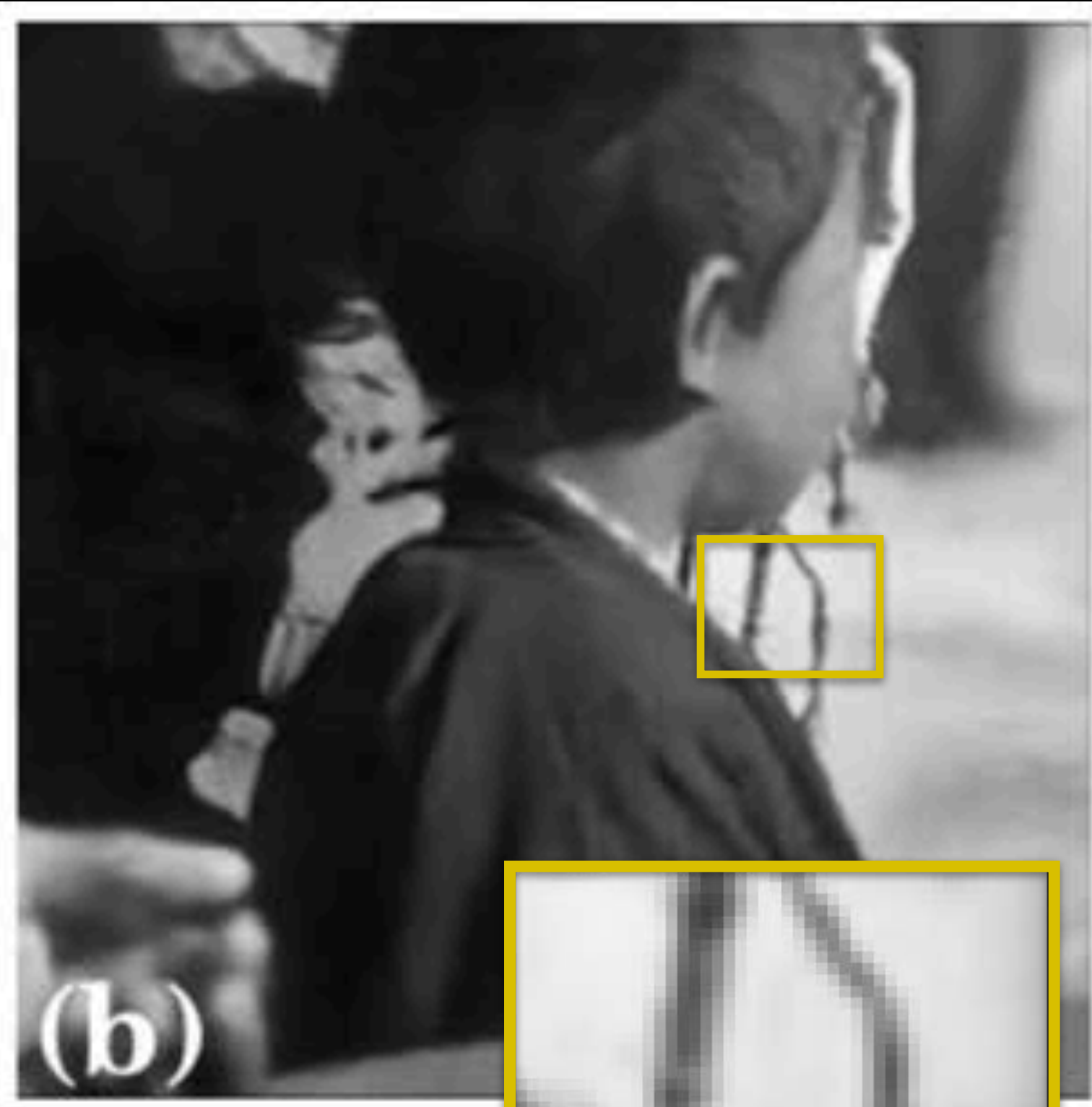
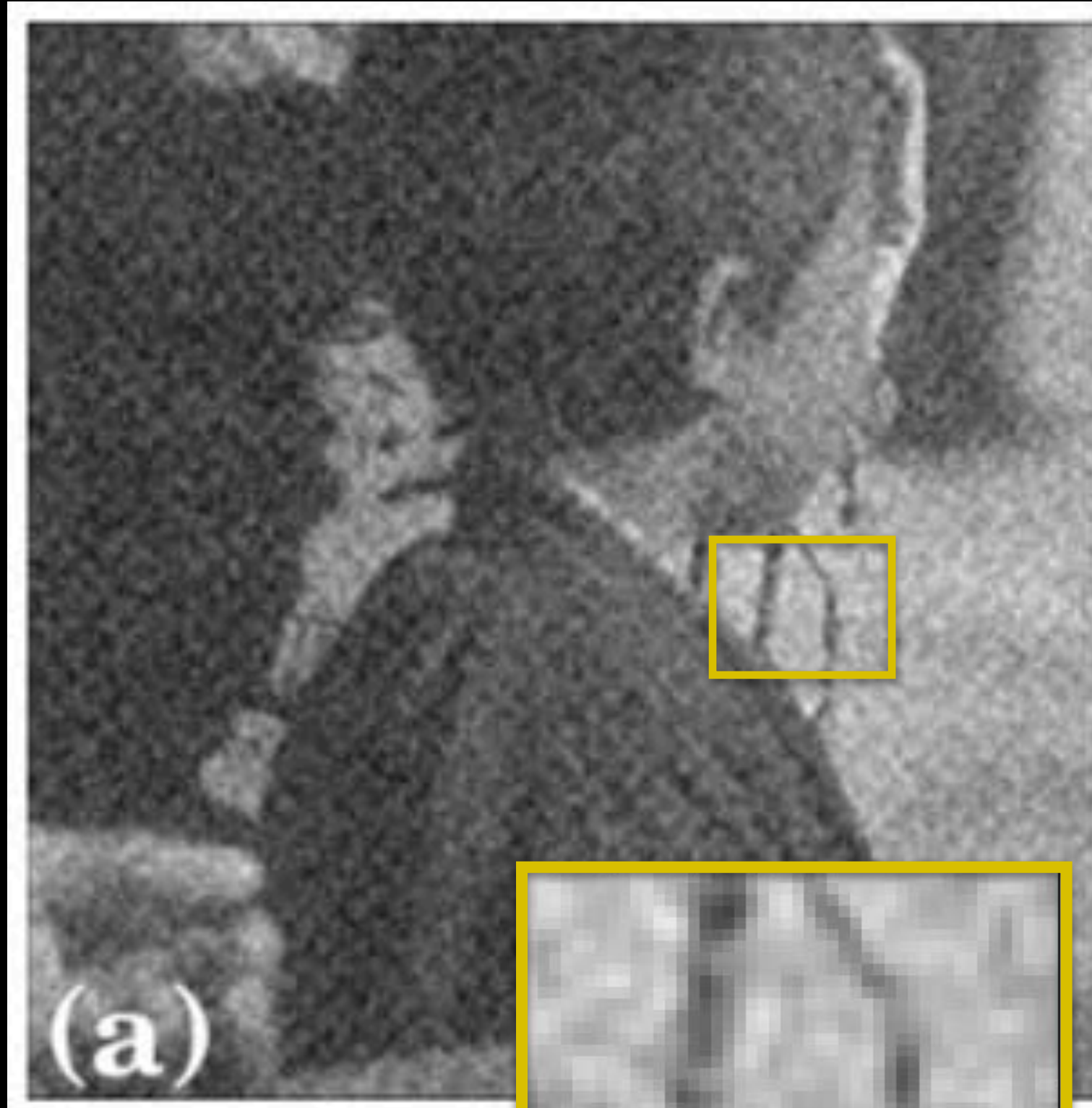
Take home message:

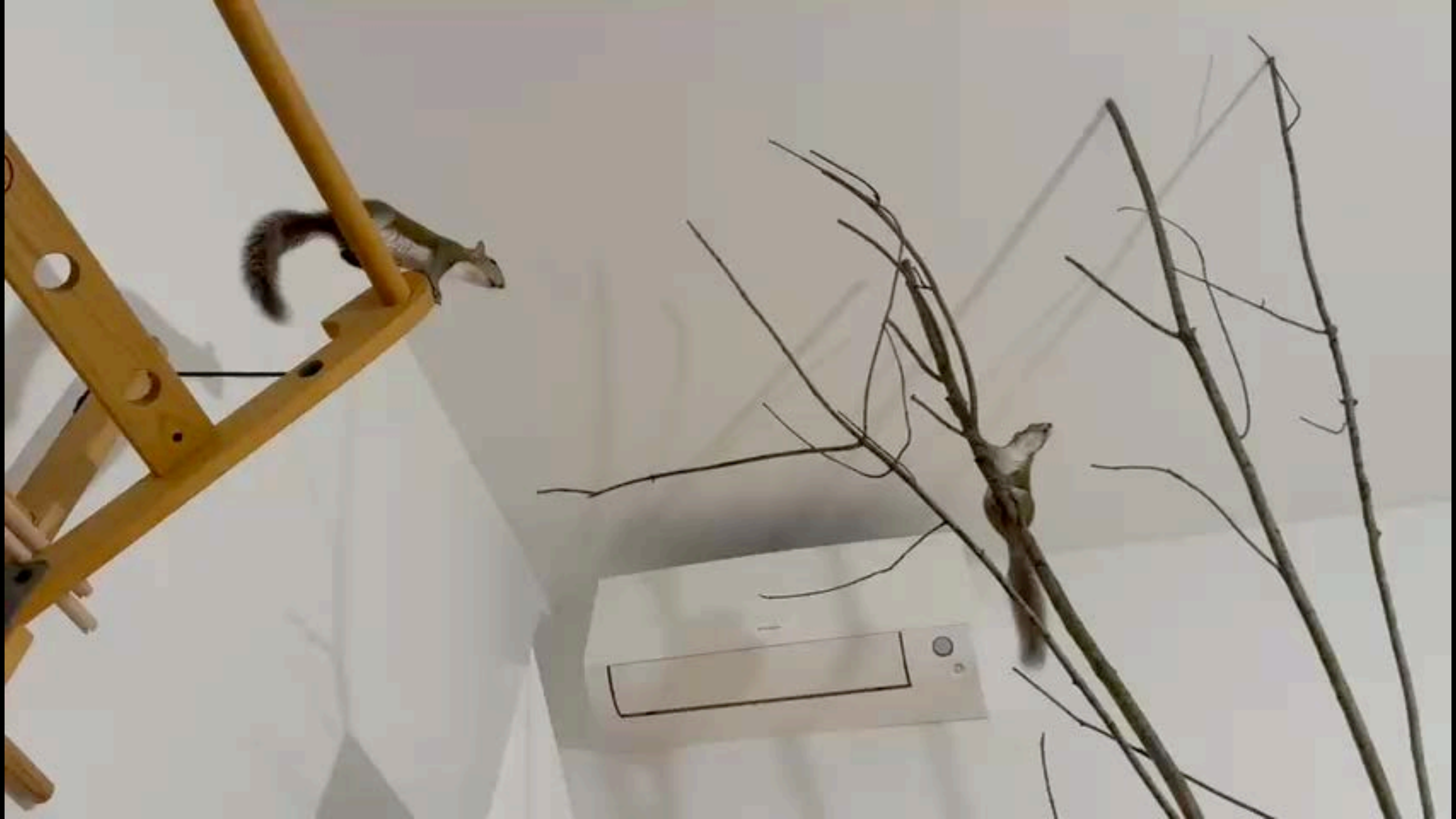
Noise2Void is as easy as it gets if reducing per-pixel noise is the goal!

INPUT

TRADITIONAL CARE

NOISE2VOID







juglab / CiliaAI-minicourse

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CiliaAI-minicourse

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T

Add file



<> Code

 fjug and claude	README: keep the program only up to Thursday's recap	941add · 2 hours ago	🕒 8 Commits
data	Add denoising exercise (BM3D and Noise2Void)	20 hours ago	
denoising	README: let students choose between Colab and local track	3 hours ago	
.gitignore	Ignore instructor-only material and Claude app outputs	19 hours ago	
README.md	README: keep the program only up to Thursday's recap	2 hours ago	
check_setup.py	Add setup check script for students	19 hours ago	

📖 README

<https://github.com/juglab/CiliaAI-minicourse>





CiliaAI mini-course: hands-on exercises

Material for the hands-on part of the Cilia-AI / CiliaConnect mini-course on image analysis and AI.

Before the course, please choose how you want to work (in the browser or on your own laptop) and get ready as described in [Before the course: get ready](#).

Program

About

Hands-on exercises for the Cilia-AI / CiliaConnect mini-course on image analysis and AI (denoising with BM3D and Noise2Void)

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